Research continuation and complete source proofs

Split-Zero Cohomology and Arithmetic Weight Control

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This reader integrates the Split-Zero carrier, finite arithmetic operator comparisons, chain retractions, homotopy control, and the current orthogonal boundary calculation. It retains the original arithmetic functions, measures, scalar actions, support fibres, and complete jets. Part I gives the local continuation with proofs. Part II preserves the complete selected source notes, their exact revisions, and their chronology.

Contents

Part I
Integrated mathematical continuation

1 Exact Split-Zero carriers and the local spectral bridge

Throughout this section, semiring homomorphisms preserve additive zero and multiplicative unit. Let R be a nonzero commutative ring and $\mathbb{B} = (\{0,1\}, \vee, \wedge)$. Define

$$G(R) = R \sqcup \{\tau\}, \qquad e = 0_R \in R \subset G(R).$$

Addition and multiplication of supported elements are those of R, including when their result is zero. The additional element satisfies

$$\tau + r = r + \tau = r, \quad \tau + \tau = \tau, \quad \tau r = r\tau = \tau, \quad \tau^2 = \tau.$$

Thus the semiring zero is τ, and $e \neq \tau$.

1.1 Scalar support and independent channel amplitudes

Proposition 1.1 (Pair carrier and scalar reflection)**.** *There is a semiring isomorphism*

$$\eta_R : G(R) \longrightarrow \{(0,0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}, \qquad \tau \mapsto (0,0), \quad r \mapsto (r,1).$$

The coordinate maps

$$p_R : G(R) \to R, \quad p_R(\tau) = 0, \quad p_R(r) = r,$$
$$\chi_R : G(R) \to \mathbb{B}, \quad \chi_R(\tau) = 0, \quad \chi_R(r) = 1$$

are semiring homomorphisms. The map p_R is universal among semiring homomorphisms from $G(R)$ to a ring.

Proof. In the product semiring, two supported pairs have sum $(r+s,1)$ and product $(rs,1)$. These identities retain support when the amplitude vanishes. The pair $(0,0)$ is an additive identity and an absorbing multiplicative zero. This proves closure and all operation identities, and the displayed map is bijective. Its coordinate projections give p_R, χ_R. If $f : G(R) \to A$ has ring target, then $e + e = e$ gives $f(e) + f(e) = f(e)$, hence $f(e) = 0_A$. Its restriction to the supported copy of R is therefore a unital ring homomorphism $h : R \to A$. The equality $f = hp_R$ holds on R and on τ, and determines h. Conversely every such h gives hp_R. □

The larger carrier $R \times \mathbb{B}$ contains $(1,0)$; its exact relation to $G(R)$ is the inclusion above. The pair presentation requires $R \times \{1\}$.

For a nontrivial bounded distributive lattice L, define

$$G_L(R) = \{(0,\lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L,$$
$$(r,\lambda) + (s,\mu) = (r+s, \lambda \vee \mu), \qquad (r,\lambda)(s,\mu) = (rs, \lambda \wedge \mu).$$

Write $z_\lambda = (0,\lambda)$, $\tau_L = z_{0_L}$, and $e_L = z_{1_L}$.

Proposition 1.2 (Lattice reflection with its original coefficient ring)**.** *The displayed carrier is a commutative semiring. The projections*

$$p_L : G_L(R) \to R, \qquad \chi_L : G_L(R) \to L$$

are respectively the universal ring reflection and the support homomorphism. In particular the amplitude codomain of $G_{\mathbb{B}^2}(\mathbb{C})$ is $\mathbb{C}$.

Proof. A summand with nonzero amplitude has top support, so its sum has top support. A factor with lower support has zero amplitude, so a product of lower support has zero amplitude. These facts prove closure. All laws restrict from the product semiring; distributivity uses that of L. Both projections preserve operations, zero and unit. If $f : G_L(R) \to A$ has ring target, $e_L + z_\lambda = e_L$ gives $f(z_\lambda) = 0$ by cancellation in A. Restriction to the supported copy of R is a ring homomorphism h and $f = hp_L$. Conversely each h gives this factorization. Surjectivity of p_L gives uniqueness. $\square$

Let $j : \mathbb{C} \to G(\mathbb{C})$ include the supported copy. It preserves addition, multiplication and unit; it takes ordinary zero to e, whereas the semiring zero is τ. Put

$$M = G(\mathbb{C})^2, \quad p^{(2)} = (p_\mathbb{C}, p_\mathbb{C}) : M \to \mathbb{C}^2, \quad \chi^{(2)} = (\chi_\mathbb{C}, \chi_\mathbb{C}) : M \to \mathbb{B}^2.$$

For $\epsilon \in \mathbb{B}$ let $u_0 = \tau$, $u_1 = e$.

Proposition 1.3 (Exact scalar-lattice and two-amplitude embeddings). *There are injective semiring homomorphisms*

$$b : G_{\mathbb{B}^2}(\mathbb{C}) \hookrightarrow M, \qquad B : G_{\mathbb{B}^2}(\mathbb{C}^2) \hookrightarrow M$$

defined by

$$\begin{aligned} b(c,(1,1)) &= (j(c), j(c)), \\ b(0,(\lambda_1,\lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}), \\ B((a,b),(1,1)) &= (j(a), j(b)), \\ B((0,0),(\lambda_1,\lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}). \end{aligned}$$

Their amplitude and support identities are

$$p^{(2)} b = \Delta_\mathbb{C} p_{\mathbb{B}^2}, \quad \chi^{(2)} b = \chi_{\mathbb{B}^2}, \qquad p^{(2)} B = p_{\mathbb{B}^2}, \quad \chi^{(2)} B = \chi_{\mathbb{B}^2},$$

where $\Delta_\mathbb{C}(c) = (c, c)$.

Proof. The pair presentation identifies M with

$$\left\{((a,b),(\lambda_1,\lambda_2)) \in \mathbb{C}^2 \times \mathbb{B}^2 : \lambda_1 = 0 \Rightarrow a = 0, \quad \lambda_2 = 0 \Rightarrow b = 0\right\},$$

with componentwise operations. The map b restricts the product semiring map $(c, \lambda) \mapsto ((c, c), \lambda)$. Lower support in its source forces $c = 0$, so it lands in the displayed subset. It retains the amplitude and both support bits and is injective. The map B restricts the identity coordinate map on $\mathbb{C}^2 \times \mathbb{B}^2$; lower support in its source forces both amplitudes to vanish, so it too lands in M and is injective. The prescriptions agree at the top-support zero. Both maps preserve zero and unit. Projection proves all four identities. $\square$

The element $(j(1), \tau) \in M$ is outside both images. There is no function $r : M \to G_{\mathbb{B}^2}(\mathbb{C}^2)$ satisfying both $p_{\mathbb{B}^2} r = p^{(2)}$ and $\chi_{\mathbb{B}^2} r = \chi^{(2)}$: it would send that element to $((1,0),(1,0))$, outside its target carrier. This identifies the precise obstruction to a reverse map preserving both projections.

The synchronized carrier has the exact map

$$G(\mathbb{C}^2) \cong G(\mathbb{C}) \times_\mathbb{B} G(\mathbb{C}) \hookrightarrow M.$$

It takes τ to (τ, τ) and a supported (a, b) to $(j(a), j(b))$. Equality of the two support characters leaves exactly those cases, proving bijectivity onto the fibre product. Operations agree componentwise. Its image has masks $(0,0)$, $(1,1)$, while the other masks in M carry independent mixed amplitudes.

The map $p^{(2)}$ is the universal ring reflection of M. Indeed, (e, e) is additively idempotent, so every ring-target map kills it. Adding (e, e) to $x \in M$ gives the fully supported element of amplitude $p^{(2)}x$. The map is consequently determined by a ring homomorphism from the supported copy of $\mathbb{C}^2$, proving the universal property. A semiring isomorphism $G_{\mathbb{B}^2}(\mathbb{C}) \cong M$ would induce a ring isomorphism $\mathbb{C} \cong \mathbb{C}^2$. This is impossible because $(1, 0)$ is a nontrivial idempotent of $\mathbb{C}^2$, and $\mathbb{C}$ has only $0, 1$. The positive relation between the original carriers is b, with diagonal amplitude projection.

1.2 Tesserine sectors and a supported critical fixed locus

Reserve τ for split absence and use κ for conjugation. Let

$$\sigma(s) = 1 - s, \quad \kappa(s) = \bar{s}, \quad K_4 = \{1, \sigma, \kappa, \sigma\kappa\}.$$

The real tesserine algebra and its complexification are

$$A_{\mathbb{R}} = \mathbb{R}[K_4] = \mathbb{R}[\sigma, \kappa]/(\sigma^2 - 1, \kappa^2 - 1), \qquad A_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} A_{\mathbb{R}}.$$

The polynomial algebra in this presentation is commutative.

Proposition 1.4 (Character algebra and quartet coordinates). *For $\epsilon, \delta \in \{1, -1\}$ let*

$$E_{\epsilon\delta} = \tfrac{1}{4}(1 + \epsilon\sigma)(1 + \delta\kappa).$$

Character evaluation gives an algebra isomorphism

$$\mathcal{F} : A_{\mathbb{C}} \to \mathbb{C}^4, \qquad x \mapsto (\mathrm{ev}_{\epsilon,\delta}(x))_{\epsilon,\delta},$$

with inverse $(d_{\epsilon\delta}) \mapsto \sum d_{\epsilon\delta} E_{\epsilon\delta}$. For the exact quartet element

$$Q(s) = s \cdot 1 + (1 - s)\sigma + \bar{s}\,\kappa + (1 - \bar{s})\sigma\kappa, \qquad s = \beta + i\gamma,$$

the four character values are

$$c_{++} = 2, \qquad c_{+-} = 0, \qquad c_{-+} = 2(2\beta - 1), \qquad c_{--} = 4i\gamma.$$

Projection to the last two specified characters is a unital algebra map $\pi : A_{\mathbb{C}} \to \mathbb{C}^2$. The composite πQ is a real affine bijection

$$\mathbb{C} \to \mathbb{R} \times i\mathbb{R}, \qquad s \mapsto (a, b) = (c_{-+}(s), c_{--}(s)), \qquad s = \tfrac{1}{2} + \tfrac{1}{4}(a + b).$$

Proof. Evaluation of $E_{\epsilon\delta}$ at (ϵ', δ') is $\frac{1}{4}(1 + \epsilon\epsilon')(1 + \delta\delta')$. It is one when both signs agree and zero otherwise. This proves the two inverse formulas and all orthogonal idempotent identities, retaining the factors of four. Character evaluation preserves the algebra operations. Directly,

$$\begin{aligned}
c_{++} &= s + (1 - s) + \bar{s} + (1 - \bar{s}) = 2, \\
c_{+-} &= s + (1 - s) - \bar{s} - (1 - \bar{s}) = 0, \\
c_{-+} &= s - (1 - s) + \bar{s} - (1 - \bar{s}) = 2(s + \bar{s}) - 2 = 4\beta - 2, \\
c_{--} &= s - (1 - s) - \bar{s} + (1 - \bar{s}) = 2(s - \bar{s}) = 4i\gamma.
\end{aligned}$$

The proposed inverse recovers $\beta + i\gamma$. Conversely every real a and purely imaginary b is recovered by substituting that inverse into the two character expressions. $\square$

The quartet image has real affine dimension two. Its real linear span has dimension three, spanned by E_{++}, E_{-+}, iE_{--}, retaining the constant character coordinate. The real tesserine algebra has dimension four. The maps $Q, \mathcal{F}, \pi$ and the displayed affine inverse specify the exact relationship of these objects.

Proposition 1.5 (Supported sector lift and fixed locus). *Define*

$$\Psi : \mathbb{C} \to M, \qquad \Psi(s) = (j(c_{-+}(s)), j(c_{--}(s))).$$

It has $\chi^{(2)}\Psi(s) = (1, 1)$ for all s. The $G(\mathbb{C})$-linear automorphisms

$$T_\sigma(x, y) = ((-1)x, (-1)y), \qquad T_\kappa(x, y) = (x, (-1)y)$$

of the free semimodule M give

$$\Psi(\sigma s) = T_\sigma \Psi(s), \qquad \Psi(\kappa s) = T_\kappa \Psi(s).$$

Here $-1 = j(-1)$ is a supported unit. Their product satisfies

$$\mathrm{Fix}(T_\sigma T_\kappa) = \{\tau, e\} \times G(\mathbb{C}), \qquad \Psi^{-1}\,\mathrm{Fix}(T_\sigma T_\kappa) = \{s : \Re s = \tfrac{1}{2}\}.$$

Proof. The sector formulas give $(a, b)(1 - s) = (-a, -b)$ and $(a, b)(\bar{s}) = (a, -b)$. Multiplication by $j(-1)$ negates supported amplitudes and fixes τ. It preserves addition, commutes with scalar multiplication and squares to identity. This proves the automorphism and equivariance statements. Its fixed points are τ and supported $j(c)$ with $c = -c$. Since $2 \neq 0$ in $\mathbb{C}$, the latter is precisely e. The product action is $((-1)x, y)$, proving its fixed locus. On the image of Ψ the first coordinate is supported and is fixed exactly when $4\beta - 2 = 0$. □

The scalar-lattice channel atoms have images $b(z_{10}) = (e, \tau)$ and $b(z_{01}) = (\tau, e)$. The separate support swap has fixed masks $(0, 0), (1, 1)$. Since $\chi^{(2)}\Psi$ is constantly $(1, 1)$, it cannot detect the critical line by that swap. The support action induced by $T_\sigma T_\kappa$ is identity; its amplitude action is $(-a, b)$. The preceding proposition supplies the exact fixed-locus test with the first-channel supported zero retained.

1.3 Exact sector, holonomy and positive-moment maps

Fix a prime p, $\ell_p = \log p > 0$, and $u \in S^1$. For an arithmetic packet, take $u = \psi(\mathrm{Frob}_p)$ with $L/\mathbb{Q}$ finite abelian, p unramified, and $\psi : \mathrm{Gal}(L/\mathbb{Q}) \to S^1$ a character. All powers use $p^w = \exp(w\ell_p)$ with the positive real logarithm.

Proposition 1.6 (Sector-to-holonomy map with constants retained). *The analytic function*

$$H_{p,u} : \mathbb{C}^2 \to \mathbb{C}^\times, \qquad H_{p,u}(a, b) = u \exp\left(\frac{\ell_p}{4}(a + b)\right)$$

gives a function $H_{p,u}p^{(2)}$ on M. On the quartet image,

$$H_{p,u}(c_{-+}(s), c_{--}(s)) = up^{s-1/2} =: z_{p,u}(s),$$

$$\frac{\log |z_{p,u}(s)|}{\ell_p} = \frac{c_{-+}(s)}{4} = \Re s - \tfrac{1}{2}, \qquad |z_{p,u}(s)|^2 - 1 = \exp\left(\frac{\ell_p}{2} c_{-+}(s)\right) - 1.$$

Proof. The exact inverse for the sector coordinates gives $s - \frac{1}{2} = (a + b)/4$. Substitution proves the exponential identity. On that image a is real and b is purely imaginary. Since $|u| = 1$, taking absolute values yields $\exp(\ell_p a/4)$. Taking its real logarithm and, separately, its square proves the two remaining identities. □

The radial exponent is $c_{-+}/4$; the exponent $c_{-+}/2$ belongs to the squared modulus. The map $H_{p,u}$ is used with its analytic-function type, without any assertion that it preserves semiring addition.

Proposition 1.7 (Rank-one packet). *For $z \in \mathbb{C}$ define*

$$v_2(z) = \begin{pmatrix} 1 \\ z \end{pmatrix}, \quad T_2(z) = v_2(z)v_2(z)^* = \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix}, \quad P_z(w) = w - \bar{z}.$$

Then $T_2(z)$ is Hermitian positive semidefinite of rank one and

$$\ker T_2(z) = \mathbb{C}(-\bar{z}, 1)^\mathsf{T}.$$

It is Toeplitz exactly when $|z| = 1$, equivalently when the unique root of P_z lies on S^1. For $z = z_{p,u}(s)$ this is equivalent to $\Re s = \frac{1}{2}$.

Proof. For $x \in \mathbb{C}^2$, $x^* T_2(z)x = |v_2(z)^* x|^2 \geq 0$. The vector $v_2(z)$ is nonzero and hence its outer product has rank one. The adjoint kernel equation is $x_0 + \bar{z}x_1 = 0$, giving the displayed space. A two-by-two matrix is Toeplitz exactly when its diagonal entries agree, here $1 = |z|^2$. The root $\bar{z}$ has modulus $|z|$. The final equivalence follows from the preceding proposition and $\ell_p > 0$. □

Theorem 1.8 (Positive finite packets and their exact diagonal defect). *Let $r \geq 1$, $\alpha_j > 0$, and $z_j \in \mathbb{C}$. For $n \geq 1$ put*

$$v_n(z) = (1, z, \ldots, z^{n-1})^\intercal, \qquad M_n = \sum_{j=1}^{r} \alpha_j v_n(z_j) v_n(z_j)^*.$$

The matrix M_n is Hermitian positive semidefinite. For $n \geq 3$, it is Toeplitz exactly when every $|z_j| = 1$. With matrix indices starting at zero,

$$\mathcal{D}(M_n) := (M_n)_{22} - 2(M_n)_{11} + (M_n)_{00} = \sum_{j=1}^{r} \alpha_j (|z_j|^2 - 1)^2.$$

For these packets and $n \geq 3$, vanishing of $\mathcal{D}$ already characterizes Toeplitzness. If the z_j are pairwise distinct and $r \leq n$, the rank is r. If $r < n$, coefficient vectors identify the kernel with

$$\left(\prod_{j=1}^{r} (w - \bar{z}_j) \right) \mathbb{C}[w]_{\leq n-1-r} \subset \mathbb{C}[w]_{\leq n-1}.$$

In particular $r = n - 1$ gives corank one and exactly the conjugated roots $\bar{z}_1, \ldots, \bar{z}_{n-1}$.

Proof. The identity

$$x^* M_n x = \sum_j \alpha_j |v_n(z_j)^* x|^2$$

proves positivity and shows that its kernel is the simultaneous kernel of all $v_n(z_j)^*$. The first three diagonal entries are $\sum \alpha_j$, $\sum \alpha_j |z_j|^2$, and $\sum \alpha_j |z_j|^4$. Their displayed linear combination gives the defect without rescaling any weight. Toeplitzness equates these entries, forcing zero defect. Every weight is positive and every square nonnegative, hence all $|z_j| = 1$. Conversely that condition gives $(M_n)_{k\ell} = \sum_j \alpha_j z_j^{k-\ell}$, a Toeplitz matrix. The same argument proves the zero-defect characterization.

Let V have columns $v_n(z_j)$. Its first r rows have Vandermonde determinant $\prod_{i<j}(z_j - z_i) \neq 0$, so V has rank r. The established kernel identity is ker $M_n =$ ker V^*, proving the rank. For $P_x(w) = \sum_{k=0}^{n-1} x_k w^k$, the kernel equations say $P_x(\bar{z}_j) = 0$. Distinct linear factors divide P_x exactly when these equations hold: division by one factor preserves the other root equations since that factor is nonzero at those roots. This proves the product and degree bound. At $r = n - 1$, the remaining factor is a scalar; for a nonzero kernel vector it is nonzero, so there are no further roots. ☐

The size bound is sharp for general positive packets. At $n = 2$, the data $(\alpha_1, \alpha_2) = (4, 1)$ and $(z_1, z_2) = (1/2, 2)$ give

$$M_2 = 4 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This is positive definite and Toeplitz, while neither atom is on S^1. At size three the diagonal is $(5, 5, 65/4)$ and the defect is $45/4$. One atom retains the two-by-two criterion. Zero-weight atoms are invisible; repeated atoms combine their weights before applying the distinct-column rank count.

The exact sector defect is therefore

$$\mathcal{D}(M_n) = \sum_{j=1}^{r} \alpha_j \left[\exp\left(\frac{\log p_j}{2} c_{-+}(s_j) \right) - 1 \right]^2.$$

The function chain

$$M \xrightarrow{p^{(2)}} \mathbb{C}^2 \xrightarrow{H_{p,u}} \mathbb{C}^\times \xrightarrow{z \mapsto v_n(z) v_n(z)^*} \mathrm{Herm}_{\geq 0}(n)$$

has explicit maps at every step. Positive finite summation supplies the preceding theorem.

1.4 The theta-packet form and its scalar comparison

Proposition 1.9 (Eigenpair form quotient and sector channel)**.** *For every complex d-by-d matrix A, Hermitian matrix G, and nonzero eigenvector v with $Av = \rho v$, let $W = A^*G + GA - G$. Then*

$$v^*Wv = (2\Re\rho - 1)v^*Gv.$$

On the domain of these data where $v^*Gv \neq 0$, the quotient

$$\omega(A, G, v, \rho) = \frac{v^*Wv}{v^*Gv} = 2\Re\rho - 1 = \frac{c_{-+}(\rho)}{2}$$

is real. Its associated holonomy satisfies

$$|z_{p,u}(\rho)|^2 = \exp(\ell_p\omega), \qquad z_{p,u}(\rho) = u\exp\left(\frac{\ell_p}{4}(c_{-+}(\rho) + c_{--}(\rho))\right).$$

Proof. Since $(Av)^* = \bar{\rho}\, v^*$,

$$v^*(A^*G + GA - G)v = (Av)^*Gv + v^*G(Av) - v^*Gv = (\bar{\rho} + \rho - 1)v^*Gv.$$

This proves the unquotiented identity even when the right factor is zero. Division on the stated domain gives ω, and $c_{-+}(\rho) = 4\Re\rho - 2$ gives the factor $1/2$. The holonomy identities follow from the exact sector exponential above. □

This is the typed map from an eigenpair of the form packet to its Split-Zero sector and local moment matrix. The moment matrix is positive for every parameter, so that positivity cannot supply a further zero-defect conclusion. For $\rho = 1$, $u = 1$,

$$z = \sqrt{p}, \qquad T_2(z) = \begin{pmatrix} 1 & \sqrt{p} \\ \sqrt{p} & p \end{pmatrix} \geq 0,$$

and its diagonal entries differ. On the form side $A = (1)$, $G = (1)$, $v = (1)$ give $W = (1)$ and $\omega = 1$; the exact maps take these data to the same off-critical packet. Any arithmetic identity forcing the actual family of W to vanish on its eigenvectors must therefore be established from that family's construction. These maps retain that requirement rather than replacing it by Gram positivity.

1.5 Projective zeta at all zeros and at its pole

A pair in $G(\mathbb{C})^2$ is unimodular exactly when at least one amplitude is nonzero. Such a coordinate is a supported unit. If both amplitudes are zero, every linear combination has zero amplitude and cannot be the unit. Let $\mathbb{P}^1_{\mathrm{um}}(G(\mathbb{C}))$ denote unimodular pairs modulo common unit multiplication.

Proposition 1.10 (Projective amplitude section and reciprocal)**.** *The map*

$$p_* : \mathbb{P}^1_{\mathrm{um}}(G(\mathbb{C})) \to \mathbb{P}^1(\mathbb{C}), \quad [x : y] \mapsto [p(x) : p(y)]$$

has an injective section

$$\iota : \mathbb{P}^1(\mathbb{C}) \to \mathbb{P}^1_{\mathrm{um}}(G(\mathbb{C})), \quad [a : b] \mapsto [j(a) : j(b)],$$

whose image is the full-support component. With

$$\mathcal{R} : G(\mathbb{C}) \to \mathbb{P}^1_{\mathrm{um}}(G(\mathbb{C})), \quad x \mapsto [1 : x], \qquad r : \mathbb{C} \to \mathbb{P}^1(\mathbb{C}), \quad c \mapsto [1 : c],$$

one has $p_*\mathcal{R} = rp_{\mathbb{C}}$ and $\mathcal{R}j = \iota r$.

Proof. Unimodularity ensures the projected pair is nonzero. A common supported unit projects to a nonzero complex scalar, so p_* is well defined. A nonzero ordinary pair becomes a unimodular supported pair; changing its representative lifts to common supported unit multiplication. Thus ι is well defined, and $p_*\iota$ is identity. Every full-support pair is the supported lift of its amplitude pair. The pair $(1, x)$ is always unimodular, and projection of its coordinates proves both remaining identities. ☐

In particular $\mathcal{R}(e) = [1 : e] = \infty_e$ and $\mathcal{R}(\tau) = [1 : \tau] = \infty_\tau$. Both project to $[1 : 0]$, while their Boolean projective supports are $[1 : 1]$ and $[1 : 0]$. Similarly $[e : 1]$ and $[\tau : 1]$ project to the ordinary zero while retaining different supports.

Proposition 1.11 (Two exact charts for the supported zeta map). *For the meromorphic zeta function with its simple pole of residue one at* 1*, the full-support lift is*

$$Z_{\mathrm{sp}} : \mathbb{C} \to \mathbb{P}^1_{\mathrm{um}}(G(\mathbb{C})), \qquad Z_{\mathrm{sp}}(s) = \begin{cases} [j(\zeta(s)) : 1], & s \neq 1, \\ [1 : e], & s = 1. \end{cases}$$

It equals $\iota\widehat{\zeta}$. *At every zeta zero* ρ *its value is* $[e : 1]$. *Near the pole it has the chart*

$$Z_{\mathrm{sp}}(s) = [1 : j(h(s))], \qquad h(s) = 1/\zeta(s), \quad h(1) = 0, \quad h'(1) = 1.$$

The reciprocal is a finite-amplitude coordinate away from the zeta zeros, including its extension at 1*. At a zero of order* m*, it has a pole of order* m*.*

Proof. Put $t = s - 1$ and write $\zeta(1 + t) = t^{-1} + g(t)$ with g holomorphic near zero. Then $h(1 + t) = t/(1 + tg(t))$. Its denominator has value one at zero, so this is holomorphic there with value zero and derivative one. On the overlap where $\zeta \neq 0$, common multiplication by $j(\zeta(s))$ gives $[1 : j(h(s))] = [j(\zeta(s)) : 1]$. This proves the transition and pole value. The first chart directly gives $[e : 1]$ at each zero. At a zero of order m, write $\zeta(\rho + t) = t^m u(t)$ with $u(0) \neq 0$. Its reciprocal is $t^{-m}u(t)^{-1}$, proving its pole order; the first chart retains the original order m. ☐

The swap $J_*[x : y] = [y : x]$ is induced by an invertible permutation matrix and commutes with ι, p_*. Hence $J_* Z_{\mathrm{sp}}$ is the global reciprocal projective map, sending zeta zeros to ∞_e and the pole to $[e : 1]$. Analytic statements use the complex structure transported to the full-support component by ι; no topology on the extra support components is needed.

1.6 Matrix packets and their full relation module

Theorem 1.12 (Block packet with explicit hypotheses). *Let* V *be a complex Hilbert space of finite dimension* $d > 0$ *with fixed positive-definite Hermitian inner product. Let* $g \in \mathrm{GL}(V)$ *and* $m \geq 1$*. Use that metric and its orthogonal direct sums for all adjoints. Define*

$$W_m(g) : V \to V^{m+1}, \quad v \mapsto (v, gv, \ldots, g^m v), \qquad \mathbb{T}_m(g) = W_m(g)W_m(g)^*,$$

$$D_m(g) : V^m \to V^{m+1}, \qquad \begin{cases} (D_m(g)y)_0 = -g^* y_1, \\ (D_m(g)y)_k = y_k - g^* y_{k+1} \quad (1 \leq k \leq m-1), \\ (D_m(g)y)_m = y_m. \end{cases}$$

The matrix $\mathbb{T}_m(g)$ *is positive semidefinite and*

$$\ker \mathbb{T}_m(g) = \mathrm{im}\, D_m(g), \qquad \mathrm{rank}\, \mathbb{T}_m(g) = d, \qquad \mathrm{corank}\, \mathbb{T}_m(g) = md.$$

It is block Toeplitz exactly when g *is unitary for the fixed metric.*

Proof. The Gram factorization gives positivity and $\ker \mathbb{T}_m = \ker W_m^*$. The first component of $W_m v$ is v, so W_m is injective and the kernel dimension is $(m+1)d - d = md$. The adjoint is

$$W_m^*(v_0, \ldots, v_m) = \sum_{k=0}^{m} (g^*)^k v_k.$$

Substitution of $D_m(y)$ gives

$$-g^* y_1 + \sum_{k=1}^{m-1} \big((g^*)^k y_k - (g^*)^{k+1} y_{k+1}\big) + (g^*)^m y_m = 0.$$

Thus its image lies in the kernel. If $D_m(y) = 0$, the last component gives $y_m = 0$, and backward substitution gives each preceding $y_k = 0$. Its image dimension is md, proving equality with the kernel. The block of indices j, k is $g^j (g^*)^k$. For unitary g this is g^{j-k}, hence block Toeplitz. Conversely block Toeplitzness equates blocks $(0,0), (1,1)$, giving $I = gg^*$. Invertibility yields $g^* = g^{-1}$ and unitarity. $\square$

At $m = 0$ the only block is I, so the bound $m \geq 1$ is essential. At $m = 1$, the full kernel is $\{(-g^* y, y) : y \in V\}$; its vector polynomials are $(wI - g^*)y$. Taking determinant gives the exact scalar map

$$wI - g^* \longmapsto \det(wI - g^*) = \overline{\det(\bar{w}I - g)}.$$

The equality follows by conjugating the determinant expansion and transposing. It retains conjugate eigenvalues and multiplicities. For the standard metric and $J_t = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ with $t > 0$, the polynomial is $(w-1)^2$ but

$$J_t J_t^* = \begin{pmatrix} 1 + t^2 & t \\ t & 1 \end{pmatrix} \neq I.$$

Its block packet consequently fails Toeplitzness, although the determinant roots lie on the unit circle. The kernel remains $\{(-J_t^* y, y) : y \in \mathbb{C}^2\}$, retaining the matrix information that determinant forgets.

2 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R-modules V_l, with linear maps $\rho_{lk} : V_l \to V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f), where $\alpha : L \to L'$ preserves bottom and joins with R-linear fibre maps

$$f_l : V_l \to W_{\alpha(l)}, \qquad f_k \rho_{lk} = \rho'_{\alpha(l),\alpha(k)} f_l. \tag{D1}$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)} f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 2.1 (Full support-index reconstruction)**.** *The diagram category just defined is equivalent to the category of $G(R)$-semimodules and their $G(R)$-linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l,x)+(k,y)=(l\vee k,\rho_{l,l\vee k}x+\rho_{k,l\vee k}y),\quad \tau(l,x)=(\perp,0),\quad r^\bullet(l,x)=(l,rx). \tag{D2}$$

Its global zero is $(\perp,0)$*. On morphisms it is* $\mathcal{T}(\alpha,f)(l,x)=(\alpha(l),f_lx)$*.*

The inverse sends a semimodule M *to*

$$\begin{gathered} L_M=eM,\quad l\vee k=l+k,\quad \perp=0_M,\\ V_l=\{m\in M: em=l\},\qquad \rho_{lk}(m)=m+k\quad (l+k=k). \end{gathered}\tag{D3}$$

The natural comparison maps are

$$\mathcal{T}(L_M,V)\longrightarrow M,\quad (l,m)\mapsto m,\quad m\mapsto (em,m), \tag{D4}$$

and the label-and-fibre isomorphism

$$L\longrightarrow e\mathcal{T}(L,V),\quad l\mapsto (l,0),\qquad x\in V_l\longmapsto (l,x). \tag{D5}$$

Proof. For (D2), threefold addition transports both parenthesizations to $l\vee k\vee j$. Linearity and composition of the ρ's make both amplitudes $\rho_lx+\rho_ky+\rho_jz$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l. Their addition law at a vector in fibre l is the ordinary R-module addition in V_l; in particular $0_R^\bullet$ gives $(l,0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha,f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2=e$, $e+e=e$, and $1+e=1$ in $G(R)$. Thus every $l\in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m\in V_l$,

$$m+l=(1+e)m=m,\qquad m+(-1_R)^\bullet m=em=l.$$

The inherited addition in V_l, with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r\in R$, $e(r^\bullet m)=em=l$, since $er^\bullet=e$. Its scalar action satisfies every R-module law inherited from the semimodule, and $0_Rm=em=l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M.

If $l+k=k$, then $e(m+k)=l+k=k$, so ρ_{lk} has the correct codomain. Its zero law is $l+k=k$; its additivity uses $k+k=k$. Also $r^\bullet k=k$, proving R-linearity. Identity follows from $m+l=m$, and composition from $k+j=j$. A split-linear map $F: M\to N$ takes eM to eN, with $\alpha(l)=F(l)$. It preserves joins and bottom, restricts to R-linear maps on each fibre, and $F(m+k)=F(m)+F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m+(l\vee k))+(n+(l\vee k))=m+n+(l\vee k)=m+n,$$

because $e(m+n)=l\vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ. It is a split-linear isomorphism. In the reconstructed total, $e(l,x)=(l,0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l,x)+(k,0)=(k,\rho_{lk}x)$, recovering the given ρ. For a morphism (α,f), (D5) takes $(l,0)$ to $(\alpha(l),0)$; for F, (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. □

2.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \to \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \tag{D6}$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N, let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \to D$ of R-module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \operatorname{im} d_C^{\text{prev}}$ inside Z_C, and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \qquad [z] \longmapsto [z] \tag{D7}$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \to \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \to N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \tag{D8}$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear map satisfying the respective condition has its unique factorization through the displayed subsemimodule. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

3 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R, while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m, including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

3.1 Source spaces and fixed arithmetic factors

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x),\ \phi(0) = 0,\ \int_{\mathbb{R}} \phi(x)\, dx = 0\},$$

$$\mathscr{B} = \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z},\ k \geq 0\}, \quad D = -x\partial_x.$$

Use the original Fourier, theta and Mellin transforms

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x\xi}\, dx, \quad \Theta\phi(x) = 2\sum_{n\geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \tag{A1}$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 3.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \to \mathscr{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \operatorname{Re} s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \qquad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{A2}$$

For the exact original source $\phi_* = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \qquad H_{P(D)\phi_*} = P, \qquad \mathcal{M}\Theta(P(D)\phi_*) = gP. \tag{A3}$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\widehat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2}\sum_{n\geq 1} \mu(n)\Theta\phi(nx) = \sum_{k\geq 1}\left(\sum_{n|k} \mu(n)\right)\phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k)|\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\operatorname{Re} s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) =$

$O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2}\{2\Gamma(s/2+2) - 3\Gamma(s/2+1)\} = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. □

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathscr{B}]$. Let $Q = \mathscr{B}/\Theta V$ and $q : \mathscr{B} \to Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathscr{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \tag{A4}$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q. This follows from injectivity of theta and $\operatorname{im} d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-}, with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

3.2 The exact operator range, including its inverse

Proposition 3.2 (Source and target Euler inverses)**.** *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1}\, dy.$$

Then

$$(D-\rho)\mathscr{B} = \ker(\mathcal{M}(\,\cdot\,)(\rho)), \qquad (D-\rho)S_\rho = 1, \qquad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s-\rho}. \tag{A5}$$

The operator $D - \rho$ is injective. For $0 < \operatorname{Re}\rho < 1$, the same formula, extended evenly, gives

$$(D-\rho)V = \ker(H_{(\,\cdot\,)}(\rho)). \tag{A6}$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \operatorname{Re}\rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N-\sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N+\sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho\phi(x) = -\int_0^1 u^{\rho-1}\phi(xu)\, du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D-\rho)S_\rho\phi = \phi$ on $\mathbb{R}$ gives $(1-\rho)\int S_\rho\phi = \int\phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V. □

For any polynomial $h(t) = \prod_i (t-\rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathscr{B} = \ker(j_h\mathcal{M}), \qquad h(D)V = \ker(j_h H), \qquad E_h = \mathbb{C}[t]/(h). \tag{A7}$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho_i y}/k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathscr{B}/h(D)\mathscr{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \tag{A8}$$

Theorem 3.3 (Finite arithmetic kernel, quotient and packet section)**.** *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \tag{A9}$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t-\rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu, and set

$$R_{\mathrm{ref}} u = S_h^{\mathscr{B}} \Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M} F, \quad \sigma = q R_{\mathrm{ref}}.$$

Then

$$\mathcal{M} R_{\mathrm{ref}} u = (g/h) R_Z(u), \quad J_Z R_{\mathrm{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \tag{A10}$$

The projector is $\sigma \overline{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H_\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h \Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\mathrm{ref}} = j_h v \, \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ. Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. □

3.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E, define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$D R_{\mathrm{ref}} - R_{\mathrm{ref}} A = f_0 \ell, \qquad f_0 = \Theta\phi_*. \tag{A11}$$

Indeed $tR_Z(u) - R_Z(tu)$ is divisible by the monic h, and its degree is at most d; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathscr{B}_Z = \Theta V + R_{\mathrm{ref}} E$ fits into the actual extension

$$0 \to V \xrightarrow{\Theta} \mathscr{B}_Z \xrightarrow{J_Z} E \to 0. \tag{A12}$$

It splits complex-linearly through R_{ref}; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_*\ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathscr{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\simeq} [V \to \mathscr{B}_Z] \longrightarrow [V \to \mathscr{B}]. \tag{A13}$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

3.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathscr{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below uses these specified metrics. Put $\phi_j = D^j\phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j\phi_j$. For $F_m = \Theta\Phi_m : \mathbb{C}^{m+1} \to \mathscr{H}$, define

$$\begin{gathered} H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \\ \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{gathered} \tag{A14}$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \to \mathbb{C}^{m+1}$,

$$\begin{gathered} (R_{\text{ref}} + F_m B)^*(R_{\text{ref}} + F_m B) = G_m + (B - B_m)^* H_m (B - B_m), \\ G_m = R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{gathered} \tag{A15}$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0\ell + S_m B_m - I_m B_m A, \qquad DR_m - R_m A = F_{m+1} K_m. \tag{A16}$$

Integration by parts in dx, with the source boundary term $[x\overline{F}H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \tag{A17}$$

In particular rank $W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \overline{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy–Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \qquad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \tag{A18}$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \tag{A19}$$

The next sections compute the actual arithmetic moment problem behind G_m, sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

4 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

4.1 The fixed arithmetic maps and their boundary

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x),\ \phi(0) = 0,\ \int_{\mathbb{R}} \phi(x)\,dx = 0\},$$

$$\mathscr{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty\ (b \in \mathbb{Z},\ k \geq 0)\}, \qquad D = -x\partial_x,$$

$$\Theta\phi(x) = 2\sum_{n\geq 1} \phi(nx), \quad Q = \mathscr{B}/\Theta V, \quad q : \mathscr{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx).$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^s\,\frac{dx}{x}, \qquad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x)e^{-2\pi i x\xi}\,dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \phi_j = D^j\phi_*, \quad f_j = \Theta\phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ, and retain

$$h_Z(t) = \prod_{\rho\in Z} (t-\rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \ldots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$R_{\text{ref}}u = S^{\mathscr{B}}_{h_Z}\Theta(R_Z(u)(D)\phi_*), \quad JF = j_h\mathcal{M}F, \quad \sigma_Z = qR_{\text{ref}},$$

$$JR_{\text{ref}} = 1_E, \qquad J\Theta = 0, \qquad DR_{\text{ref}} - R_{\text{ref}}A = f_0\ell_Z, \qquad \ell_Z(u) = [t^{d-1}]R_Z(u). \tag{TC1}$$

Here $S^{\mathscr{B}}_\rho F(x) = x^{-\rho}\int_x^\infty F(y)y^{\rho-1}\,dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S^{\mathscr{B}}_{h_Z}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^{L} c_j\phi_j, \qquad F_L = \Theta\Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathscr{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \qquad C : E \longrightarrow \mathbb{C}^{L+1}. \tag{TC2}$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \to \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1}. Directly,

$$DR - RA = \Theta K_R, \qquad K_R = \phi_*\ell_Z + \Phi_{L+1}(S_LC - I_LCA) : E \longrightarrow V. \tag{TC3}$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^*R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathscr{B}$, integration by parts gives

$$\langle DF, H\rangle + \langle F, DH\rangle = \langle F, H\rangle,$$

because $[x\overline{F(x)}H(x)]_0^\infty = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^*G_R + G_RA - G_R = -\big(R^*\Theta K_R + (\Theta K_R)^*R\big). \tag{TC4}$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

4.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \qquad (\mathcal{U}F)(y) = e^{y/2}F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\operatorname{Im} z| < \pi/4$. On every closed narrower strip, $\operatorname{Re}(e^{2z}) \geq c_a e^{2\operatorname{Re} z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\operatorname{Re} z \to +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1}f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2}f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\operatorname{Im} z = a \operatorname{sign} t$ has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \tag{TC5}$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \overline{v}\, d\mu$ has a smaller positive exponential moment, by Cauchy–Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2+it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g. It follows that

$$\overline{\bigcup_{L\geq 0} F_L\mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \tag{TC6}$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

4.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^*F_m, \quad C_m = F_m^*R_{\text{ref}}, \quad B_m = -H_m^{-1}C_m, \quad \Pi_m = F_mH_m^{-1}F_m^*, \quad R_m = R_{\text{ref}} + F_mB_m. \tag{TC7}$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \qquad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \tag{TC8}$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \to 1$ strongly. Since E is finite dimensional, $R_m \to 0$ in $\mathrm{Hom}(E, \mathcal{H})$, and $G_m \to 0$ in matrix norm.

Exact fixed-level image. At every specified level m, the entire image of the source-Gram map is

$$\big\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \to \mathbb{C}^{m+1}\big\} = \{G_m + \Delta : \Delta \succeq 0,\ \operatorname{rank} \Delta \le m+1\}. \tag{TC8a}$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \tag{TC8b}$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m+1$ has a factorization $\Delta = T^*T$ with $T : \mathbb{C}^d \to \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ, retain its nonnegative eigenvalues, and insert their square roots into $m+1$ rows. Taking $C = B_m + H_m^{-1/2} T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E, there exist a finite L and a coefficient map $C : E \to \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \qquad JR = 1, \qquad qR = \sigma_Z, \qquad R^* R = G. \tag{TC9}$$

The construction is as follows. Choose $m \ge d-1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \to \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1}. Put

$$\Delta = G - G_m > 0, \qquad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \tag{TC10}$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E. The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \qquad R_m^* F_m = 0,$$

so its Gram is

$$R^* R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ. This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\mathrm{Ext}^1_{\mathbb{C}[t]}(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E. The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \qquad [p(t)] \longmapsto [p(D)\phi_*]. \tag{TC10a}$$

It is well-defined and $\mathbb{C}[t]$-linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \qquad [\psi] \longmapsto j_h\left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*}\right). \tag{TC10b}$$

The denominator is

$$\mathcal{M}\phi_*(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\operatorname{Re} s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z; this proves well-definedness of κ_h and $\kappa_h\iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z. For the first factor $t-\rho$, its source inverse on $x > 0$ is

$$S_\rho\psi(x) = x^{-\rho}\int_x^\infty \psi(y)y^{\rho-1}\,dy = -\int_0^1 \psi(tx)t^{\rho-1}\,dt.$$

The second expression and $0 < \operatorname{Re}\rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\operatorname{Re}\rho-1}$. The first expression gives Schwartz decay at infinity. Integration of $(D-\rho)S_\rho\psi = \psi$ on the full real line gives $(1-\rho)\int S_\rho\psi = \int\psi = 0$, so $S_\rho\psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s-\rho)$, so the remaining zero jets permit iteration through every factor of h_Z. Consequently $\ker(j_h\mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$, where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h\kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta\big(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)\big),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D-equivariant section: an intertwiner $T : E \to \mathscr{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathscr{B}$. To see this injectivity, $(D-\rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathscr{B}$ force $c = 0$, and one iterates over the factors of h_Z. The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

4.4 Sharp control with all nilpotent jets present

For a positive definite G, set

$$W_A(G) = A^*G + GA - G, \qquad \epsilon_A(G) = \sup_{u\neq 0}\frac{|u^*W_A(G)u|}{u^*Gu}, \qquad \eta_Z = \max_{\rho\in Z}|2\operatorname{Re}\rho - 1|. \tag{TC11}$$

For every eigenvector $Au = \rho u$,

$$u^*W_A(G)u = (2\operatorname{Re}\rho - 1)u^*Gu. \tag{TC12}$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G. We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t}e^{tA^*}G_0e^{tA},\\ G_\epsilon &= \int_{-\infty}^{\infty} e^{-\epsilon|t|}F(t)\,dt,\\ W_A(G_\epsilon) &= \epsilon\int_{-\infty}^{\infty}\operatorname{sign}(t)e^{-\epsilon|t|}F(t)\,dt. \end{aligned} \tag{TC13}$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A-\rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho\in Z} e^{t\rho}\sum_{j=0}^{m_\rho-1}\frac{t^j}{j!}N_\rho^jP_\rho. \tag{TC14}$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\overline{\rho} + \sigma - 1$ has absolute value at most η_Z. This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^*F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \tag{TC15}$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \tag{TC16}$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at}\, dt = k! a^{-k-1}$ show that

$$G_\epsilon = \sum_{\rho,\sigma \in Z} \sum_{i=0}^{m_\rho - 1} \sum_{j=0}^{m_\sigma - 1} \frac{I_{i+j}(\overline{\rho} + \sigma - 1, \epsilon)}{i! j!} \cdot (N_\rho^i P_\rho)^* G_0 (N_\sigma^j P_\sigma). \tag{TC17}$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\boxed{\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \to \mathbb{C}^{L+1}}} \epsilon_A\big((R_{\text{ref}} + F_L C)^*(R_{\text{ref}} + F_L C)\big) = \eta_Z.} \tag{TC18}$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA} u\|_G \leq e^{(1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA} u\|_G \leq e^{(-1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t, hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z, and use (TC13) on that summand. Its control is strictly smaller than η_Z. The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

4.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^{n} I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \qquad W_{A_n,n}(G) = A_n^* G + G A_n - nG. \tag{TC19}$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \cdots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \operatorname{Spec} A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \tag{TC20}$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \ldots, \rho)$ for a zero attaining η_Z. Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\boxed{\inf_{G>0} \sup_{v \neq 0} \frac{|v^* W_{A_n,n}(G) v|}{v^* G v} = n\eta_Z.} \tag{TC21}$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

4.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \overline{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \to \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \to E.$$

The source boundary of R_m lies in the span of $f_0, \ldots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \operatorname{rank} W_m \leq 2. \tag{TC22}$$

Reflection stability gives

$$\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0. \tag{TC23}$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m-self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\boxed{\epsilon_m \;\geq\; \sum_{\rho \in Z} m_\rho \max(2 \operatorname{Re} \rho - 1, 0) = \sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}|.} \tag{TC24}$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A-invariant flag, keeping all generalized eigenvectors. Choose a G_m-orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \to E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ, repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal entries $\delta_j = 2\,\mathrm{Re}\,\rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m. Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \le P \le I$ and $H \le H_+$,

$$\sum_{\delta_j>0} \delta_j = \mathrm{Tr}(PH) \le \mathrm{Tr}(PH_+) \le \mathrm{Tr}(H_+) = \epsilon_m. \tag{TC25}$$

Each trace inequality follows by inserting the positive square root of P or H_+, so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+, the same argument gives $\sum_j \max(\delta_j, 0) \le r_+\epsilon_A(G)$. In the present source construction $r_+ \le 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\mathrm{Im}\, c_m)^2. \tag{TC26}$$

Indeed the two nonzero eigenvalues are those of $-\left(\begin{smallmatrix} c_m & v_m \\ u_m & \overline{c_m} \end{smallmatrix}\right)$, and (TC23) gives $\mathrm{Re}\, c_m = 0$. If $a_{m+1} = (1-\Pi_m) f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \qquad u_m = \eta_{m+1}\left(1 - \frac{\det G_{m+1}}{\det G_m}\right). \tag{TC27}$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1}\left(1 - \frac{\det G_{m+1}}{\det G_m}\right) v_m - (\mathrm{Im}\, c_m)^2 \;\ge\; \left(\sum_{\rho\in Z} m_\rho |\,\mathrm{Re}\,\rho - \tfrac{1}{2}|\right)^2. \tag{TC28}$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

4.7 Original supported maps and the precise result delivered

The passage from a finite source correction C to its representative, Gram, and boundary is the explicit chain of maps

$$C \longmapsto R_{\mathrm{ref}} + \Theta\Phi_L C \longmapsto \big(qR,\ JR,\ R^*R,\ K_R\big) = \big(\sigma_Z,\ 1_E,\ G_R,\ K_R\big). \tag{TC29}$$

For every active support label λ, the linear maps lift as $(\lambda, u) \mapsto (\lambda, Ru)$ and $(\lambda, u) \mapsto (\lambda, K_R u)$. The quotient of the boundary is $(\lambda, \Theta K_R u) \mapsto (\lambda, 0_Q)$; external absence remains the external zero τ. For the two-leg differential $d(\phi, \psi) = \Theta(\phi - \widehat{\psi})$, the two primitives are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$. Their difference $(K_R u, \widehat{K_R u})$ is the retained degree-zero cycle. Thus all source changes in the optimization have actual typed cochain lifts with the same support synchronization as the input programme.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

5 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

5.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x),\ \phi(0) = 0,\ \textstyle\int_{\mathbb{R}} \phi = 0\}, \qquad D = -x\partial_x.$$

Let $\mathscr{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b|D^kF(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta\phi(x) = 2\sum_{n\geq 1}\phi(nx), \qquad \mathcal{M}F(s) = \int_0^\infty F(x)x^s\,\frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0 = \Theta\phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j\phi_*$ and $f_j = D^jf_0 = \Theta\phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \qquad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2}F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity}. Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathscr{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g, including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho\in Z}(s-\rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \ldots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \qquad \varepsilon = (j_hv)^{-1} \in E^\times, \qquad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h, including the factor 2 in $g = 2\xi$. It commutes with A. For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu. Write $R_{\text{ref}} : E \to \mathscr{B}$ for the source note's fixed finite section R_0. This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m. The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \qquad J_ZR_{\text{ref}} = I_E, \qquad J_Z\Theta = 0. \tag{TB.1}$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2+it)|^2\,\frac{dt}{2\pi}, \qquad d\nu_Z(t) = |v(1/2+it)|^2\,\frac{dt}{2\pi}. \tag{TB.2}$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

5.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\mathrm{ref}} u + \sum_{j=0}^{m} f_j (Bu)_j, \qquad B : E \longrightarrow \mathbb{C}^{m+1}. \tag{TB.3}$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N, with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \qquad C_N P = j_h P. \tag{TB.4}$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j \, d\nu_Z(t), \qquad 0 \leq i, j \leq N. \tag{TB.5}$$

It is positive definite by (TB.2). Define

$$K_N = C_N (H_N^\nu)^{-1} C_N^*. \tag{TB.6}$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 5.1 (The actual source minimum)**.** *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \qquad \mathcal{M} R_m u = v \, p_{N,u}, \tag{TB.7}$$

$$\boxed{G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon.} \tag{TB.8}$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \tag{TB.9}$$

has degree at most m. Its coefficients give exactly the map $B_m : E \to \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M} R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^{m} (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \tag{TB.10}$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h, and the degree bound makes its quotient a polynomial of degree at most m. Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p+z)^* H_N^\nu (p+z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \tag{TB.11}$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $qR_m = \sigma_Z$ and $J_Z R_m = I_E$. □

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

5.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z, with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E, no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 5.2 (The retained vertical-line recurrence)**.** *There are $b_n \in \mathbb{C}$, with $b_n + \overline{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \qquad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \ge 1). \tag{TB.12}$$

Proof. On the original line $s = 1/2 + it$, one has $\overline{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \le n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1-s) q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \overline{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. □

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \tag{TB.13}$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^{N} \frac{v_n v_n^*}{\kappa_n}. \tag{TB.14}$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z,w)=\sum_{n=0}^{N}\frac{q_n(z)\overline{q_n(w)}}{\kappa_n}. \tag{TB.15}$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0\le k<m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 5.3 (Confluent kernel boundary formula)**.** *For every $N\ge d-1$,*

$$\boxed{AK_N+K_NA^*-K_N=\frac{v_{N+1}v_N^*+v_Nv_{N+1}^*}{\kappa_N}.} \tag{TB.16}$$

Before taking any jets the corresponding polynomial identity is

$$(z+\overline{w}-1)\mathcal{K}_N(z,w)=\frac{q_{N+1}(z)\overline{q_N(w)}+q_N(z)\overline{q_{N+1}(w)}}{\kappa_N}. \tag{TB.17}$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n, because $j_h(sq_n)=A(j_hq_n)$. Substitute these relations into

$$AK_N+K_NA^*-K_N=\sum_{n=0}^{N}\frac{(Av_n)v_n^*+v_n(Av_n)^*-v_nv_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n+\overline{b}_n=1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncancelled term is the $n=N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z,w. This is a polynomial identity; no division by $z+\overline{w}-1$ is performed at its zero set, and differentiating it proves the full confluent version. ☐

5.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m=A^*G_m+G_mA-G_m=-\{R_m^*(DR_m-R_mA)+(DR_m-R_mA)^*R_m\}. \tag{TB.18}$$

The boundary term $[x\overline{F}H]_0^\infty$ vanishes for $F,H\in\mathscr{B}$. Put

$$L_N=AK_N+K_NA^*-K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m=U_\varepsilon^*K_N^{-1}L_NK_N^{-1}U_\varepsilon.} \tag{TB.19}$$

For clarity, multiplication gives

$$K_N^{-1}L_NK_N^{-1}=K_N^{-1}A+A^*K_N^{-1}-K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N=v_{N+1}^*K_N^{-1}v_{N+1},\qquad \beta_N=v_N^*K_N^{-1}v_N,\qquad \gamma_N=v_N^*K_N^{-1}v_{N+1}. \tag{TB.20}$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\boxed{\epsilon_m = \frac{|\operatorname{Re}\gamma_N| + \sqrt{\alpha_N\beta_N - (\operatorname{Im}\gamma_N)^2}}{\kappa_N}.} \tag{TB.21}$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1}U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y/(y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N}\begin{pmatrix}\gamma_N & \beta_N \\ \alpha_N & \overline{\gamma_N}\end{pmatrix}.$$

Its trace and determinant give

$$\lambda_\pm = \frac{\operatorname{Re}\gamma_N \pm \sqrt{\alpha_N\beta_N - (\operatorname{Im}\gamma_N)^2}}{\kappa_N}.$$

Cauchy–Schwarz in K_N^{-1} gives $|\gamma_N|^2 \le \alpha_N\beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re}\gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \overline{\rho}$, with the multiplicities retained, $2\operatorname{Re}\operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1}L_N) = \frac{2\operatorname{Re}\gamma_N}{\kappa_N}. \tag{TB.22}$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2\operatorname{Re}\rho - 1. \tag{TB.23}$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \ge \max_{\rho\in Z} |2\operatorname{Re}\rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

5.5 Exact change when one more theta direction is admitted

The step $m \mapsto m+1$ means $N \mapsto N+1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \ge 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{ww^*}{k}. \tag{TB.24}$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1}ww^*K_N^{-1}}{k+\theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1}ww^*K_N^{-1}U_\varepsilon}{k+\theta}, \tag{TB.25}$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k+\theta} > 0. \tag{TB.26}$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \ne 0$, the determinant formula follows by extending $K_N^{-1}w$ to a basis, in which the rank-one map $K_N^{-1}ww^*/k$ has one possible nonzero eigenvalue θ/k. Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h, as in (TB.9). This gives a source polynomial P_u of degree at most $m+1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1}, where $W_{m+1} = \operatorname{span}(\phi_*, D\phi_*, \ldots, D^{m+1}\phi_*)$. The quotient map $\mathscr{B}/\Theta W_m \to \mathscr{B}/\Theta W_{m+1}$ has kernel $\Theta W_{m+1}/\Theta W_m$; injectivity of Θ identifies this kernel with the original one-dimensional space W_{m+1}/W_m. The scalar e sends a class to the zero in its same support fibre, while τ sends it to external absence. None of (TB.24)–(TB.26) changes those maps.

5.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make this measure invariant under $s \mapsto 1-s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1-s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$sQ_n = Q_{n+1} + \tfrac{1}{2}Q_n - \frac{h_n}{h_{n-1}}Q_{n-1} \quad (n \geq 1). \tag{TB.27}$$

For $n=0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \qquad d_n u = \frac{1}{h_n}\int_{\mathbb{R}} \overline{Q_n(1/2+it)}\, r(u)(1/2+it)\, d\mu(t). \tag{TB.28}$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g\sum_{n>m} Q_n d_n u, \qquad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n.} \tag{TB.29}$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n, and use the adjoint relation $s^* = 1-s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n}d_{n+1} = d_{n-1} + d_n(\tfrac{1}{2}I - A) - \delta_{n0}\ell_Z.} \tag{TB.30}$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1}d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1}(d_{m+1}^* d_m + d_m^* d_{m+1}).} \tag{TB.31}$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 5.4 (The full source defect and its observation of the packet)**.** *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{\begin{aligned} DR_m - R_m A = {} & -\frac{h_{m+1}}{h_m}Q_m(D)f_0\, d_{m+1} \\ & - Q_{m+1}(D)f_0\, d_m. \end{aligned}} \tag{TB.35}$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \qquad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \tag{TB.36}$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^{m} Q_j d_j$ and compute

$$sr_m - r_m A = \ell_Z - \sum_{j=0}^{m} (sQ_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j. For $0 \le j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z. The coefficient at Q_m is $-h_{m+1} d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed primitive belongs to V, since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D) f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \ge 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k. Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \ge 0$. In particular every term of the positive series in (TB.29) vanishes on u, so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. □

5.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1-\overline{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \to E_{Z^\dagger}$ defined in Section 7. The original arithmetic residue matrix S, satisfying $S^* = -S$ and $A^* S + SA = S$, is still related to its dual metric by

$$G_m^{\mathrm{D}} = S^* G_m^{-1} S, \qquad W_m^{\mathrm{D}} = -S^* G_m^{-1} W_m G_m^{-1} S. \tag{TB.32}$$

The original arithmetic Weil form, with $J_g = M_{j_h g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \tag{TB.33}$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \to 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \longrightarrow +\infty, \qquad \prod_{m=m_0}^{M} \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \longrightarrow 0. \tag{TB.34}$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \to 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z-moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, https://dlmf.nist.gov/18.2. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

6 Which completion retains the arithmetic jets?

This section extends the Hilbert/jet calculation of the September 12 orthogonal-boundary note. All changes of topology are specified maps on the same arithmetic functions. The theta relation space, the operator $D = -x\partial_x$, and $g = 2\xi$ are retained.

6.1 Logarithmic coordinates and the original measure

Let $\mathscr{B}, V, \Theta$ have the definitions in Section 3. Put $\mathscr{H} = L^2(\mathbb{R}_+, dx)$. The map

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \qquad \mathcal{L}^{-1}f(x) = x^{-1/2}f(\log x)$$

is unitary from $\mathscr{H}$ to $L^2(\mathbb{R}, dy)$, since $dx = e^y\,dy$. Direct differentiation gives

$$\mathcal{L}D\mathcal{L}^{-1} = \tfrac{1}{2} - \partial_y, \qquad \mathcal{M}F(\rho) = \int_{\mathbb{R}} (\mathcal{L}F)(y)e^{(\rho-1/2)y}\,dy. \tag{JT1}$$

On $\mathscr{H}$, D denotes the closed maximal distributional operator which $\mathcal{L}$ identifies with $1/2 - \partial_y$ on the Sobolev domain $H^1(\mathbb{R})$. For an integer $r \geq 0$, let $\mathscr{H}^{(r)}$ be the domain of the first r powers of D in $\mathscr{H}$, with norm $\|F\|_r^2 = \sum_{k=0}^r \|D^kF\|_{\mathscr{H}}^2$. The Mellin-line transform makes this the norm with density $w_r(t) = \sum_{k=0}^r |1/2 + it|^{2k}$ relative to $dt/(2\pi)$.

Theorem 6.1 (Finite differential graph norms and the complete jet graph)**.** *Let Z be a nonempty finite packet of actual nontrivial zeros with full multiplicities, $E_Z = \mathbb{C}[t]/h_Z$, and $J_Z : \mathscr{B} \to E_Z$ the complete Mellin-jet map. For every finite r,*

$$\overline{\operatorname{span}\{D^j f_0 : j \geq 0\}}^{\mathscr{H}^{(r)}} = \mathscr{H}^{(r)}, \quad f_0 = \Theta\phi_*, \tag{JT2}$$

$$\overline{\{(F, J_ZF) : F \in \mathscr{B}\}}^{\mathscr{H}^{(r)}\oplus E_Z} = \mathscr{H}^{(r)} \oplus E_Z. \tag{JT3}$$

Thus J_Z is not closable in this norm. The completed theta-boundary graph is $\mathscr{H}^{(r)} \oplus 0$; its quotient in (JT3) is E_Z through $[(F, u)] \mapsto u$.

Proof. For each $0 < c < \pi/4$, the actual source gives $|g(1/2 + it)| \leq C_ce^{-c|t|}$. Indeed $e^{z/2}f_0(e^z)$ is holomorphic in $|\operatorname{Im} z| < \pi/4$: its Gaussian series and derivatives converge uniformly on compact subsets since $\operatorname{Re}(e^{2z}) > 0$. On smaller horizontal strips it decays faster than every exponential at the positive end. Poisson gives $Jf_0 = f_0$, so analytic continuation gives $e^{z/2}f_0(e^z) = e^{-z/2}f_0(e^{-z})$; this controls the negative end. Shift the Fourier contour to height c or $-c$. The vertical integrals vanish by that decay, proving the bound.

The measure $d\mu_r = w_r(t)|g(1/2+it)|^2dt/(2\pi)$ has a positive exponential moment. If $u \in L^2(\mu_r)$ is orthogonal to every polynomial, $d\nu = \overline{u}\,d\mu_r$ is a finite complex measure with a smaller positive exponential moment, by Cauchy–Schwarz. Its Fourier transform is analytic in a strip, and every derivative at zero vanishes. Analytic continuation makes it zero there. Fourier uniqueness gives $\nu = 0$, hence $u = 0$. For this uniqueness step, convolve with Gaussian approximate identities: each convolution is in L^2, since $\|\nu * \gamma_\epsilon\|_2 \leq \|\nu\|_{\mathrm{TV}}\|\gamma_\epsilon\|_2$; its transform is zero, so Plancherel makes it zero, and weak convergence then gives $\nu = 0$. Polynomials are therefore dense in $L^2(\mu_r)$. Multiplication by $g(1/2 + it)$

is an isometry onto $L^2(w_r\,dt/(2\pi))$: divide almost everywhere, since this nonzero analytic function vanishes on a discrete, measure-zero set. The Mellin transform of $D^j f_0$ is $(1/2+it)^j g(1/2+it)$, proving (JT2).

Every theta polynomial has zero J_Z-jet. Thus the closure in (JT3) contains $(h,0)$ for every h. The actual packet section $R_{\mathrm{ref}} : E_Z \to \mathscr{B}$ satisfies $J_Z R_{\mathrm{ref}} = 1$. Approximate $R_{\mathrm{ref}}u$ in $\mathscr{H}^{(r)}$ by theta polynomials $p_n(D)f_0$. Then $(R_{\mathrm{ref}}u - p_n(D)f_0, u) \to (0,u)$. This proves (JT3). For $u \neq 0$ it prevents closability. The boundary and quotient statements follow from the coordinate projections. □

Corollary 6.2 (Differential on the completed graph)**.** *The continuous map*

$$(D,A) : \mathscr{H}^{(r+1)} \oplus E_Z \longrightarrow \mathscr{H}^{(r)} \oplus E_Z, \quad (F,u) \mapsto (DF, Au) \tag{JT4}$$

extends the original $(F, J_Z F) \mapsto (DF, J_Z DF)$. On the quotient by the boundary closure it is exactly $A = M_t$.

Proof. The norm definitions bound $D : \mathscr{H}^{(r+1)} \to \mathscr{H}^{(r)}$; the jet identity $J_Z D = A J_Z$ follows by retaining the complete Taylor polynomial of $s\mathcal{M}F(s)$. Apply (JT3). □

Its split observation records $(e_{\mathscr{H}}, u^\bullet)$ when the Hilbert component tends to its supported zero and the jet survives. The projections compare this point to $e_{\mathscr{H}}$ and to $u^\bullet$; they do not identify it with (τ,τ).

6.2 Exponential weights and exact jet norms

For $a \geq 0$, retain the original functions and introduce

$$\mathscr{H}_a = L^2(\mathbb{R}_+, e^{2a|\log x|}\,dx), \qquad \|F\|_a^2 = \int_{\mathbb{R}} |\mathcal{L}F(y)|^2 e^{2a|y|}\,dy. \tag{JT5}$$

The identity inclusions $\mathscr{B} \hookrightarrow \mathscr{H}_a \hookrightarrow \mathscr{H}_0$ are continuous. Source seminorms with powers exceeding $a+1/2$ control both ends. Smooth functions compactly supported in log coordinates prove density.

Theorem 6.3 (Exact threshold, including the boundary)**.** *Put $\ell_{\rho,k}(F) = (\mathcal{M}F)^{(k)}(\rho)/k!$ and $b = \operatorname{Re}\rho - 1/2$, with $k \geq 0$. The functional on $\mathscr{B}$ extends continuously to $\mathscr{H}_a$ if and only if $|b| < a$. Its exact squared norm is*

$$\|\ell_{\rho,k}\|^2 = \frac{(2k)!}{(k!)^2}\left(\frac{1}{[2(a-b)]^{2k+1}} + \frac{1}{[2(a+b)]^{2k+1}}\right). \tag{JT6}$$

Proof. Inner products are conjugate-linear in their first argument. The Riesz vector in log coordinates must be

$$r_{\rho,k}(y) = \frac{y^k}{k!} e^{(\overline{\rho}-1/2)y} e^{-2a|y|}.$$

Its norm squared is $(k!)^{-2}\int y^{2k} e^{2by-2a|y|}\,dy$. The two half-lines converge exactly when $|b| < a$; their gamma integrals give (JT6). If $|b| \geq a$, testing any putative Riesz representative against compactly supported smooth functions forces it to be this same non-square-integrable density, a contradiction. The equality case is included: one end has no exponential decay. □

Theorem 6.4 (Complete confluent Cauchy Gram)**.** *For distinct centres ρ_i with $|\operatorname{Re}\rho_i - 1/2| < a$ and orders $0 \leq k < m_i$, let $J_a : \mathscr{H}_a \to \mathbb{C}^d$ be their Taylor-jet map. In the original Taylor coordinates its Gram $C_a = J_a J_a^*$ is*

$$(C_a)_{(i,k),(j,l)} = \frac{(k+l)!}{k!\,l!}\left(\frac{1}{(2a - z_{ij})^{k+l+1}} + \frac{(-1)^{k+l}}{(2a+z_{ij})^{k+l+1}}\right), \quad z_{ij} = \rho_i + \overline{\rho}_j - 1. \tag{JT7}$$

It is positive definite. The exact right inverse $S_a = J_a^* C_a^{-1}$ satisfies

$$J_a S_a = 1, \quad \|S_a u\|_a^2 = u^* C_a^{-1} u, \quad \|F\|_a^2 = \|(1 - S_a J_a)F\|_a^2 + (J_a F)^* C_a^{-1}(J_a F). \tag{JT8}$$

Proof. The entry is $\langle r_{\rho_i,k}, r_{\rho_j,l}\rangle_a$; integrating $y^{k+l}e^{z_{ij}y-2a|y|}/(k!l!)$ on both half-lines proves (JT7) with its stated sign. The vectors are independent. After removing their common nonzero factor, a dependence is an exponential polynomial $\sum_i P_i(y)e^{\overline{\rho}_i y} = 0$. For each i, apply $\prod_{j\neq i}(\partial_y - \overline{\rho}_j)^{m_j}$. On the i-th polynomial factor every nonzero constant plus derivative is invertible on the finite-degree polynomial space; hence $P_i = 0$. Thus $C_a > 0$. Now $S_a J_a$ is the orthogonal projection onto range $J_a^* = (\ker J_a)^\perp$. Multiplying the specified matrices proves all of (JT8). $\square$

For an actual packet the unchanged theta subspace lies in $\ker J_a$. Consequently there is an onto continuous quotient map

$$\mathscr{H}_a/\overline{\Theta V}^{\mathscr{H}_a} \longrightarrow \mathbb{C}^d, \quad [F] \mapsto J_a F, \qquad \ker = (\ker J_a)/\overline{\Theta V}^{\mathscr{H}_a}. \tag{JT9}$$

Its surjectivity follows from (JT8). Define the retained coordinate map

$$T : E_Z \longrightarrow \mathbb{C}^d, \qquad T[p] = \big(p^{(k)}(\rho_i)/k!\big)_{i,k}.$$

Then $J_a|_{\mathscr{B}} = TJ_Z$, and the exact representative difference is $R_{\text{ref}}T^{-1} - S_a : \mathbb{C}^d \to \ker J_a$. Here R_{ref} is the original source map; S_a is a Hilbert map, with no claim it lies in $\mathscr{B}$. In E_Z coordinates the observation Gram is $C_E = T^{-1}C_a(T^{-1})^*$ and the right inverse is S_aT.

6.3 The energy term carried by the weighted topology

Theorem 6.5 (Weighted boundary identity on the actual representatives). *On $\mathscr{B}$, with $(\Sigma H)(x) = \operatorname{sgn}(\log x)H(x)$,*

$$\langle DF, H\rangle_a + \langle F, DH\rangle_a - \langle F, H\rangle_a = 2a\langle F, \Sigma H\rangle_a. \tag{JT10}$$

For an actual packet representative $R : E_Z \to \mathscr{B}$ with $DR - RA = B = \Theta k$, put $G_a = R^*R$, $T_a = R^*\Sigma R$. Adjoints use exactly $\mathscr{H}_a$. Then

$$A^*G_a + G_aA - G_a = 2aT_a - (B^*R + R^*B), \qquad -G_a \le T_a \le G_a. \tag{JT11}$$

For $Au = \rho u \neq 0$,

$$2\operatorname{Re}\rho - 1 = 2a\frac{u^*T_au}{u^*G_au} - 2\operatorname{Re}\frac{\langle Ru, Bu\rangle_a}{\|Ru\|_a^2}. \tag{JT12}$$

Proof. In log coordinates integrate $-(\overline{f}h)'e^{2a|y|}$ on the two half-lines. Source decay kills the terms at infinity; those at zero cancel because the weight is continuous there. The weak derivative of the weight is $2a\operatorname{sgn}(y)e^{2a|y|}$. This proves (JT10) with its positive sign. Substitute $DR = RA + B$ and move both boundary terms to obtain (JT11). Multiplication by sign is a self-adjoint contraction, proving the form bounds. Evaluation on u proves (JT12). $\square$

At $a = 0$, (JT11) is the source boundary-control formula. At $a > 0$, the topology making a jet continuous has its explicitly retained energy term $2aT_a$. Off-line jets cannot remain continuous as $a \downarrow 0$. For line-centred jets, all $a > 0$ work but (JT6) diverges as $a \downarrow 0$. These are exact comparisons on the same Mellin map and differential, without a weight theorem assumed.

6.4 The tesserine coordinate in the actual weight form

Retain $c_{-+} = 4(\beta - 1/2)$, $c_{--} = 4i\gamma$ for $\rho = \beta + i\gamma$. For $p > 1$ and $|\psi| = 1$,

$$z = \psi\exp\left(\frac{\log p}{4}(c_{-+} + c_{--})\right), \qquad \frac{u^*(A^*G + GA - G)u}{u^*Gu} = \frac{c_{-+}}{2}, \qquad |b| < a \iff |c_{-+}| < 4a. \tag{JT13}$$

The first equality is exactly $z = \psi p^{\rho - 1/2}$; the second is evaluation on $Au = \rho u$. Thus the original sector, local holonomy radius, arithmetic weight form and jet-continuity strip have specified comparison maps. The scalar evaluations do not replace the original matrix or its support-labelled source.

7 The current arithmetic route and its next calculation

The source chain now supplies an actual $\mathbb{C}[t]$-module cohomology Q, a finite invariant packet inside that cohomology, a specified nonzero extension class, an original theta boundary for its representative defect, and an exact arithmetic formula for the relative weight form. The local continuation makes that last formula an explicit confluent interpolation problem. These constructions are logically prior to any purity conclusion.

7.1 A single chain of exact comparison maps

For each actual finite packet, retaining every full order, the maps proved in this reader are

$$E = \mathbb{C}[t]/h \xrightarrow{R_{\mathrm{ref}}} \mathscr{B} \xrightarrow{q} Q, \qquad J_Z R_{\mathrm{ref}} = 1, \qquad DR_{\mathrm{ref}} - R_{\mathrm{ref}}A = \Theta\phi_*\ell.$$

Every admitted polynomial correction keeps these same quotient and jet maps. At its distinguished orthogonal level let $B_{N,u}(t) = (p_{N,u}(t) - R_Z(u)(t))/h(t)$, a polynomial of degree at most m. The complete calculation is

$$\begin{gathered} d\nu_Z(t) = \left|\frac{g(1/2+it)}{h(1/2+it)}\right|^2 \frac{dt}{2\pi}, \quad N = d+m, \quad C_N P = j_h P, \quad K_N = C_N(H_N^\nu)^{-1}C_N^*, \\ p_{N,u} = (H_N^\nu)^{-1}C_N^* K_N^{-1} U_\varepsilon u, \qquad R_m u = R_{\mathrm{ref}}u + \Theta(B_{N,u}(D)\phi_*), \\ G_m = U_\varepsilon^* K_N^{-1} U_\varepsilon, \qquad W_m = U_\varepsilon^* K_N^{-1}(AK_N + K_N A^* - K_N)K_N^{-1}U_\varepsilon. \end{gathered} \tag{R1}$$

The middle line evaluates the specified polynomial at the original D; every source vector lies in V.

Write $v_n = j_h q_n$ and $\kappa_n = \|q_n\|_{\nu_Z}^2$. The original arithmetic measure now determines the exact quantity

$$\epsilon_m = \frac{|\operatorname{Re}(v_N^* K_N^{-1} v_{N+1})| + \sqrt{(v_{N+1}^* K_N^{-1} v_{N+1})(v_N^* K_N^{-1} v_N) - (\operatorname{Im}(v_N^* K_N^{-1} v_{N+1}))^2}}{\kappa_N}. \tag{R2}$$

The proofs in Section 5 establish (R1)–(R2) before passing to cohomology. For a reflection-stable packet the first absolute-value term is zero by the exact trace identity. The same sequence obeys

$$\epsilon_m \ \geq \sum_{\rho \in Z} m_\rho |\operatorname{Re}\rho - \tfrac{1}{2}|. \tag{R3}$$

Thus the rank-two boundary detects the combined displacement of the entire packet, not only its largest individual displacement. The estimate concerns the distinguished R_m sequence. The larger family of all polynomial source corrections has the separately proved metric image and sharp infimum in Section 4; those results do not identify its arbitrary member with R_m.

The full two-row source observation in Section 5 is injective after the original A-iterates are retained. It therefore offers a finite arithmetic calculation preserving every nilpotent jet: compute the source moments, the adjacent monic polynomials and their complete residue vectors, rather than discard a component whose scalar trace vanishes.

7.2 Topology, duality and the two sides of a weight estimate

The theta-polynomial span is dense in every finite differential graph norm. The completed jet graph, however, is $\mathscr{H}^{(r)} \oplus E$, with quotient E, not a quotient of $\mathscr{H}^{(r)}$ in which the arithmetic component has vanished. This supplies the exact comparison between the shrinking Hilbert representative and its unchanged cohomological class.

The exponential observation norm supplies a second exact comparison. A Mellin jet at ρ is bounded precisely in the strip

$$|\operatorname{Re}\rho - \tfrac{1}{2}| < a,$$

with the factorial norm in Theorem 6.3. The original differential then has the energy term $2a\ \mathrm{sgn}(\log x)$ in Theorem 6.5. Consequently stronger topology and weight control are related by that explicit term; choosing a positive observation norm does not remove it.

For any packet Z, reflection gives the conjugate-linear ring isomorphism

$$\dagger_Z : E_Z \longrightarrow E_{Z^\dagger}, \qquad f^\dagger(s) = \overline{f(1-\overline{s})}, \quad Z^\dagger = \{1-\overline{\rho} : \rho \in Z\},$$

with all multiplicities retained. In fact $h_Z^\dagger = (-1)^d h_{Z^\dagger}$; this proves well-definedness, and applying the same operation twice is the identity. Restrict the following internal residue pairing to reflection-stable Z, so that this map is an involution on E_Z. The arithmetic residue form is the source form

$$\mathcal{R}_Z(f,u) = \sum_{\rho\in Z} \operatorname{Res}_\rho \frac{f^\dagger u}{g}\, ds.$$

Its Jacobian contraction is

$$\mathcal{R}_Z(f,g'u) = \sum_{\rho\in Z} m_\rho \overline{f(1-\overline{\rho})} u(\rho). \tag{R4}$$

Indeed locally $g = z^{m_\rho} w(z)$ gives $g'/g = m_\rho/z + w'/w$, proving (R4) with multiplicities. The trace form pairs reflected centres and can retain nontrivial null directions from nilpotents. In the retained basis $1, t, \ldots, t^{d-1}$, define

$$S_{ij} = \sum_{\rho\in Z} \operatorname{Res}_\rho \frac{(s^i)^\dagger s^j}{g(s)}\, ds, \quad 0 \le i,j < d, \qquad J_g = M_{j_h g'}.$$

Changing representatives by h makes the local quotient by g holomorphic, proving that this matrix represents the intrinsic form. By (R4) the trace pairing has matrix SJ_g. Its exact comparison to the positive source metric is therefore $G_m^{-1} S J_g$, since $f^* G_m (G_m^{-1} S J_g) u = f^* S J_g u$. Its positivity is an arithmetic question about this operator, not a consequence of the positive Gram construction.

The finite-field two-sided weight theorem remains a source for methods, with its compact-support comparison, duality and tensor hypotheses. This reader has not constructed a realization meeting those geometric hypotheses for the entire arithmetic quotient. It has instead calculated the current characteristic-zero objects and their original maps, including the local-cohomology spectral extraction and meromorphic complement documented in the complete finite-comparison appendix.

7.3 Provenance and mathematical status

The September 11 support, Rees, comparison and audit packages are retained in the companion source directory. The current analytic baseline is the September 12 sequence: finite operator comparison (PR 8), chain descent (PR 9), global retraction (PR 10), homotopy control (PR 11), formal tau recovery (PR 12) and orthogonal boundary control (PR 13). Part II includes the complete selected source proof notes with their exact revisions. Historical statements that a map remains unconstructed belong to their stated source dates; a later explicit construction in this reader or the source chain supersedes that particular statement.

The new local proofs are the full support-index equivalence, the corrected carrier and sector comparisons, the arithmetic confluent-kernel minimum and complete two-row source observation, the classification of source-metric images and aggregate control, and the graph/weighted-jet comparison. Their elementary algebraic and analytic steps are written here. No claim of global mathematical priority follows from finding or proving an identity in this research programme.

The finite validation records distinguish exact symbolic identities, rational calibration models, numerical quadratures and intentionally rejected negative controls. They do not count test methods as theorems or imply analytic certification from an L^2 numerical experiment. The supplied formalization records are also kept separate: successful exact-head CI metadata does not turn the unformalized analytic theta arguments into Lean proofs. No Lean run was performed for this continuation.

The current research calculation is (R2) on the actual $|g/h|^2$-moments, with its determinant update and two-row recurrence, together with the topology constants of Theorem 6.3. No uniform arithmetic estimate forcing (R2) to vanish has been established in this reader. RH, GRH and a global purity theorem are not conclusions of it. The exact maps and source constraints above specify the live route on which tandem work can now continue.

Part II

Complete source proof notes and chronology

A PR 8: Finite operator comparison

Source attribution. Web contribution, pull request #8, Zeta research repository.

Exact revision. 25689c5376166df37801229bc275ff5a896007d9

Source file. `workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 57502c7fe6a7e1de239c94c3479abbb6a70b4718b551ad1bb1e26e2d3e4f4c22

A.1 Exact finite operator quotients of the tau-base theta cohomology

Owner-directed continuation, 12 September 2026. Mathematical derivation and research draft; no new analytic theorem below is claimed to have been checked by Lean.

A.1.1 0. Inputs, provenance, and the new result

This note continues TAU_BASE_MODEL.md, rather than moving the programme back to a finite-field variety. Its arithmetic coefficient object is the original split semiring

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \qquad e = 0_R^\bullet.$$

The geometric base remains the one-point absolute pointed base described there. The quotient

$$p_{\mathbb{Z}} : G(\mathbb{Z}) \longrightarrow \mathbb{Z}$$

and its square with $G(\mathbb{C}) \to \mathbb{C}$ remain part of the marked object. No ideal norm, Euler factor, scaling parameter, Gaussian, or completion factor is changed here.

The supplied derived formalization (PR #6, 5f6ee6575550d8a20c6fc3990359cb4f67855f80) provides the actual reconstruction, internal quotient, homology-kernel equivalence, and compatible kernel actions. Its finite-jet module supplies the balanced tensor-quotient comparison with the split-reflection square. Those results are reused; this note does not duplicate their Lean implementation.

The new analytic step computes the previously only surjective finite Mellin-jet observation as an exact operator quotient. For every polynomial h whose roots lie in the open critical strip, it constructs

$$Q/h(D)Q \xrightarrow{\sim} E_h/g_hE_h, \qquad Q[h(D)] \xrightarrow{\sim} \ker(g_h : E_h \to E_h),$$

with $E_h = \mathbb{C}[t]/(h)$ and $g = 2\xi$. It then constructs finite spectral projectors on the original Q, not only quotients of Q, and identifies the torsion subsheaf of the balanced analytic realization with $\mathcal{O}/(2\xi)$. Their kernels, original supported zeros, nilpotent actions, and trace-producing residue pairing are all retained.

The general existence of spectral realizations of zeta zeros is prior literature (not a novelty claim here): see R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. The contribution of this note is the explicit comparison in the specified workbench, together with its original SplitZero quotients and its complete finite kernel calculation.

A.1.2 1. The exact arithmetic complex and its operator

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x),\ \phi(0) = 0,\ \int_{\mathbb{R}} \phi(x)\, dx = 0\},$$

$$\mathscr{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z},\ k \geq 0\}.$$

The source maps, with their original factors, are

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n\in\mathbb{Z}\setminus\{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}.$$

The Fourier transform is $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x)e^{-2\pi i x\xi}\, dx$. Poisson and the two moment conditions put ΘV in $\mathscr{B}$. Also $D\Theta = \Theta D$, and D preserves V.

Set

$$A = \mathbb{C}[t], \qquad t \cdot F = DF, \qquad t \cdot \phi = D\phi,$$

$$C_+ = [V \xrightarrow{\Theta} \mathscr{B}] \quad \text{in degrees } 0, 1, \qquad Q = \mathscr{B}/\Theta V.$$

The existing injectivity proof of Θ gives $H^0(C_+) = 0$ and $H^1(C_+) = Q$.

This is the one-leg subcomplex of the actual tau-base chart complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta\phi - \Theta\widehat{\psi}} \mathscr{B}].$$

Its inclusion is $\phi \mapsto (\phi, 0)$ in degree zero and the identity in degree one. The quotient complex is the second copy of V in degree zero. The explicit section is $\psi \mapsto (\widehat{\psi}, \psi)$. The operator on that second copy is $1 - D$, because $D\widehat{\psi} = \widehat{(1 - D)}\psi$. Thus the additional joint-support degree-zero cohomology is retained in this split exact sequence; it is not included in Q and then silently discarded.

The actual source multiplier is

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$H_\phi(s) = c(s)M_+\phi(s), \quad c(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}, \quad M_+\phi(s) = \int_0^\infty \phi(x)x^s \frac{dx}{x}.$$

On the open strip $\mathscr{S} = \{0 < \Re s < 1\}$, $c(s)$ is holomorphic and nowhere zero. The source identity is

$$\mathcal{M}\Theta\phi = gH_\phi.$$

The specific function

$$\phi_*(x) = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}$$

has $H_{\phi_*} = 1$. Consequently

$$H_{P(D)\phi_*}(s) = P(s), \qquad \mathcal{M}\Theta(P(D)\phi_*) = g(s)P(s).$$

These are identities with the stated Gaussian and $g = 2\xi$; no rescaling is performed.

A.1.3 2. A continuous inverse to the actual Euler differential

For any $\rho \in \mathbb{C}$, define on the subspace $\{H \in \mathscr{B} : \mathcal{M}H(\rho) = 0\}$

$$\boxed{S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1}\, dt.}$$

All powers use the real logarithm on $x, t > 0$. Direct differentiation gives

$$(D - \rho) S_\rho H = H.$$

Endpoint control. Write $x = e^y$, $h(y) = H(e^y)$, and $\sigma = \Re\rho$. At $y \geq 0$, the displayed tail and a bound $|h(t)| \leq C_N e^{-Nt}$ give

$$|S_\rho H(e^y)| \leq \frac{C_N}{N - \sigma} e^{-Ny} \quad (N > \sigma).$$

At $y \leq 0$, the zero moment changes the same expression into the negative lower tail. A bound $|h(t)| \leq C'_N e^{Nt}$ gives

$$|S_\rho H(e^y)| \leq \frac{C'_N}{N + \sigma} e^{Ny} \quad (N + \sigma > 0).$$

Differentiating uses $(S_\rho H)'_y = -\rho S_\rho H - H(e^y)$, so every derivative has the same family of estimates. This proves membership in $\mathscr{B}$ and continuity in its stated seminorms. The homogeneous solution $x^{-\rho}$ fails the required rapid decrease unless its coefficient is zero. Thus $D - \rho$ is injective, and

$$\boxed{(D - \rho)\mathscr{B} = \ker(F \mapsto \mathcal{M}F(\rho)).}$$

Mellin intertwining gives

$$\mathcal{M}(S_\rho H)(s) = \frac{\mathcal{M}H(s)}{s - \rho},$$

with the removable value filled at $s = \rho$.

There is also a source inverse. For $\rho \in \mathscr{S}$ and $\phi \in V$ with $H_\phi(\rho) = 0$, use the same tail for $x > 0$ and extend evenly. The nonvanishing of $c(\rho)$ gives $M_+\phi(\rho) = 0$. Near zero the formula is

$$S_\rho \phi(x) = -\int_0^1 u^{\rho-1} \phi(xu)\, du.$$

Differentiation under this integral proves smoothness at zero in every order. The function is even and its value at zero is zero. The upper-tail formula proves Schwartz decay at infinity. Finally

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi(x)\, dx = \int_{\mathbb{R}} \phi(x)\, dx = 0,$$

so the second moment condition is preserved. This proves the source range identity

$$\boxed{(D - \rho)V = \ker(\phi \mapsto H_\phi(\rho)) \quad (\rho \in \mathscr{S}).}$$

At the level of Taylor coefficients $\phi(x) = \sum_{j\geq 1} a_{2j} x^{2j}$, the local expression is $-\sum_{j\geq 1} a_{2j} x^{2j}/(\rho + 2j)$, interpreted to each finite Taylor order with its smooth remainder. All denominators and the minus sign are retained. Endpoints $0, 1$ are not included in this source range theorem.

A.1.4 3. Every finite operator test, without choosing zeros first

Choose any finite set of distinct centres $\rho_1, \ldots, \rho_r \in \mathscr{S}$ and positive integers m_i. Define the particular polynomial

$$h(t) = \prod_{i=1}^{r} (t - \rho_i)^{m_i}, \quad d = \deg h, \quad E_h = A/(h).$$

This defines the test; it does not presume that any centre is a zeta zero. The Chinese-remainder map is

$$\theta_h : E_h \xrightarrow{\sim} \prod_i \mathbb{C}[z_i]/(z_i^{m_i}), \qquad [P(t)] \mapsto [P(\rho_i + z_i)].$$

For a holomorphic germ family f at the centres, $j_h f$ denotes its Taylor jets transported back by θ_h^{-1}. In particular $g_h = j_h g$ retains the original local units of g.

Iterate Section 2 through the factors of h. Uniqueness makes the inverse independent of their chosen order. This gives the continuous inverse

$$S_h : \ker(j_h \mathcal{M}) \longrightarrow \mathscr{B}, \qquad h(D) S_h = \mathrm{id}.$$

The exact range identities are

$$h(D)\mathscr{B} = \ker(j_h \mathcal{M}), \qquad h(D)V = \ker(j_h H).$$

Surjectivity on the source follows from $H_{P(D)\phi_*} = P$. Surjectivity of the target jet map follows directly from finite Mellin interpolation: on an interval in $y = \log x$, the functions $e^{\rho_i y} y^j / j!$ are linearly independent. For a nonnegative smooth bump η positive on a subinterval, the Gram matrix

$$G_{(i,j),(k,l)} = \int \eta(y) e^{(\rho_i + \bar{\rho}_k) y} \frac{y^{j+l}}{j! l!} \, dy$$

is positive definite. Multiplying its inverse into the prescribed jet vector produces a compactly supported smooth preimage. This is only an interpolation device; it does not define or alter the arithmetic Weil pairing.

Consequently the vertical arrows in

$$\boxed{\begin{array}{ccc} V/h(D)V & \xrightarrow{\overline{\Theta}} & \mathscr{B}/h(D)\mathscr{B} \\ \alpha_h \downarrow \wr & & \wr \downarrow \beta_h \\ E_h & \xrightarrow{\times g_h} & E_h \end{array}}$$

are actual A-module isomorphisms:

$$\alpha_h[\phi] = j_h H_\phi, \qquad \beta_h[F] = j_h \mathcal{M} F.$$

An inverse to α_h sends the unique polynomial representative P of degree less than d to $[P(D)\phi_*]$. An inverse to β_h is any of the above explicit interpolation lifts, taken modulo $h(D)\mathscr{B}$; the class is independent of that choice by the range theorem.

A.1.5 4. The kernel and quotient of the global arithmetic operator

Both V and $\mathscr{B}$ are torsion-free A-modules: every factor $D-\rho$ is injective, since its homogeneous solution cannot be Schwartz at infinity. Over the PID A, they are flat. The bounded complex C_+ is therefore a valid flat model for derived tensoring. The displayed isomorphism computes

$$\boxed{C_+ \otimes_A^{\mathbf{L}} E_h \;\simeq\; [E_h \xrightarrow{\times g_h} E_h]}$$

in degrees zero and one. It is the base-change of the actual theta complex, not a newly declared spectral complex with an assigned differential.

The original exact sequence $0 \to V \xrightarrow{\Theta} \mathscr{B} \to Q \to 0$ also gives

$$\boxed{0 \to Q[h(D)] \xrightarrow{\kappa_h} E_h \xrightarrow{\times g_h} E_h \longrightarrow Q/h(D)Q \to 0.}$$

Here $Q[h(D)] = \ker(h(D) : Q \to Q)$. The maps are explicit:

- Given $[F] \in Q[h(D)]$, uniquely write $h(D)F = \Theta\phi$. Then $\kappa_h[F] = j_h H_\phi$.
- The quotient observation sends $[F]$ to $j_h \mathcal{M} F$ modulo $g_h E_h$.
- For $P \in \ker(g_h : E_h \to E_h)$, choose its polynomial representative of degree less than d and set

$$\boxed{\partial_h(P) = [S_h\Theta(P(D)\phi_*)] \in Q[h(D)].}$$

The last expression is defined because its theta input has zero jets modulo h. Replacing P by $P+hR$ changes the result by $\Theta(R(D)\phi_*)$. Thus it is independent of representatives and is the inverse of κ_h onto its kernel. The two injectivity and surjectivity assertions also follow directly by lifting representatives in the preceding exact sequence; no dimension-only argument is needed.

All maps commute with D, with $U_aF(x) = F(x/a)$, and with compactly supported multiplicative convolution. On E_h, the scaling action is multiplication by $j_h(a^s)$, including its entire finite nilpotent Taylor polynomial.

At one centre, retain $g(\rho+z) = u(z)z^r$, where $u(0) \neq 0$ (and $r=0$ is allowed). For test order m, the two cohomology modules have dimension $\min(m,r)$. When $r=0$, multiplication by g_h is invertible. The derived amplitude complex is then acyclic, while its reconstructed supported homology still has each labelled internal zero.

The degree-zero term is $Q[h(D)]$; the degree-one term is $Q/h(D)Q$. Their finite alternating traces cancel. The arithmetic packet trace below is the degree-one action, or equivalently its explicitly constructed direct summand in the original Q. The canonical truncation triangle retains the degree-zero term rather than relabelling it as zero.

A.1.6 5. Exact finite spectral projectors on the original cohomology

Now let Z be any finite set of actual nontrivial zeros and retain their full multiplicities r_ρ. Put

$$h_Z(t) = \prod_{\rho\in Z}(t-\rho)^{r_\rho}, \quad E_Z = A/(h_Z), \quad v_Z(s) = g(s)/h_Z(s).$$

The quotient v_Z is entire and its germs are nonzero at every point of Z. Hence $v_{Z,h} = j_{h_Z} v_Z$ is an invertible element of E_Z. This inverse retains the actual Taylor coefficients of g/h_Z; it is not set to one.

Here $g_{h_Z} = 0$, and the exact sequence becomes

$$Q[h_Z(D)] \xrightarrow[\kappa_Z]{\sim} E_Z, \qquad Q/h_Z(D)Q \xrightarrow[\beta_Z]{\sim} E_Z.$$

There is a further map that must be calculated, not assumed: the inclusion of the kernel into Q, followed by the quotient. For $P \in E_Z$,

$$\mathcal{M}\big(S_{h_Z}\Theta(P(D)\phi_*)\big) = \frac{gP}{h_Z} = v_Z P.$$

Therefore this kernel-to-quotient comparison is precisely

$$\boxed{\beta_Z\, q_Z\, \partial_Z = M_{v_{Z,h}} : E_Z \xrightarrow{\sim} E_Z.}$$

Define

$$\sigma_Z = \partial_Z \circ M_{v_{Z,h}^{-1}} : E_Z \longrightarrow Q,$$

$$\boxed{\Pi_Z = \sigma_Z J_Z : Q \longrightarrow Q, \qquad J_Z[F] = j_{h_Z}\mathcal{M}F.}$$

Then

$$\boxed{J_Z\sigma_Z = \mathrm{id}, \quad \Pi_Z^2 = \Pi_Z, \quad \operatorname{im}\Pi_Z = Q[h_Z(D)], \quad \ker\Pi_Z = h_Z(D)Q.}$$

This gives the actual equivariant direct sum

$$\boxed{Q = Q[h_Z(D)] \oplus h_Z(D)Q.}$$

A representative formula is completely determined. For $F \in \mathscr{B}$, let $P_{Z,F}$ be the unique polynomial of degree less than $\deg h_Z$ representing $v_{Z,h}^{-1} j_{h_Z}\mathcal{M}F$. Then

$$\boxed{\Pi_Z[F] = [S_{h_Z}\Theta(P_{Z,F}(D)\phi_*)].}$$

For a simple actual zero ρ, this specializes to

$$\Pi_{\{\rho\}}[F] = \frac{\mathcal{M}F(\rho)}{g'(\rho)} \left[x^{-\rho}\int_x^\infty \Theta\phi_*(t)t^{\rho-1}\,dt\right].$$

The displayed denominator is the actual $g'(\rho) = 2\xi'(\rho)$; it is not omitted. The bracketed class is an actual nonzero eigenclass in Q, proved by the exact sequence, even though its representative is only an eigenvector modulo the theta relation.

For finite packets the projectors satisfy

$$\Pi_Z\Pi_W = \Pi_{Z\cap W}, \qquad \Pi_{Z\cup W} = \Pi_Z + \Pi_W \quad (Z \cap W = \varnothing).$$

These follow from the direct sums, coprimality of disjoint packet polynomials, and commutation with D. Inclusions of finite packets therefore give compatible invariant subspaces and quotient observations. No convergence of an infinite sum of projectors is asserted.

A.1.7 6. Carry every projector through the original split quotient

For any vector space M the single-active-fibre lift is $M^\tau = \{\tau\} \sqcup \{m^\bullet : m \in M\}$. For a submodule N use exactly

$$q_N^{\mathrm{sp}} : M^\tau \to (M/N)^\tau, \quad m^\bullet \mapsto [m]^\bullet, \quad \tau \mapsto \tau.$$

This is the original coequalizer of the relation inclusion and its internal-zero companion. Its universal property is the one implemented in PR #6, including arbitrary noncancellative target semimodules.

Consequently the finite observation has the split-linear equivalence

$$\boxed{(Q/h(D)Q)^\tau \xrightarrow{\sim} G(E_h/g_hE_h), \qquad [F]^\bullet \mapsto (j_h\mathcal{M}F \bmod g_hE_h)^\bullet.}$$

The domain is being used as a split semimodule, not as an unconstructed ring. A class in $h(D)Q$ maps to the supported zero, and external absence alone maps to external absence.

Lift the projector as

$$\widetilde{\Pi}_Z(\lambda, [F]) = (\lambda, \Pi_Z[F]).$$

This formula applies to the actual nonempty support faces of the tau-base theta model; their degree-one transports are the stated identities of Q. At bottom it is the original zero map of the bottom module. The full reconstruction interface permits a nonzero bottom fibre when a subsequent diagram requires one; it is not silently specialized away on an opposite-index dual diagram.

For disjoint packets,

$$\boxed{\widetilde{\Pi}_Z\widetilde{\Pi}_W = \varepsilon, \qquad \varepsilon(\lambda, x) = (\lambda, 0).}$$

This is the supported lift of the zero linear operator, not the constant external-zero map. More uniformly, take $\Pi_\varnothing = 0_Q$ and lift it to ε. The scalar representation

$$G(\mathrm{End}_{\mathbb{C}}\, Q) \longrightarrow \mathrm{End}_{G(\mathbb{C})}(Q^\tau)$$

sends $f^\bullet$ to its split lift and sends τ to the constant-τ map. It is a unital semiring morphism and sends the supported scalar zero to ε.

Retain smaller admitted source labels For a subspace $W \subset V$, first use its specified enlargement $W_D = \mathbb{C}[D]W$ and keep the map

$$\mathscr{B}/\Theta W \longrightarrow Q_{W_D} := \mathscr{B}/\Theta W_D, \qquad \ker = \Theta W_D/\Theta W.$$

For a D-stable W, define the ideal

$$J_{W,h} = \{j_h H_\phi : \phi \in W\} \subseteq E_h.$$

It is an ideal because $H_{P(D)\phi} = PH_\phi$ and every element of E_h has a polynomial representative. The same target range calculation gives

$$\boxed{Q_W/h(D)Q_W \cong E_h/(g_hJ_{W,h}).}$$

Enlarging to $W + \mathbb{C}[D]\phi_*$ makes $J_{W,h} = E_h$. The exact finite comparison kernel is

$$\boxed{g_h E_h/(g_h J_{W,h}).}$$

Thus the finite equality for the full arithmetic relation space has a precise smaller-label comparison. It is not asserted for every original W without this kernel.

A.1.8 7. The Hausdorff observation and the remaining global kernel

The projectors just constructed have continuous finite-rank representatives on $\mathscr{B}$: J_Z is continuous, E_Z is finite dimensional, and S_{h_Z} is continuous on its zero-jet domain. They annihilate ΘV, hence also its closure. They therefore descend to

$$Q^{\mathrm{H}} = \mathscr{B}/\overline{\Theta V}.$$

The finite observation still has kernel exactly $h_Z(D)Q^{\mathrm{H}}$. Indeed

$$\ker J_Z = h_Z(D)\mathscr{B} + \Theta V,$$

and continuity gives $\overline{\Theta V} \subset \ker J_Z$. Hence adding the closure to the right side leaves it unchanged. The same finite section yields a continuous direct sum

$$Q^{\mathrm{H}} = \sigma_Z(E_Z) \oplus h_Z(D)Q^{\mathrm{H}}.$$

The algebraic packet $Q[h_Z(D)]$ injects into this Hausdorff quotient: its composite with J_Z is already an isomorphism onto E_Z. No conclusion about vanishing of the entire algebraic-to-Hausdorff kernel follows from this finite statement.

Let

$$\mathcal{R}_\infty = \bigcap_{Z \text{ finite}} \ker \Pi_Z = \bigcap_{Z \text{ finite}} h_Z(D)Q.$$

The map $Q \to \prod_\rho \mathcal{O}_\rho/(g)$ has exactly this kernel. Every nonzero polynomial whose roots are in $\mathscr{S}$ acts bijectively on $\mathcal{R}_\infty$. To verify this, its kernel on Q lies in the finite packet at its zero centres; its cokernel is supported there as calculated in Section 4. On the complementary summand it is bijective, and its inverse preserves all other commuting finite projectors. This proves injectivity and surjectivity on the intersection.

The continuation retains $\mathcal{R}_\infty$ and its scaling action. Finite packets do not prove it zero. A topological spectral-synthesis theorem or a global cohomological trace argument must address that specific invariant module rather than infer its disappearance from finite-jet isomorphisms.

7.1 The global balanced kernel has a computed local-cohomology complement Work now on the analytic strip $\mathscr{S}$, with its sheaf $\mathcal{O}$ and sheaf $\mathcal{K}$ of meromorphic functions. Form the sheafified balanced realization

$$\mathcal{M}_Q = \mathcal{O} \otimes_A Q, \qquad \beta : \mathcal{M}_Q \longrightarrow \mathcal{O}/(g), \quad b \otimes [F] \longmapsto [b\,\mathcal{M}F].$$

Let $\mathrm{tors}_{\mathcal{O}}(\mathcal{M}_Q)$ denote its torsion subsheaf: a germ is in it when some nonzero holomorphic germ annihilates it. This is torsion over the specified analytic coefficient ring; its connection to the original split support is the reconstruction of the inclusion and quotient maps below.

Theorem. The actual Mellin map restricts to an isomorphism

$$\boxed{\beta|_{\mathrm{tors}} : \mathrm{tors}_{\mathcal{O}}(\mathcal{M}_Q) \xrightarrow{\sim} \mathcal{O}/(g).}$$

Moreover, $\mathcal{N} = \ker\beta$ has a canonical $\mathcal{K}$-module structure, and the displayed map canonically splits the sequence:

$$\boxed{\mathcal{M}_Q \cong \mathcal{O}/(g) \oplus \mathcal{N}, \qquad \mathcal{K} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{N}.}$$

Proof at every stalk. Fix $\rho \in \mathscr{S}$ and set $x = s - \rho$. If $g(\rho) \neq 0$, Section 4 with $h = t - \rho$ proves that $D - \rho$ is bijective on Q. Consequently x is invertible on $\mathcal{M}_{Q,\rho}$, and this module is a module over the meromorphic field $\mathcal{O}_\rho[1/x]$. Its torsion is zero, as is $\mathcal{O}_\rho/(g)$.

If $r = \operatorname{ord}_\rho g > 0$, the finite projector for this one full packet gives

$$Q = Q[(D-\rho)^r] \oplus (D-\rho)^r Q.$$

On the second summand, $(D-\rho)^r$ is bijective: it is injective by the direct-sum intersection, and

$$(D-\rho)^r Q = (D-\rho)^{2r} Q$$

follows by applying $(D-\rho)^r$ to the direct sum. Hence $D - \rho$ itself is bijective there. Tensoring the direct sum with $\mathcal{O}_\rho$ gives a first summand $\mathcal{O}_\rho/(x^r)$, on which β is the isomorphism calculated in Section 5, and a second summand on which x is invertible. The map β kills the second summand because it is in $x^r\mathcal{M}_{Q,\rho}$, while x^r annihilates its target. Every nonzero germ of $\mathcal{O}_\rho$ is a unit times a power of x, so that second summand is a module over $\mathcal{K}_\rho$ and is torsion-free.

This proves the stated torsion isomorphism at every stalk. It glues without choosing neighbourhood projectors: both the torsion subsheaf and the restricted map β are intrinsic. The inverse of that isomorphism, followed by the torsion inclusion, is the global section of β. Local inverses to nonzero holomorphic scalars on $\mathcal{N}$ are unique, so they glue to its $\mathcal{K}$-module structure. This proves the theorem.

7.2 The extraction is an actual local-cohomology map At the same stalk the principal-ideal local-cohomology complex is the explicit localization complex

$$R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) = [\mathcal{M}_{Q,\rho} \longrightarrow \mathcal{M}_{Q,\rho}[1/x]], \quad \text{degrees } 0, 1.$$

The decomposition just proved computes it:

$$\boxed{R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) \xrightarrow{\sim} (\mathcal{O}_\rho/(g))[0].}$$

The finite torsion summand maps to zero upon localization; the meromorphic summand maps isomorphically to itself. The map to the displayed target is β in degree zero and zero in degree one. Thus the supported arithmetic divisor is the actual cohomological image of the localization kernel, not a prescribed deletion of a list of spectral points.

Before analytic extension there is the corresponding operator-localization map

$$Q \longrightarrow A[h_Z^{-1}] \otimes_A Q,$$

with kernel $Q[h_Z(D)]$ and cokernel zero, because $h_Z(D)$ is invertible on its complementary summand. Its supported lift sends that kernel to the internal supported zero, not to external absence.

At every original label λ, reconstruct the localization complex using

$$d(\lambda, m) = (\lambda, m/1), \qquad \varepsilon(\lambda, m) = (\lambda, 0).$$

Its internal degree-zero cycles are exactly the reconstructed torsion module. Its internal first cohomology is the support skeleton, since the ordinary first cohomology vanishes. The map $p_{\mathbb{Z}}$ remains in the structural coefficient square throughout; x is the analytic image of the operator $t-\rho$, not a replacement for the scalar e or τ.

The theorem does not assert $\mathcal{N}=0$. It computes what local cohomology does to it: localization is an isomorphism on that module, so it contributes no cohomology to the stated local-support complex. Its possible contribution to a different global trace functor has not been declared absent.

A.1.9 8. The residue duality is now inside an actual derived finite comparison

Assume the centres and their orders in h are stable under $\iota(s)=1-\bar{s}$. For germs define

$$f^{\dagger}(s)=\overline{f(1-\bar{s})}.$$

The chosen monic polynomial satisfies $h^{\dagger}=(-1)^{d}h$. Retain that factor. Define the perfect finite algebra pairing

$$R_h(f,u)=\sum_i \operatorname{Res}_{s=\rho_i}\frac{f^{\dagger}(s)u(s)}{h(s)}\,ds.$$

It satisfies $\overline{R_h(u,f)}=(-1)^{d+1}R_h(f,u)$. No phase is inserted to change this sign. Since $g^{\dagger}=g$, multiplication by g_h is self-adjoint for this pairing. Therefore it induces a perfect pairing

$$\boxed{\overline{\operatorname{coker} g_h}\ \otimes\ \ker g_h\longrightarrow\mathbb{C}.}$$

Well-definedness is the identity $R_h(ga,u)=R_h(a,gu)$; nondegeneracy follows by taking the annihilator of the image in the finite perfect algebra pairing. Through Section 4 this pairs the two actual cohomology groups of $C_+\otimes^{\mathbf{L}}_A E_h$.

For an exact packet h_Z, set $v_{Z,h}=j_h(g/h_Z)$ as above and

$$\jmath_Z: A_Z\to\ker g_h=E_Z,\qquad [f]\mapsto[j_h(h_Z/g)f].$$

Here $A_Z=\prod_{\rho\in Z}\mathcal{O}_\rho/(g)\cong E_Z$ by the specified local quotient maps; h_Z/g is holomorphic at every centre. The local units are not replaced by one. The exact contraction is

$$\boxed{R_h(f,\jmath_Z k)=\sum_{\rho\in Z}\operatorname{Res}_\rho\frac{f^{\dagger}k}{g}\,ds=:R_Z(f,k).}$$

Thus the previously constructed arithmetic residue form is the duality pairing of the actual finite operator comparison after the displayed unit morphism.

The Jacobian map is still $J_g[k]=[g'k]$. Its contraction is

$$\boxed{R_h(f,\jmath_Z J_g k)=R_Z(f,J_g k)=\sum_{\rho\in Z}r_\rho\overline{f(1-\bar{\rho})}k(\rho)=\operatorname{Tr}(M_{f^{\dagger}k}|A_Z).}$$

In particular the factor 2 in $g=2\xi$ is retained in both the denominator and the Jacobian. The trace contraction is invariant under their combined explicitly specified transformation, not by deleting that factor in only one place.

Every map lifts to the original support fibres. Dual support arrows are precomposition maps on the opposite index category. A vanishing pairing value is the supported scalar $e_{\mathbb{C}}$; an absent argument gives τ.

A.1.10 9. Cohomological placement and arithmetic traces

On the tau-base chart use the one-leg sheaf $\mathcal{T}_+$ and its A-action. The bounded projective resolution from TAU_BASE_MODEL.md gives the canonical map

$$\mathsf{A}_\tau(\mathcal{T}_+ \otimes^{\mathbf{L}}_A E_h) \xrightarrow{\sim} \mathsf{A}_\tau(\mathcal{T}_+) \otimes^{\mathbf{L}}_A E_h \xrightarrow{\sim} [E_h \xrightarrow{g_h} E_h].$$

These isomorphisms use finite evaluation sums and the displayed free/flat resolutions. They are not compact-support base-change assertions for a different arithmetic scheme.

The packet projector is now an actual finite-rank endomorphism of the original $H^1_\tau = Q$ (and of each of its retained support fibres). For an admissible smooth compactly supported multiplicative test k,

$$H_k F(x) = \int_0^\infty k(a) F(x/a)\, \frac{da}{a}$$

commutes with the projector. Its ordinary finite-rank trace is

$$\boxed{\mathrm{Tr}(H_k \Pi_Z | Q) = \sum_{\rho \in Z} r_\rho \mathcal{M}k(\rho).}$$

This is a finite-rank trace on the original cohomology, using its explicit finite invariant image. It is stronger than declaring that a surjective finite observation has a trace. Infinite trace summation still requires the specified topology and admissible test class.

On r-fold products over the tau-point, the already calculated external tensor map gives $H^r = Q^{\otimes r}$. The operator $\Pi_Z^{\otimes r}$ is a finite projector onto $Q[h_Z]^{\otimes r}$. The full nilpotent scaling action there is

$$a^{\rho_1 + \cdots + \rho_r} \exp\left((\log a) \sum_{j=1}^{r} N_{\rho_j, j} \right).$$

For a repeated exponent the exact modulus comparison remains

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re\rho - 1)\log a.$$

This supplies actual finite invariant objects on which a Deligne-style tensor estimate would have to act. It does not prove that estimate. The source's opposite weight bounds, its global dualizing theory, and the admissibility of its geometric hypotheses have not been inferred from these finite isomorphisms.

A.1.11 10. Status and publication boundary

Established in this note, with written proofs: the differential range identities on the stated spaces; the exact finite operator quotient and torsion kernel; the derived finite comparison; the analytic torsion-subsheaf and local-cohomology identification with $\mathcal{O}/(2\xi)$; finite spectral direct summands with explicit projectors; the original split lifts and smaller-label kernels; the residue/Jacobian contraction on that comparison; and compatible finite-rank arithmetic traces and tensor actions.

Not claimed: a global injectivity theorem for the full balanced analytic realization; vanishing of $\mathcal{R}_\infty$; identification of the entire four-point chart category with an arithmetic curve; a full compact-support/ordinary-cohomology realization satisfying the hypotheses of Weil II; positivity of the unshifted Weil form; RH or GRH. The finite tests are not a Lean or analytic proof certificate.

The prior base note and this continuation are published as additive research material beside PR #6. The original scalar core, formal modules, frozen reader and all existing branches remain unchanged. New derivations are AI-assisted work under the owner's programme; the structural source and its Lean implementation retain their own attribution. No copyrighted source corpus or private conversation is included.

References and source interfaces

- Owner's SplitZero programme: the original scalar, support fibre and quotient constructions; PR #6 at 5f6ee6575550d8a20c6fc3990359cb4f67855f80, especially SplitZeroHomology.lean, SplitZeroComplex.lean, SplitZeroInternalQuotient.lean, and SplitZeroFiniteJets.lean.
- TAU_BASE_MODEL.md: the actual characteristic-zero chart, cohomology, right-adjoint and tensor constructions used here. It remains an authored research model, not an accepted geometric RH realization.
- Deligne, P., *La conjecture de Weil. II* (1980), sections 3.3.4-3.3.6 and 6.2.1-6.2.7. Selected supplied TeX passages were reread for the intended comparison/duality architecture; no source proof is republished or claimed fully audited.
- Meyer, R., *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. Prior spectral-realization context; no theorem from it is silently used to identify the test spaces here.
- The Stacks Project, Derived tensor product, Tag 06XY. The general bounded-flat-complex construction used in Section 4.

B PR 9: Chain descent

Source attribution. Web contribution, pull request #9, Zeta research repository.

Exact revision. 8f99b94d306c3b1aaa817d52c57fa6fd298d511c

Source file. `workbenches/tau-chain-descent/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 1ab370cf2db9fce70020bebbab76d3dbd3bd37d39e28c83114eeb0eccfa2f0b3

B.1 Chain-level arithmetic descent over the tau base

Owner-directed research continuation, 12 September 2026. Draft proofs for review; no new Lean certification or RH claim.

B.1.1 0. Fixed inputs and publication scope

This is an additive continuation of Zeta PR #8 at 25689c5376166df37801229bc275ff5a896007d9, particularly workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md (Git blob 68f59d098d60708f8e899bb960abbd1135ebb3b4). Its sections 1–5 supply the stated analytic range theorem and the finite packet maps. The new result here is their chain-level descent, its actual extension class, and the connection of that class to residue duality and the arithmetic trace.

The source programme retains

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$p_R(\tau) = 0_R, \qquad p_R(r^\bullet) = r.$$

The absolute pointed base and the coefficient square remain

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \qquad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the first vertical map remains the infinite ring of integers. Every new arrow below is a coefficient or cochain construction over this marked base. No base field, ideal norm, Euler factor, or completion constant is changed.

The supplied derived note and PR #6 provide Recovery.recoverEquiv, Relations.coequalizer_universal, the internal-homology coequalizer, ChainMap.homologyKernelEquiv, and homology_intertwines. Their algebraic conclusions are used at their stated types. No existing formal module is modified. Analytic inputs from PR #8 remain written research arguments; they are not promoted to Lean theorems.

B.1.2 1. The actual arithmetic source and packet

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x),\ \phi(0) = 0,\ \int_{\mathbb{R}} \phi(x)\, dx = 0\},$$

$$\mathscr{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Retain

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n\in\mathbb{Z}\setminus\{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x\xi}\, dx, \qquad A = \mathbb{C}[t], \quad t \cdot F = DF.$$

The one-leg tau-chart complex is

$$C_+ = [V \xrightarrow{\Theta} \mathscr{B}], \quad |V| = 0, \quad |\mathscr{B}| = 1, \qquad Q = H^1(C_+).$$

The injectivity of theta gives $H^0(C_+) = 0$. The two-leg complex is

$$C_{+-} = [V \oplus V \longrightarrow \mathscr{B}], \qquad (\phi, \psi) \longmapsto \Theta\phi - \Theta\widehat{\psi}.$$

Its source operator is $(D, 1-D)$; the inclusion $C_+ \to C_{+-}$ is $\phi \mapsto (\phi, 0)$ and the identity in degree one. It leaves the extra degree-zero copy of V in C_{+-} present.

Use the unchanged arithmetic multiplier

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The input function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

and the established identities are

$$\mathcal{M}\Theta\phi = gH_\phi, \qquad H_{\phi_*} = 1, \qquad H_{P(D)\phi_*}(s) = P(s).$$

Fix a nonempty finite packet Z of actual nontrivial zeros, with full orders m_ρ. Set

$$h(t) = \prod_{\rho\in Z} (t-\rho)^{m_\rho}, \quad d = \deg h, \quad E = A/(h), \quad v(s) = g(s)/h(s).$$

The specific monic polynomial defines this finite test; it does not rescale g. Write j_h for the complete Taylor-jet/Chinese-remainder map at these centres, and retain the invertible element

$$v_h = j_h v \in E^\times, \qquad \varepsilon_Z = v_h^{-1}.$$

For $u \in E$, let $R_Z(u) \in \mathbb{C}[t]_{<d}$ be the unique polynomial representing $\varepsilon_Z u$.

The input range theorem supplies the inverse $S_h^{\mathscr{B}}$ of $h(D)$ on the zero-jet range, and the analogous source inverse S_h^V. For one factor,

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1}\, dy.$$

The stated source inverse uses $0 < \Re\rho < 1$ and the zero moment. Iteration supplies S_h. Define the actual maps

$$J : \mathscr{B} \to E, \quad JF = j_h \mathcal{M} F,$$

$$s : E \to \mathscr{B}, \quad s(u) = S_h^{\mathscr{B}} \Theta(R_Z(u)(D)\phi_*).$$

The inverse is defined because $gR_Z(u)$ has all required zero jets. Directly,

$$\mathcal{M}s(u) = vR_Z(u), \quad Js = 1_E, \quad J\Theta = 0, \quad h(D)s(u) = \Theta(R_Z(u)(D)\phi_*). \tag{1}$$

These equations fix the representative map, including all derivatives of g/h. On Q, the existing projector is $\Pi_Z = \bar{s}\bar{J}$, with image $Q[h(D)]$ and kernel $h(D)Q$.

B.1.3 2. The exact equivariant extension and its class

Define an actual subspace of the original test-function space:

$$\mathscr{B}_Z = \{F \in \mathscr{B} : [F] \in Q[h(D)]\}.$$

Then

$$\mathscr{B}_Z = \Theta V \oplus s(E) \quad \text{as complex vector spaces},$$

with maps

$$(\phi, u) \longmapsto \Theta\phi + s(u), \qquad F \longmapsto \big(\Theta^{-1}(F - sJF), JF\big). \tag{2}$$

For the inverse, $F - sJF$ has zero class in Q by the packet projector theorem; injectivity of theta makes its source unique. The intersection is zero by $Js = 1$ and $J\Theta = 0$.

The original operator D preserves $\mathscr{B}_Z$, and there is a short exact sequence of A-modules

$$\boxed{0 \longrightarrow V \xrightarrow{\Theta} \mathscr{B}_Z \xrightarrow{J} E \longrightarrow 0.} \tag{3}$$

The arrows in (3), not just the three modules, specify its extension class. Compute it using the resolution

$$0 \longrightarrow A \xrightarrow{\times h} A \longrightarrow E \longrightarrow 0.$$

The resulting isomorphism $\mathrm{Ext}^1_A(E, V) \cong V/h(D)V$ sends an extension to $h(D)b$ in the source, where b is any lift of $1 \in E$. Fixing this explicit rule fixes its sign. Under the existing map $[\phi] \mapsto j_h H_\phi$, the class of (3) is

$$\boxed{[\mathscr{B}_Z] = \varepsilon_Z = j_h(h/g) \in E^\times.} \tag{4}$$

Proof: use the lift $s(1)$ and (1). Its source boundary is $R_Z(1)(D)\phi_*$, whose H-jet is ε_Z.

In particular, (3) has no A-linear section. This also follows immediately from the injectivity of $h(D)$ on $\mathscr{B}$: an A-linear map $E \to \mathscr{B}$ has image killed by $h(D)$ and is therefore zero. The map s in (2) is the explicit complex-linear section, and the next section computes exactly how it fails to intertwine the arithmetic action. This non-splitting occurs for every nonempty packet, including critical-line packets; it is not an RH-failure criterion.

The general use of extension classes is standard homological algebra (Stacks, Tags 010I and 06XP). Formula (4) calculates the specific extension arising from this theta complex.

B.1.4 3. The scaling defect is an original theta boundary

For $P \in A$ and $u \in E$, define

$$q_P(u) = \frac{P(t)R_Z(u) - R_Z(Pu)}{h(t)} \in A, \qquad k_P(u) = q_P(u)(D)\phi_* \in V. \tag{5}$$

The numerator is divisible by h, since both of its residues are $P\varepsilon_Z u$. Using (1) and the injectivity of $h(D)$ on $\mathscr{B}$ gives

$$\boxed{P(D)s - sM_P = \Theta k_P.} \tag{6}$$

No quotient has yet been applied in (6). The right side is an actual theta function. The composition law is

$$\boxed{k_{PQ} = P(D)k_Q + k_P M_Q.} \tag{7}$$

It follows either by expanding the numerator of (5), or by expanding the commutator in (6) and using injectivity of theta.

For $P = t$, the numerator of (5) has degree at most d. Hence

$$\ell_Z(u) = [t^{d-1}]R_Z(u), \quad q_t(u) = \ell_Z(u), \quad \boxed{Ds(u) - s(tu) = \ell_Z(u)\Theta\phi_*.} \tag{8}$$

Thus, in the exact coordinates (2), the original operator is

$$\boxed{D_{\mathscr{B}_Z} = \begin{pmatrix} D_V & \phi_*\ell_Z \\ 0 & M_t \end{pmatrix}.} \tag{9}$$

Its off-diagonal map is rank one and its coefficient is determined by the retained local arithmetic unit. It is not chosen to improve a spectral estimate.

For one simple zero,

$$h = t - \rho, \quad \varepsilon_Z = 1/g'(\rho),$$

$$s(1) = \frac{1}{g'(\rho)} x^{-\rho} \int_x^\infty \Theta\phi_*(y) y^{\rho-1}\, dy,$$

$$\boxed{(D - \rho)s(1) = \frac{1}{g'(\rho)}\Theta\phi_*.} \tag{10}$$

The coefficient is $1/(2\xi'(\rho))$, with the original factor two retained.

The full dilation group For $a > 0$, retain $U_a F(x) = F(x/a)$ and $A_a = M_{j_h(a^s)}$ on E. Define

$$k_a(u) = S_h^V\big(U_a(R_Z(u)(D)\phi_*) - R_Z(A_a u)(D)\phi_*\big). \tag{11}$$

The source of S_h^V has zero H-jets because $a^s\varepsilon_Z u$ and $\varepsilon_Z A_a u$ have equal jets. All powers use the real logarithm of a. The exact identities are

$$U_a s - sA_a = \Theta k_a, \qquad k_{ab} = U_a k_b + k_a A_b, \qquad k_1 = 0. \tag{12}$$

The same construction applies to a smooth compactly supported multiplicative test f: replace U_a by $H_f = \int f(a)U_a\, da/a$ and A_a by $M_{j_h\mathcal{M}f}$. The source integral belongs to V; its H-jets give the same cancellation. These are actual linear maps, not a choice of a formal matrix on E alone.

B.1.5 4. A derived equivariant inclusion with its full homotopy

Put $P_Z = sJ : \mathscr{B} \to \mathscr{B}$. It is a finite-rank idempotent, and $P_Z\Theta = 0$. Consequently

$$\mathsf{p}_Z = (0, P_Z) : C_+ \to C_+ \tag{13}$$

is a cochain idempotent. Its image is the two-term complex $[0 \to s(E)]$, complex-linearly isomorphic to $E[-1]$.

For the operator $P(D)$ on C_+, define a degree-minus-one map with sole component

$$\mathsf{H}_P^1 = k_P J : \mathscr{B} \to V.$$

Equation (6) gives

$$\boxed{P(D)\mathsf{p}_Z - \mathsf{p}_Z P(D) = d\mathsf{H}_P + \mathsf{H}_P d.} \tag{14}$$

In degree one this is $\Theta k_P J$; in degree zero it is zero because $J\Theta = 0$. For dilation use $\mathsf{H}_a^1 = k_a J$. Equation (12) is the coherence law of those homotopies.

There is also a strictly A-linear derived presentation. Set

$$C_Z = [V \xrightarrow{\Theta} \mathscr{B}_Z].$$

In the coordinates (2) its differential is $\phi \mapsto (\phi, 0)$ and its action is (9). The maps

$$\iota : C_Z \hookrightarrow C_+, \qquad \pi : C_Z \to E[-1], \quad \pi^0 = 0,\ \pi^1 = J \tag{15}$$

are strictly A-linear cochain maps, and π is a quasi-isomorphism. Thus the actual equivariant inclusion is the roof

$$\boxed{E[-1] \underset{\simeq}{\xleftarrow{\pi}} C_Z \xrightarrow{\iota} C_+.} \tag{16}$$

The complex-linear deformation retraction also has an exact contracting homotopy: on $\mathscr{B}_Z$, send $\Theta\phi + s(u)$ to $\phi \in V$. Its equation is $1 - s\pi = dH + Hd$ (with s interpreted as the degree-one cochain section). This homotopy is not claimed A-linear; the A-linear replacement is the roof (16).

The strictly A-linear map $j : C_+ \to E[-1]$ is J in degree one and zero in degree zero. It satisfies $j\iota = \pi$. This gives a split retract in $D(A)$ without falsely asserting an A-linear section into $\mathscr{B}$.

On the two-leg complex use $\mathsf{p}_Z^0 = 0$ on $V \oplus V$, $\mathsf{p}_Z^1 = P_Z$, and $\mathsf{H}_a^1(F) = (k_a JF, 0)$. The source dilation is $(U_a, aU_{1/a})$, so (14) holds with its original two-leg differential. The inclusion $C_+ \to C_{+-}$ commutes with these cochain projectors. The extra degree-zero cohomology of C_{+-} is killed by this projector; it remains in the complementary complex.

B.1.6 5. The mixed-support homotopies require the actual synchronization map

Let $L = \mathcal{P}(\{+, -\})$ be the original chart-support lattice. Its three active complexes are

$$C_+ = [V \xrightarrow{\Theta} \mathscr{B}], \quad C_- = [V \xrightarrow{-\Theta\mathcal{F}} \mathscr{B}], \quad C_{+-} = [V \oplus V \xrightarrow{\Theta(\phi - \widehat{\psi})} \mathscr{B}].$$

Here $\mathcal{F}$ is the specified Fourier transform, with $\mathcal{F}^2 = 1$ on even functions. The source inclusions into the joint fibre are $v \mapsto (v, 0)$ and $v \mapsto (0, v)$; degree-one transports are identities. The bottom complex is zero in this particular model. The new formalization allows a nonzero bottom fibre in the general reconstruction; no assertion about all diagrams is obtained by specializing this model.

The reconstructed operations are

$$(\lambda, x) + (\mu, y) = (\lambda \vee \mu, \rho_{\lambda,\lambda\vee\mu}x + \rho_{\mu,\lambda\vee\mu}y),$$

$$e(\lambda, x) = (\lambda, 0), \qquad \tau(\lambda, x) = (\perp, 0). \tag{17}$$

The cochain projector $\widetilde{\mathfrak{p}}_Z$ is zero-in-its-fibre in degree zero and P_Z in degree one. It commutes with the support transports. So does the actual action: U_a on the plus source, $aU_{1/a}$ on the minus source, their direct sum on the joint source, and U_a at every active overlap.

The objectwise homotopies are, however, different maps. With $b = k_a JF$, they are

$$\mathsf{H}^1_{a,+}(F) = b, \qquad \mathsf{H}^1_{a,-}(F) = -\widehat{b}.$$

Their images in the joint source are $(b, 0)$ and $(0, -\widehat{b})$. The difference is the explicit cycle

$$\boxed{(b, \widehat{b}) \in \ker d_{+-} = H^0(C_{+-}).} \tag{18}$$

For nonzero k_a, there is no label-preserving family of homotopies compatible with both source inclusions: the joint component would have to equal both $(b, 0)$ and $(0, -\widehat{b})$. This says exactly which naturality square fails. It does not discard the cohomology or the arithmetic action.

The replacement is the programme's actual synchronization. Define the support map

$$r_L(\perp) = \perp, \qquad r_L(\lambda) = \{+, -\} \quad (\lambda \neq \perp).$$

In degree zero use literal multiplication by $E = (1, e)$ on the original two-coordinate support module:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea).$$

In degree one define $\mathfrak{r}^1(\lambda, F) = (r_L\lambda, F)$ for active labels and preserve global zero. An absent source coordinate becomes its supported vector zero, so the source differential has unchanged amplitude. Hence

$$\boxed{d\mathfrak{r}^0 = \mathfrak{r}^1 d, \qquad \mathfrak{r}^2 = \mathfrak{r}.} \tag{19}$$

This uses the operation verified in the synchronization attachment, now on the actual complex. The fixed active fibre is exactly C_{+-}. On degree-one homology, the map is

$$(\lambda, [F]) \longmapsto (\{+, -\}, [F]).$$

It has no nonzero amplitude kernel, because all degree-one transports are the identity of Q. The original mask is retained through

$$x \longmapsto (\mathrm{supp}(x), \mathfrak{r}x).$$

This map is injective in each degree of this chart complex: in degree zero the coordinate inclusions are injective, and in degree one the transport is the identity. This injectivity is a proved property of these maps, not an assumption for general diagrams with noninjective transports.

There are now two actual split-linear homotopies with the same support map r_L. Define

$$\widetilde{\mathsf{H}}^{\mathrm{left},1}_a(\lambda, F) = (\{+, -\}, (k_a JF, 0)),$$

$$\widetilde{\mathsf{H}}_a^{\text{right},1}(\lambda, F) = (\{+,-\}, (0, -\mathcal{F}k_a JF)),$$

and send global zero to global zero. Both satisfy

$$\boxed{\mathfrak{r}\big(\widetilde{U}_a\widetilde{\mathsf{p}}_Z + (-1)^{\bullet}\widetilde{\mathsf{p}}_Z\widetilde{U}_a\big) = \widetilde{d}\widetilde{\mathsf{H}}_a + \widetilde{\mathsf{H}}_a\widetilde{d}.} \tag{20}$$

In degree one this is (6); in degree zero it follows from $Jd = 0$. The two choices differ by the retained cycle (18). No averaging or removal of that cycle is performed. This is the precise role of synchronization here: it permits a common supported comparison fibre for the homotopy, while the original mask and the difference of homotopies remain recorded.

In the simple-zero case (10), the defect in any source support is the nonzero chain amplitude $\Theta(\phi_*/g'(\rho))$. Its internal homology class is the zero of that support, not external absence.

For every chain comparison $f : C \to C'$, the supplied formalized kernel equivalence remains

$$\{z \in Z(C) : f(z) \in B(C')\}/B(C) \xrightarrow{\sim} \ker H(f).$$

Applied to π in (15), it computes the zero kernel in degree one from its actual exact sequence. Applied to support synchronization, it yields the degree-one assertion just proved and the separately retained degree-zero data. It is not applied to a different global comparison without carrying that comparison's maps.

For any smaller D-stable admitted source $W \subset V$, the map $Q_W \to Q$ has kernel $\Theta V/\Theta W$. A homotopy value may not already lie in W. The explicit replacement is enlargement by the actual source values, with this quotient map and its kernel retained. The present chart uses full V, where every displayed homotopy is defined.

B.1.7 6. The same extension class produces the arithmetic duality factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$. Then

$$f^\dagger(s) = \overline{f(1-\bar{s})}, \qquad h^\dagger = (-1)^d h.$$

The perfect finite pairing is

$$R_h(f, u) = \sum_{\rho \in Z} \operatorname{Res}_{s=\rho} \frac{f^\dagger(s)u(s)}{h(s)}\, ds,$$

with exact symmetry $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$.

Identify E with the actual divisor algebra $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho/(g)$ by polynomial evaluation and the complete Chinese-remainder jets. The nonzero local units in g/h make this an isomorphism. The operator represented by the extension class is

$$M_{\varepsilon_Z} : E \to E.$$

Substituting (4) into the residue formula gives

$$\boxed{R_h(f, M_{\varepsilon_Z} u) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger u}{g}\, ds = R_Z(f, u).} \tag{21}$$

This is independent of representatives: changing ε_Z by h times a germ adds a holomorphic differential. Thus the actual extension class is precisely the unit morphism converting the polynomial-resolution residue form into the arithmetic residue form.

Let $J_g = M_{j_h g'}$. Then

$$\boxed{R_h(f, M_{\varepsilon_Z} J_g u) = \sum_{\rho\in Z} m_\rho \overline{f(1-\bar\rho)} u(\rho) = \mathrm{Tr}(M_{f^\dagger u} \mid E).} \tag{22}$$

Locally $g = z^m w(z)$ gives $g'/g = m/z + w'/w$. This proves the formula and retains every multiplicity. The product $\varepsilon_Z J_g$ carries both factors associated to $g = 2\xi$.

The pairing-valued support map sends a zero value to $e_{\mathbb{C}}$ and an absent argument to τ. Its original input support and relation fibre remain available upstream. Formula (22) is a calculated trace, not a positivity assertion.

Reflection is strict for this specified section Define conjugate-linear involutions

$$\mathsf{j}_B F(x) = x^{-1}\overline{F(1/x)}, \quad \mathsf{j}_V \phi = \overline{\widehat{\phi}}, \quad j_E u = u^\dagger.$$

They satisfy $\Theta \mathsf{j}_V = \mathsf{j}_B \Theta$ and $J \mathsf{j}_B = j_E J$. The actual source function is Fourier-fixed: writing $\phi_* = D(D-1)e^{-\pi x^2}$, Fourier sends the two factors to $(1-D)(-D) = D(D-1)$ and fixes the Gaussian. It is real, so $\mathsf{j}_V \phi_* = \phi_*$.

Since $\varepsilon_Z^\dagger = (-1)^d \varepsilon_Z$ and reflection preserves polynomial degree,

$$R_Z(j_E u) = (-1)^d (R_Z(u))^\dagger.$$

Uniqueness of the inverse of $h(D)$ on its range gives

$$\mathsf{j}_B S_h^{\mathscr{B}} = (-1)^d S_h^{\mathscr{B}} \mathsf{j}_B.$$

Both factors $(-1)^d$ are retained in the substitution, and their product is one. Therefore

$$\boxed{\mathsf{j}_B s = s j_E, \qquad \mathsf{j}_B P_Z = P_Z \mathsf{j}_B.} \tag{23}$$

This is an actual equality for this section; it would not follow for an arbitrary choice of representatives. Its compatibility with the nonzero dilation homotopy is

$$\boxed{\mathsf{j}_V k_a = a k_{1/a} j_E.} \tag{24}$$

Indeed, apply j_B to (12), use $\mathsf{j}_B U_a = a U_{1/a} \mathsf{j}_B$ and (23), then use injectivity of theta. The split lift is semilinear through $r^\bullet \mapsto \bar r^\bullet$, which fixes both e and τ.

B.1.8 7. A finite-rank Lefschetz trace on the actual chain complex

For $f \in C_c^\infty(\mathbb{R}_{>0})$, let U_f be its convolution action on C_+. The cochain endomorphism

$$\mathsf{p}_Z \mathsf{U}_f = (0, sJH_f)$$

has finite-rank components. Its cochain supertrace is therefore defined without an infinite-dimensional regularization:

$$\boxed{\mathrm{Str}(\mathsf{p}_Z \mathsf{U}_f \mid C_+) = -\,\mathrm{Tr}(sJH_f \mid \mathscr{B}) = -\sum_{\rho\in Z} m_\rho \mathcal{M}f(\rho).} \tag{25}$$

The finite-rank identity $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ reduces the middle expression to $\mathrm{Tr}(JH_f s \mid E)$; this is multiplication by $j_h \mathcal{M} f$. Equation (25) equals the supertrace of the induced operator on homology.

On C_{+-} its degree-zero contribution is still zero, since the same cochain projector has zero degree-zero component.

The boundary representatives in (6), (14), and (20) are retained even though their contribution to homology is zero. The finite-rank trace does not delete their complex or assign its zero to external absence.

For the n-fold tensor complex over the tau-base product, use the original coefficient tensor over $\mathbb{C}$ and its differential

$$d(x_1 \otimes \cdots \otimes x_n) = \sum_j (-1)^{\sum_{i<j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_n.$$

The projector $\mathsf{p}_Z^{\otimes n}$ has image $E^{\otimes n}[-n]$. For independently specified tests $f_1, \ldots, f_n$,

$$\mathsf{Str}\left(\bigotimes_j (\mathsf{p}_Z \mathsf{U}_{f_j})\right) = (-1)^n \prod_j \sum_{\rho \in Z} m_\rho \mathcal{M} f_j(\rho). \tag{26}$$

Let $A = U_a \mathsf{p}_Z$ and $B = \mathsf{p}_Z U_a$ as cochain maps. A homotopy for $A^{\otimes n} - B^{\otimes n}$ is

$$\sum_{j=1}^{n} B^{\otimes(j-1)} \otimes \mathsf{H}_a \otimes A^{\otimes(n-j)},$$

with the displayed Koszul sign on homogeneous inputs when the degree-minus-one map crosses the preceding factors. The telescoping identity and (14) prove it. The full support mask maps to the product of its input masks; vanishing products of represented amplitudes stay supported.

B.1.9 8. What now enters the Weil II programme

The structural base, the infinite quotient, and the actual theta complex are unchanged. The new objects are the extension (3), its unit class (4), its synchronized homotopy (20), its strict equivariant derived roof (16), and the chain trace (25). The arithmetic residue unit in (21) is now supplied by that very extension rather than appended without its source.

Deligne's inspected comparison-image argument (Weil II 3.3.4–3.3.6) requires two weight bounds on the same cohomological image. The inspected derived formulation (6.2.1–6.2.7) retains the dualizing operation, its degree and Tate shifts, and its exchange of direct-image functors. This continuation supplies coherent arithmetic maps and finite trace operations for the tau model; it does not transfer the finite-field hypotheses or assert those two bounds.

The retained error is explicit: it is the original boundary Θk_a on representatives, with the nonzero Ext class (4). It is removed at homology by the original supported quotient, not by making its input absent. A geometric comparison can now be tested for compatibility with this class and the residue map (21). A complex-linear section that ignores (6) fails that test.

No infinite sum of packet projectors is asserted to converge. The existing meromorphic/kernel summand from PR #8 remains in its own exact sequence, and a different global trace functor must retain or analyze its contribution. The source's finite-jet isomorphisms and the new finite-rank chain trace do not by themselves set that summand to zero.

B.1.10 9. Verification and references

The new finite checker tests polynomial remainder identities, the extension class, rank-one scaling defects, cochain homotopies, dilation cocycles, residue contraction, tensor signs, trace identities, and original support operations. These finite algebraic models are not numerical tests of zeta zeros and not certificates for the analytic range theorem. No Lean execution is claimed for this continuation.

Read source interfaces:

- PR #8 at the pinned revision above, FINITE_OPERATOR_COMPARISON.md, sections 1–10; TAU_BASE_MODEL.md is its already-published model.
- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1–7. In particular its allowance of a nonzero bottom fibre and the exact homology-kernel/action theorems are retained.
- formal/splitzero/SplitZeroHomology.lean, lines 175–end, read at the same revision; blob 7b9e5c49a494dd0c751133e0234b3e01048f7b67.
- Supplied Deligne TeX excerpts: French segment 2, original lines 882–950; segment 3, original lines 675–756. The used equations are the image bounds and duality operations, not every assertion in the historical transcription. No source corpus is republished with this contribution.
- The Stacks Project, Tags 010I and 06XP, for extension classes and their projective-resolution interpretation.
- R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277, for prior spectral-realization context. No global priority claim is made here.

New calculations are AI-assisted work under the owner's programme. Existing mathematical files, formalizations, licensing, and source attribution remain untouched.

C PR 10: Global retraction

Source attribution. Web contribution, pull request #10, Zeta research repository.

Exact revision. d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7

Source file. `workbenches/tau-global-retraction/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859

C.1 A global retraction of the original theta complex over the tau base

Owner-directed continuation of Zeta PR #9, 12 September 2026. New derivations are submitted as research proofs for review, not as Lean-certified analytic theorems.

C.1.1 0. Fixed source, result, and provenance

The base is PR #9 at 8f99b94d306c3b1aaa817d52c57fa6fd298d511c; its chain-descent note has Git blob 86db838e79d10a8fe7d4376303706c4ea64570a1. Its source and the owner's derived SplitZero attachment remain the inputs. This continuation constructs an inverse on the entire original theta image, not only on a chosen finite spectral packet.

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \qquad e_R = 0_R^\bullet,$$

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \qquad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \tag{1}$$

The first vertical target is still infinite. The geometric base, the original additive laws, and the arithmetic quotient are unchanged. The auxiliary cutoffs below are parameters of an explicit continuous representative map; they do not change an Euler factor, the theta relation space, or the completed zeta function.

Result. For every pair of the cutoffs specified in section 3 there is an explicitly constructed continuous complex-linear map

$$\Lambda_{\alpha,\beta} : \mathscr{B} \longrightarrow V, \qquad \boxed{\Lambda_{\alpha,\beta}\Theta = \mathrm{id}_V\,.} \tag{2}$$

Consequently the actual image ΘV is closed and complemented in the stated strong-Schwartz topology. The original quotient $Q = \mathscr{B}/\Theta V$ is Hausdorff. The note constructs its global section, the full supported cochain homotopy, the exact change of the finite sections in #9, its continuous dual sequence, and the completed tensor powers. The proof does not assume RH, simplicity of any zero, or vanishing of the separate balanced-Mellin kernel.

Spectral realizations on nuclear Frechet spaces are prior work, notably R. Meyer, arXiv:math/0412277. That paper's abstract was consulted for scope, but no theorem from an unread body of that paper is imported. The proof below is given for the workbench's precise spaces. No global novelty claim is made.

C.1.2 1. Spaces and all transforms

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x),\ \phi(0) = 0,\ \int_{\mathbb{R}} \phi(x)\, dx = 0\},$$

$$\mathscr{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) := \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for every } b \in \mathbb{Z},\ k \geq 0\}, \qquad D = -x\partial_x. \tag{3}$$

These seminorms give $\mathscr{B}$ its Frechet topology; under $x = e^y$ they are the weighted uniform derivative seminorms on smooth functions of y. Completeness follows by uniform convergence on compact sets of every derivative, followed by the weighted bounds. V has its closed-subspace Schwartz topology.

The original maps are

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z}\setminus\{0\}} \phi(nx) = 2\sum_{n\geq 1} \phi(nx),$$

$$\mathcal{F}\phi(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x\xi}\, dx, \qquad \mathcal{F}^{-1}u(x) = \int_{\mathbb{R}} u(\xi) e^{2\pi i x\xi}\, d\xi,$$

$$JF(x) = x^{-1}F(1/x), \qquad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \, \frac{dx}{x}. \tag{4}$$

Poisson summation, with both source moments zero, gives

$$\Theta\mathcal{F} = J\Theta, \qquad J^2 = 1, \qquad DJ = J(1-D). \tag{5}$$

The theta sum and each derivative converge absolutely on closed subsets of $x > 0$. At infinity, Schwartz estimates give all $p_{b,k}$ bounds. At zero use (5) and the same estimates on $\widehat{\phi}$. This proves continuity $\Theta : V \to \mathscr{B}$. Also J is continuous on $\mathscr{B}$: $p_{b,k}(JF)$ is bounded by the binomial sum of $p_{1-b,j}(F), 0 \leq j \leq k$.

The original complex is $C_+ = [V \xrightarrow{\Theta} \mathscr{B}]$ in degrees $0, 1$, over $A = \mathbb{C}[t]$ with t acting as D. The two-leg tau chart retains $C_{+-} = [V^2 \to \mathscr{B}]$ with differential $\Theta(\phi - \widehat{\psi})$ and source action $(D, 1-D)$.

C.1.3 2. Arithmetic inversion on the two exterior regions

Let μ be the ordinary Mobius function, characterized by

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For $F \in \mathscr{B}$ define, for $x \neq 0$,

$$\boxed{\mathcal{U}F(x) = \frac{1}{2}\sum_{n\geq 1} \mu(n) F(n|x|).} \tag{6}$$

The factor $1/2$ is the inverse of the full nonzero-integer theta sum in (4). The map is complex-linear into smooth even functions on $\mathbb{R} \setminus \{0\}$. It is not asserted to be smooth at zero for arbitrary F.

Here are the required estimates. For $j \geq 0$, let $P_j(T) = (-1)^j T(T+1)\cdots(T+j-1)$, with $P_0 = 1$. Then $x^j \partial_x^j = P_j(D)$. If $C_{N,j}(F)$ is the sum of the $p_{N,k}(F)$ weighted by the absolute coefficients of P_j, then, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2}\zeta(N)C_{N,j}(F)x^{-N-j}. \tag{7}$$

Indeed each summand is bounded by $C_{N,j}(F)n^{-N}x^{-N-j}/2$. This proves convergence of every derivative uniformly for $|x| \geq \delta > 0$, and gives rapid decay there. It also proves continuity of the cut-off exterior maps used below.

For every $\phi \in V$,

$$\boxed{\mathcal{U}\Theta\phi(x) = \phi(x), \qquad \mathcal{U}J\Theta\phi(x) = \widehat{\phi}(x) \quad (x \neq 0).} \tag{8}$$

For the first equality the double series is absolutely convergent for each $x > 0$: its absolute value is bounded by $\sum_{k\geq 1} d(k)|\phi(kx)|$, which is finite by Schwartz decay. Rearrange by $k = mn$ and use the displayed divisor identity. Evenness handles negative x. The second equality follows from (5) and the first equality applied to $\widehat{\phi} \in V$.

In the split lift, every $\mu(n)$ is the supported integer $\mu(n)^\bullet$, including $\mu(n) = 0$. A convergent sum within a specified active fibre begins at that fibre's zero and is taken in that fibre's vector-space topology. Formula (8) thus lifts to a support-preserving equality. A computed cancellation is not external absence.

C.1.4 3. The reconstruction operator, with the strict contraction proved

Choose and retain real, even $\alpha, \beta \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq \alpha, \beta \leq 1$ and equal to 1 on neighborhoods of zero. Their supports and all values are part of the auxiliary data. Work on $\mathcal{H} = L^2(\mathbb{R}, dx)$ with the Fourier convention in (4).

Define bounded operators

$$A_\alpha = M_\alpha, \qquad B_\beta = \mathcal{F}^{-1}M_\beta\mathcal{F}, \qquad T_{\alpha,\beta} = A_\alpha B_\beta. \tag{9}$$

Lemma. $T_{\alpha,\beta}$ is compact and $c_{\alpha,\beta} := \|T_{\alpha,\beta}\| < 1$.

Proof. Its kernel is $\alpha(x)\check{\beta}(x-y)$, whose squared integral is $\|\alpha\|_2^2\|\beta\|_2^2$. Hence it is Hilbert-Schmidt. Both factors are contractions. If the norm were 1, compactness would supply a unit vector f attaining it. Equality in

$$\|A_\alpha B_\beta f\| \leq \|B_\beta f\| \leq \|f\|$$

forces $\widehat{f}$ to be supported where $\beta = 1$; there $B_\beta f = f$. The first equality forces f to be supported where $\alpha = 1$. Both supports are compact. The inverse Fourier transform of compactly supported $\widehat{f} \in L^2$ is entire, by differentiation under its finite-interval integral. Its continuous restriction agrees with the compactly supported f, so it vanishes on an open real interval and hence identically. This contradicts $\|f\| = 1$. Thus $c_{\alpha,\beta} < 1$.

Consequently the exact Neumann inverse exists:

$$\boxed{(1 - T_{\alpha,\beta})^{-1} = \sum_{k\geq 0} T_{\alpha,\beta}^k, \qquad \|(1 - T_{\alpha,\beta})^{-1}\| \leq \frac{1}{1 - c_{\alpha,\beta}}.} \tag{10}$$

No uniform bound as the cutoffs vary is claimed. In particular the gap $1 - c_{\alpha,\beta}$ is retained, not replaced by a positive universal constant.

The operator $T_{\alpha,\beta}$ maps L^2 continuously to the Schwartz space. For each derivative, compact Fourier support and Cauchy-Schwarz bound $\partial^j B_\beta f$ uniformly by a constant times $\|f\|_2$; multiplication by the smooth compactly supported α gives every Schwartz seminorm.

Define the two exterior data and their combination:

$$a_F = (1-\alpha)\mathcal{U}F, \qquad b_F = (1-\beta)\mathcal{U}JF,$$

$$Y_{\alpha,\beta}F = a_F + \alpha\mathcal{F}^{-1}b_F. \tag{11}$$

The exterior terms are extended by zero near the origin, where their cutoff is identically zero. Estimates (7) prove that both are even Schwartz functions and depend continuously on F. Therefore $Y : \mathscr{B} \to \mathcal{S}(\mathbb{R})^{\text{even}}$ is continuous.

Put

$$\Phi_{\alpha,\beta}F = (1-T_{\alpha,\beta})^{-1}Y_{\alpha,\beta}F. \tag{12}$$

This initially defines an L^2 vector, but the identity $\Phi F = YF + T\Phi F$ proves it is Schwartz. If q is any Schwartz seminorm and $q(Tu) \le C_q\|u\|_2$, then

$$q(\Phi F) \le q(YF) + \frac{C_q}{1-c_{\alpha,\beta}}\|YF\|_2. \tag{13}$$

This is an explicit continuity estimate. Both sides are even because the cutoff operators preserve parity.

For $F = \Theta\phi$, equations (8) give

$$Y\Theta\phi = (1-A_\alpha)\phi + A_\alpha(1-B_\beta)\phi = (1-T_{\alpha,\beta})\phi,$$

hence

$$\boxed{\Phi_{\alpha,\beta}\Theta\phi = \phi.} \tag{14}$$

C.1.5 4. Keep both original moment conditions

For arbitrary F, (12) need not satisfy the two source moments. The correction is the following specified projection, not an alteration of V.

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \qquad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int\phi)$ are $(1,1)$ and $(0,1)$. Define

$$\mathsf{m}(\phi) = \big(\phi(0), \int_{\mathbb{R}}\phi\big), \qquad j_0(a,b) = a\gamma_0 + (b-a)\gamma_2,$$

$$P_V\phi = \phi - j_0\mathsf{m}(\phi) = \phi - \phi(0)\gamma_0 - \left(\int\phi - \phi(0)\right)\gamma_2. \tag{15}$$

Then $\mathsf{m}j_0 = 1$, $P_V^2 = P_V$, $\operatorname{im} P_V = V$, and $P_V|_V = 1$. All maps are continuous.

The promised map is

$$\boxed{\Lambda_{\alpha,\beta} = P_V(1 - T_{\alpha,\beta})^{-1} Y_{\alpha,\beta} : \mathscr{B} \to V.} \tag{16}$$

Equation (14) and $P_V|_V = 1$ prove (2). The discarded moment vector has the recorded map $\mathsf{m}\Phi : \mathscr{B} \to \mathbb{C}^2$ and the explicit correction $j_0\mathsf{m}\Phi$. It vanishes on the actual theta image. No endpoint constant was absorbed into the zeta function.

C.1.6 5. The full quotient is closed and has a continuous section

Write $\Lambda = \Lambda_{\alpha,\beta}$, and define

$$P_\Theta = \Theta\Lambda, \qquad K = 1_{\mathscr{B}} - \Theta\Lambda. \tag{17}$$

Direct substitution gives

$$P_\Theta^2 = P_\Theta, \quad K^2 = K, \quad \ker K = \Theta V, \quad \operatorname{im} K = \ker \Lambda. \tag{18}$$

Since K is continuous, its kernel is closed. Thus $Q = \mathscr{B}/\Theta V$ with its actual quotient topology is Frechet and Hausdorff. If $q : \mathscr{B} \to Q$ is the original quotient map, the continuous section is

$$\boxed{s : Q \to \mathscr{B}, \qquad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0.} \tag{19}$$

The explicit topological direct-sum isomorphism is

$$\boxed{V \oplus Q \xrightarrow{\sim} \mathscr{B}, \quad (\phi, u) \mapsto \Theta\phi + su, \quad F \mapsto (\Lambda F, qF).} \tag{20}$$

For a convergent sequence $\Theta\phi_n \to F$ in $\mathscr{B}$, one can check the closedness without any abstract theorem: $\phi_n = \Lambda\Theta\phi_n \to \Lambda F$ in V, so $F = \Theta\Lambda F$ by continuity of theta.

This computes the earlier Hausdorff arrow:

$$Q^{\text{alg}} = \mathscr{B}/\Theta V \longrightarrow Q^{\text{H}} = \mathscr{B}/\overline{\Theta V}, \qquad [F] \mapsto [F], \tag{21}$$

as an isomorphism with zero amplitude kernel. It does not say the map from Q into its product of spectral jets is injective. The latter map and the balanced analytic map still have their own named kernels:

$$Q \to \prod_\rho \mathcal{O}_\rho/(g), \qquad \mathcal{O} \otimes_{\mathbb{C}[t]} Q \xrightarrow{\beta} \mathcal{O}/(g).$$

No claim that $\mathcal{R}_\infty$ or $\ker \beta$ vanishes is inferred from (21).

Smaller admitted relations For any specified subspace $W \subset V$, retain

$$Q_W = \mathscr{B}/\Theta W \longrightarrow Q, \qquad \ker = \Theta V/\Theta W \cong V/W. \tag{22}$$

The isomorphism sends $[\phi] \mapsto [\Theta\phi]$. It is algebraic for arbitrary W; when W is closed, (20) makes it a topological isomorphism onto the kernel. The pullback of the global section has the representative relation kernel in (22); the full-source result is not silently asserted with W in place of V.

C.1.7 6. Perform the original internal quotient and synchronization

On the single-leg map, or a fixed-index diagram of that same map, the split lifts are

$$\widetilde{\Theta}(\lambda, \phi) = (\lambda, \Theta\phi), \quad \widetilde{\Lambda}(\lambda, F) = (\lambda, \Lambda F), \quad \widetilde{K}(\lambda, F) = (\lambda, KF). \tag{23}$$

They preserve external bottom and commute with scalar reflection. In particular

$$\widetilde{K}\widetilde{\Theta}(\lambda, \phi) = (\lambda, 0_{\mathscr{B}}) = z_{V,\mathscr{B}}(\lambda, \phi).$$

Here $z_{V,\mathscr{B}}$ is the label-preserving zero map from the source fibre to the target fibre. The mixed chart, whose minus restriction is $-\Theta\mathcal{F}$, is treated separately by the explicit maps (25)-(28), without changing that restriction. The actual chart has zero bottom fibre; a general nonzero bottom fibre is acted on by its stated linear map, rather than identified with the single global zero.

The supported image projection is idempotent, and the quotient map is exactly the coequalizer of the theta relation inclusion and its supported-zero companion. The implementation's universal property applies against all split-linear targets. A theta boundary is sent to $e_\lambda = (\lambda, 0)$; it is not sent to external absence.

For the single-leg complex there is an actual continuous contraction

$$C_+ \underset{s}{\overset{q}{\rightleftharpoons}} Q[-1], \qquad H^1 = \Lambda,$$

$$\boxed{qs = 1, \qquad 1_{C_+} - sq = dH + Hd.} \tag{24}$$

In degree zero this is $\Lambda\Theta = 1_V$; in degree one it is $\Theta\Lambda = 1 - K$. In split notation the subtraction means addition of the supported scalar $(-1)^\bullet$ times the second map. It yields the correct labelled zero wherever the amplitude vanishes.

For the two-leg chart set $H_0 = V[0] \oplus Q[-1]$ with zero differential, and define

$$p_L^0(\phi, \psi) = \psi, \quad p_L^1 = q, \qquad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s,$$

$$H_L^1(F) = (\Lambda F, 0). \tag{25}$$

Then $p_L i = 1$, and $1 - ip_L = dH_L + H_L d$. The degree-zero identity is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$. The other leg gives

$$p_R^0(\phi, \psi) = \widehat{\phi}, \quad p_R^1 = q, \qquad H_R^1(F) = (0, -\widehat{\Lambda F}). \tag{26}$$

It obeys $1 - ip_R = dH_R + H_R d$. Their homotopies differ by

$$\boxed{H_L^1 F - H_R^1 F = (\Lambda F, \widehat{\Lambda F}),} \tag{27}$$

the actual joint degree-zero cycle, which is retained.

On the complete original support diagram, use literal multiplication by $E_{\mathsf{sync}} = (1, e)$ in degree zero and the source's active-label-to-joint map in degree one:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea), \qquad \mathfrak{r}^1(\lambda, F) = (\{+, -\}, F) \quad (\lambda \neq \perp).$$

The absent input remains absent. The original differential obeys $d\mathfrak{r} = \mathfrak{r}d$ and $\mathfrak{r}^2 = \mathfrak{r}$. Consequently (25) lifts to the fully typed equation

$$\boxed{\mathfrak{r} - \widetilde{i}\widetilde{p}_L\mathfrak{r} = \widetilde{d}\,\widetilde{H}_L + \widetilde{H}_L\widetilde{d},} \tag{28}$$

where $\widetilde{H}_L(\lambda, F) = (\{+,-\}, (\Lambda F, 0))$ for each active label. The right-leg equation is analogous with (26). The original mask is recorded with the injection $x \mapsto (\operatorname{supp} x, \mathfrak{r}x)$ already proved for this chart. Neither synchronization nor the contraction identifies the three supported origins with τ.

C.1.8 7. Global scaling defects, and their agreement with the finite extensions

Let $U_aF(x) = F(x/a)$, $a > 0$, and denote its induced continuous action on Q by $\overline{U}_a$. The section (19) gives a global continuous boundary map

$$\boxed{k_a = \Lambda U_a s : Q \to V, \qquad U_a s - s\overline{U}_a = \Theta k_a.} \tag{29}$$

The proof applies $K + \Theta\Lambda = 1$ to $U_a s$, then uses $qU_a s = \overline{U}_a$. Therefore

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \qquad k_1 = 0. \tag{30}$$

The same calculation works for D and compactly supported multiplicative convolution, with $k_D = \Lambda D s$. In the explicit coordinates (20),

$$\boxed{U_a|_{\mathscr{B}} = \begin{pmatrix} U_a|_V & k_a \\ 0 & \overline{U}_a \end{pmatrix}, \qquad D|_{\mathscr{B}} = \begin{pmatrix} D|_V & k_D \\ 0 & D_Q \end{pmatrix}.} \tag{31}$$

Thus the topological splitting is not substituted for an equivariant splitting. Equation (29) is the precise comparison between them.

For two cutoff choices, let s_0, s_1 be their sections and set $b_{10} = \Lambda_0 s_1 : Q \to V$. Then

$$s_1 - s_0 = \Theta b_{10}, \qquad k_a^{(1)} = k_a^{(0)} + U_a b_{10} - b_{10}\overline{U}_a. \tag{32}$$

All dependence on the chosen representative map is carried by this displayed coboundary. No canonical cutoff or average over choices is declared.

Restrict back to PR #9 without changing its extension class Use its existing maps $s_Z : E_Z \to \mathscr{B}$, $\sigma_Z = qs_Z : E_Z \to Q$, and $J_Z : \mathscr{B} \to E_Z$. Put $b_Z = \Lambda s_Z$. The new continuous representative of the same finite arithmetic class is

$$s_Z^{\text{new}} = s\sigma_Z = Ks_Z = s_Z - \Theta b_Z. \tag{33}$$

If $P(D)s_Z - s_Z M_P = \Theta k_{P,Z}$, direct substitution gives

$$k_{P,Z}^{\text{new}} = k_{P,Z} - P(D)b_Z + b_Z M_P. \tag{34}$$

For $P = h_Z$, this changes the source boundary by $-h_Z(D)b_Z$. It is zero in $V/h_Z(D)V$, so the extension class remains exactly

$$\varepsilon_Z = j_{h_Z}(h_Z/g), \qquad g = 2\xi. \tag{35}$$

The polynomial residue form still becomes the arithmetic one by M_{ε_Z}; its Jacobian contraction remains $M_{\varepsilon_Z} M_{j_h g'}$. No unit, derivative, sign, or factor of 2 is changed.

The two finite cochain projectors differ by the explicit homotopy $-b_Z J_Z$ in degree minus one. Their finite-rank trace difference is zero because $J_Z\Theta = 0$. Hence their cochain Lefschetz trace is the unchanged

$$-\sum_{\rho\in Z} m_\rho \mathcal{M}f(\rho). \tag{36}$$

C.1.9 8. Continuous duality and completed tensor powers of the full object

Use continuous duals, denoted E'_b with the strong topology. Transpose the actual topological splitting (20). It gives

$$\boxed{0 \to Q'_b \xrightarrow{q'} \mathscr{B}'_b \xrightarrow{\Theta'} V'_b \to 0, \qquad \Theta'\Lambda' = 1.} \tag{37}$$

The continuous cochain-dual complex is $[\mathscr{B}'_b \xrightarrow{\Theta'} V'_b]$ in degrees $-1, 0$. Its cohomology is Q'_b in degree -1 and zero in degree 0. The transpose contraction proves this directly; no general exactness theorem for arbitrary locally convex quotients is assumed.

The actual finite residue map into that continuous dual is

$$\overline{E_Z} \longrightarrow Q'_b, \qquad \bar{f} \longmapsto \big(u \mapsto R_Z(f, j_Z u)\big), \tag{38}$$

where $j_Z q = J_Z$. Continuity follows from the finite Mellin jets and finite dimensionality of E_Z. It is injective by their surjectivity and perfect residue duality. Multiplication by g' then gives the arithmetic finite trace pairing, with its previously calculated radical retained.

Let $\widehat{\otimes}_\pi$ denote completed projective tensor product. The comparison from the previous algebraic tensor is its canonical completion map. All chain operators in (24) are continuous. Thus their finite tensor products and homotopies extend to the completed products. With $P = sq$ on C_+, the homotopy

$$H_n = \sum_{j=1}^{n} P^{\widehat{\otimes}(j-1)} \widehat{\otimes} H \widehat{\otimes} 1^{\widehat{\otimes}(n-j)} \tag{39}$$

includes $(-1)^{\sum_{i<j}|x_i|}$ on homogeneous inputs. It satisfies

$$dH_n + H_n d = 1 - P^{\widehat{\otimes} n}.$$

Consequently, for the completed external coefficient complex on the same tau-base chart product,

$$\boxed{H^k(C_+^{\widehat{\otimes} n}) = \begin{cases} Q^{\widehat{\otimes} n}, & k = n, \\ 0, & k \neq n. \end{cases}} \tag{40}$$

This follows from a split continuous contraction, not from a blanket claim that completed tensor product preserves every exact sequence.

For the joint chart use (25). Its full cohomology in degree k is the direct sum, indexed by subsets $J \subseteq \{1, \dots, n\}$ with $|J| = k$, of the ordered completed tensors having Q in the slots of J and V in the other slots. In those V slots the scaling action is $aU_{1/a}$, exactly as in the original minus-coordinate representative. The degree-zero contribution is not discarded. Split reconstruction retains each product mask and each internal zero.

C.1.10 9. The pairing calculation is now a specified global boundary problem

The section $s : Q \to \mathscr{B} \subset L^2(dx)$ supplies a positive definite auxiliary form

$$\mathcal{H}_s(u, v) = \langle su, sv \rangle_{L^2(dx)}.$$

This is not assigned to the arithmetic Weil form. The explicit weight-covariance defect is calculated using (29) and $\langle U_a f, U_a g\rangle = a\langle f, g\rangle$:

$$\boxed{\begin{aligned}\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a\mathcal{H}_s(u, v) = & - \langle U_a su, \Theta k_a v\rangle - \langle \Theta k_a u, U_a sv\rangle \\ & + \langle \Theta k_a u, \Theta k_a v\rangle.\end{aligned}} \tag{41}$$

The original source boundary and both of its cross-pairings remain. Being a boundary in Q does not assert these pairings are zero in the representative Hilbert space. In a split lift each computed zero is its supported value e.

For a reduced eigenline of the actual dilation representation with exponent ρ, the left side at $u = v$ is $(a^{2\Re\rho} - a)\mathcal{H}_s(u, u)$. Formula (41) locates that defect in the full global boundary cocycle, with the original operator retained.

A concrete comparison explaining which topology must be kept In fact ΘV is dense in $L^2(\mathbb{R}_{>0}, dx)$, although it is closed in $\mathscr{B}$. Here is a proof retaining the map between them. Set $b_* = \Theta\phi_*$ and

$$W : L^2(dx) \xrightarrow{\sim} L^2(dy), \qquad WF(y) = e^{y/2}F(e^y).$$

The Fourier transform in y, with kernel $e^{-2\pi i\nu y}$, gives

$$\mathcal{F}_y W b_*(\nu) = g(1/2 - 2\pi i\nu).$$

This is nonzero almost everywhere, because g is a nonzero entire function. The span of all translates of Wb_* is dense in $L^2(dy)$: a vector orthogonal to all translates has Fourier transform whose product with $\overline{\mathcal{F}_y W b_*}$ is an L^1 function with zero Fourier transform, hence vanishes almost everywhere. The nonvanishing just stated then makes that vector zero. Since $WU_a b_* = a^{1/2}(Wb_*)(y - \log a)$ and $U_a b_* \in \Theta V$, density follows.

The exact quotient comparison is therefore

$$\boxed{Q = \mathscr{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2} = 0.} \tag{42}$$

Its kernel is all of Q. Its split lift sends every supported input to the supported zero in $G(0) = \mathbb{B}$, while keeping external absence separate. The constructive replacement is the Frechet quotient and the explicit global section (19), not a claim that the ordinary Hilbert quotient secretly retains those classes.

C.1.11 10. Scope of the progress toward the Deligne comparison

The selected supplied Weil II passages 3.3.4-3.3.6 put two bounds on the same compact-to-ordinary image; 6.2 retains the dualizing operation and its shifts. This continuation does not apply those bounds over a new base without their geometric proof. It makes the existing characteristic-zero arithmetic complex globally split in its precise topological category, gives its continuous dual and completed products, and computes the operator defects on those same objects.

The formerly unresolved algebraic-to-Hausdorff kernel is now zero by (16)-(21). The finite extension class and its residue/Jacobian trace are carried by (33)-(36). The full scaling boundary is (29), its dependence on the auxiliary cutoffs is (32), and its metric defect is (41). These are concrete maps available for a global comparison.

Still retained: the common kernel of all finite spectral observations, the meromorphic summand of the balanced analytic realization, and the requirement to identify the correct arithmetic polarization/weight control on the global geometric image. The density map (42) explains why discarding the stronger topology would remove the entire arithmetic cohomology, not why the supported programme should be abandoned.

The reconstruction estimate contains $1 - c_{\alpha,\beta}$; no asymptotic uniform estimate for it is proved. No global positivity, RH, GRH, or infinite trace-class theorem is claimed. No theorem from the earlier finite-range work is upgraded to a Lean certificate. The new global retraction proof itself uses only the original theta/Poisson identities, absolute Mobius inversion, the displayed Fourier compactness argument, and the two retained moment maps.

C.1.12 11. Verification and references

The written analytic proofs are the evidence for the new analytic statements. The accompanying unittest script tests exact finite identities: divisor cancellation with supported zeros, cutoff-resolvent identities in a finite unitary model, moment corrections, contractions, synchronization masks, operator cocycles, changes of sections, preservation of extension classes, finite trace changes, and completed-tensor algebra before completion. It does not numerically certify a Fourier operator norm, an analytic convergence statement, or a Riemann zero.

Source interfaces read:

- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1-7; the source permits nonzero bottom fibres in its general interface.
- PR #9, workbenches/tau-chain-descent/RESEARCH_NOTE.md, at the commit above; local bytes match its Git blob.
- PR #8's finite comparison as retained and invoked in #9; the sections using s_Z and J_Z are explicitly dependencies of (33)-(38), not inputs to the proof of (2).
- Supplied Weil II TeX: the actual 3.3.1-3.3.6 excerpt, including its finite-type base and duality assumptions. It was read as TeX, not obtained by OCR. No full source-proof audit is claimed.
- Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277: prior-context abstract only in this continuation. A source-archive retrieval failed; no unread detailed theorem is cited as a premise.

The contribution is AI-assisted under the owner's mathematical direction. Original mathematical files, formal sources, attribution and licenses are unchanged. Only authored exposition, test code and validation records are proposed for public addition; no private transcript or source-literature corpus is published.

D PR 11: Homotopy control

Source attribution. Web contribution, pull request #11, Zeta research repository.

Exact revision. d37eade4f9e6a82c1a88aeff710a3b851a34eb49

Source file. `workbenches/tau-homotopy-control/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9924b358c0365a8cbdcddb3a7496f610c65d64617cd1860679b42e371d2fe88d

D.1 Homotopy choices and source-preserving arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #10, d26c283c2a58da57bb 5a5a4d4bc8669019f6ebb7. This integration note records the new derivations; the conversation delivery also contains a longer equation-by-equation exposition. No original scalar definitions or formal modules are changed.

D.1.1 1. Reviewed theorem and analytic specialization

The supplied other-session proof classifies all homotopies for

$$C = [V \oplus V \xrightarrow{d} B], \qquad d(v, w) = \Theta(v - \mathcal{F}w), \qquad Q = B/\Theta V,$$

where Theta is injective and F is an R-linear automorphism. For a fixed kappa:B->V with kappa Theta=0, every solution of BOTH equations

$$dH = \Theta\kappa, \qquad Hd = 0$$

has the unique form

$$H_\alpha(b) = (\kappa b + \alpha q b, \mathcal{F}^{-1}\alpha q b), \qquad \alpha : Q \to_R V.$$

Proof: writing H=(u,v), injectivity gives u-Fv=kappa. Since d(x,0)=Theta x, Hd=0 gives H Theta=0. Hence Fv factors uniquely through q, and supplies alpha. Conversely substitution proves both equations. The reference solution is H_0=(kappa,0); the family is an affine space under Hom_R(Q,V), relative to that reference.

Let gamma(v)=(v,F^-1 v). Then

$$H_\alpha - H_\beta = \gamma(\alpha - \beta)q.$$

The difference is a map H1(C)->H0(C). There are no degree-minus-two endomorphisms of this two-term coefficient complex. Thus it is a genuine degree-minus-one Hom-complex class, not an erased difference.

The supplied theorem is mathematically correct. Its Lean draft explicitly remains uncompiled, and its earlier failed chart build is not relabelled successful. No Lean or Lake executable is installed in the review environment.

For the actual analytic spaces instantiate R=C. With the quotient topology on Q, the continuous version follows: alpha q=F pr2 H is continuous, so the quotient property makes alpha continuous. Conversely the displayed formula sends continuous alpha to continuous H.

For the polynomial action A0=C[t], Fourier has the exact type

$$\mathcal{F} : V^{\jmath} \xrightarrow{\sim} V, \qquad t \cdot_{V^{\jmath}} v = (1-D)v, \qquad \jmath(t) = 1-t,$$

because F D=(1-D)F. Thus d is A0-linear from V+V^j to B. A complex-linear homotopy is not automatically A0-linear. The reflected module above is the actual comparison, not a change to the Fourier definition.

D.1.2 2. Original split support on every homotopy

Retain the base b_tau and the square G(Z)->G(C) over Z->C. The quotient G(Z)->Z remains infinite. Both lifted coefficient maps preserve tau and the supported scalar e separately.

Use the original support lattice L={bottom,+,-,J}, J={+,-}. Degree-zero fibres are 0,V,V,V+V with the coordinate inclusions. Degree-one active fibres are all B, with identity transports. Their differentials are Theta, -Theta F and d. Retain zero modules in the other degrees; their reconstructed zero map is z(lambda,x)=(lambda,0).

Literal multiplication by E_sync=(1,e) induces

$$r^0(a,b) = (a+eb, b+ea), \qquad r_L(\perp) = \perp, \quad r_L(\lambda) = J \ (\lambda \neq \perp).$$

Every homotopy lifts by

$$\widetilde{H}_\alpha(\lambda, b) = \big(J, (\kappa b + \alpha q b, \mathcal{F}^{-1}\alpha q b)\big), \quad \lambda \neq \perp; \qquad \widetilde{H}_\alpha(\tau) = \tau.$$

This is split-linear: two active inputs add at their joined label, both outputs add at J, and their amplitudes obey the two displayed linear formulas. A supported scalar zero gives (J,(0,0)); external tau gives the global zero. Therefore

$$\widetilde{d}\widetilde{H}_\alpha + \widetilde{H}_\alpha\widetilde{d} = r\widetilde{\Delta}.$$

The difference is (J,gamma(alpha-beta)qb), retaining its degree-zero class. For this chart, the pair (support(x),r(x)) is injective. No such injectivity is assumed for arbitrary diagrams with noninjective transports. For a smaller relation space W, keep Q_W->Q with kernel Theta V/Theta W; q in the formulas means the displayed quotient by the full admitted V.

D.1.3 3. Coherence for all dilations

Let S_a=U_a on V, T_a=U_a on B, U_a F(x)=F(x/a), and Tbar_a the quotient action. The actual cochain action is

$$\mathsf{T}_a^0 = (S_a, \mathcal{F}^{-1}S_a\mathcal{F}) = (U_a, aU_{1/a}), \qquad \mathsf{T}_a^1 = T_a.$$

It satisfies T_a^0 gamma=gamma S_a and qT_a=Tbar_aq. With the PR #10 section s, take P=(0,sq), and write

$$[\mathsf{T}_a, P]^1 = \Theta\kappa_a, \qquad \kappa_a = k_a q, \qquad T_a s - s\overline{T}_a = \Theta k_a.$$

The commutator product identity gives kappa_ab=S_a kappa_b+kappa_a T_b. The family of homotopies is composition-coherent,

$$H_{ab} = \mathsf{T}_a H_b + H_a \mathsf{T}_b,$$

exactly when

$$\boxed{\alpha_{ab} = S_a\alpha_b + \alpha_a\overline{T}_b.}$$

Proof: the reference family (kappa_a,0) is coherent. Subtract it and use injectivity of gamma and surjectivity of q. No action or boundary is omitted.

For a smoothly parameterized continuous-linear family with L=(d/dt)|*0 alpha*(e^t), the exact formula is

$$\alpha_{e^t} = \int_0^t S_{e^{t-r}} L\overline{T}_{e^r}\, dr.$$

This is a vectorwise integral in the complete Frechet source, with the original oriented integral for negative t. Differentiating the composition equation gives the formula; splitting the integral proves its converse. In particular alpha_a=S_a b-b Tbar_a has infinitesimal parameter D_V b-bD_Q.

D.1.4 4. A homotopy parameter leaves the control form fixed

Fix an actual finite arithmetic packet with full multiplicities,

$$h_Z(t) = \prod_{\rho\in Z}(t-\rho)^{m_\rho}, \qquad E_Z = \mathbb{C}[t]/(h_Z), \qquad A = M_t.$$

Use the existing R:E_Z->B and J_Z:B->E_Z with

$$qR = \sigma_Z, \quad J_Z R = 1, \quad J_Z\Theta = 0, \quad J_Z D = AJ_Z, \quad DR - RA = \Theta k.$$

The cochain projector P_R=(0,RJ_Z) has commutator (0,Theta kJ_Z), so the reviewed theorem applies with kappa=kJ_Z. The quotient jet Jbar_Z satisfies Jbar_Z sigma_Z=1, hence any alpha_E:E_Z->V extends to alpha_E Jbar_Z on Q. Restriction to R gives

$$\boxed{H_{\alpha_E}R = (k+\alpha_E, \mathcal{F}^{-1}\alpha_E), \qquad dH_{\alpha_E}R = \Theta k.}$$

Take H=L2(R_+,dx) with the original inner product, conjugate-linear in the first variable. Define

$$G_R = R^*R, \qquad B_R = \Theta k, \qquad W_R = -(R^*B_R + B_R^*R).$$

The composite of actual maps

$$\mathrm{Hom}_{\mathbb{C}}(E_Z, V) \longrightarrow \mathrm{Hom}(E_Z, V^2) \xrightarrow{d\circ(-)} \mathrm{Hom}(E_Z, H) \xrightarrow{B\mapsto -(R^*B+B^*R)} \mathrm{Herm}(E_Z)$$

is constant at W_R. Its last map is real-linear. The retained H0 ambiguity does not change this boundary contraction.

Integration by parts on the actual source functions gives

$$\boxed{A^*G_R + G_RA - G_R = W_R.}$$

The boundary term [x conjugate(F)H]_0^infinity vanishes by the stated decay. No claim that a theta boundary is orthogonal to a representative is used.

D.1.5 5. The actual representative-change family

Every representative of the same inclusion sigma_Z is uniquely

$$R_b = R + \Theta b, \qquad b : E_Z \to_{\mathbb{C}} V.$$

Indeed q(R'-R)=0 gives values in Theta V, and the PR #10 retraction gives the unique b=Lambda(R'-R). Finite source dimension gives continuity. Global continuous sections have the identical formula with b:Q->V continuous.

The complete changes are

$$k_b = k + D_V b - bA,$$

$$\delta G_b = R^* \Theta b + (\Theta b)^* R + (\Theta b)^* \Theta b,$$

$$\boxed{W_{R_b} - W_R = A^* \delta G_b + \delta G_b A - \delta G_b.}$$

Their extension-coordinate map is U_b(v,u)=(v+bu,u), and

$$U_b^{-1} \begin{pmatrix} D_V & k \\ 0 & A \end{pmatrix} U_b = \begin{pmatrix} D_V & k + D_V b - bA \\ 0 & A \end{pmatrix}.$$

Lift that automorphism on the common active fibre (V+E_Z)^tau. For the original masks, first keep the pair (support(x),r(x)); apply the shear on the joint fibre, and recover the original amplitudes by U_b^-1 and the original mask by its recorded component. No inverse to the unrecorded support collapse is asserted.

The projectors obey P_(R_b)-P_R=partial L_b with L_b^1(F)=(bJ_ZF,0). Their finite cochain traces agree because

$$\mathrm{Tr}(\Theta b J_Z H_f) = \mathrm{Tr}(J_Z H_f \Theta b) = 0.$$

The cochain supertrace sign remains -1 in degree one. The arithmetic extension class changes by h_Z(D)b(1), so its class in V/h_Z(D)V remains epsilon_Z=j_h(h_Z/(2xi)). The residue/Jacobian trace form is consequently unchanged.

D.1.6 6. Finite actual source ansatz and one master Gram matrix

Retain phi_*=(4pi^2 x^4-6pi x^2)exp(-pi x^2), phi_j=D^j phi_**, and

$$\Phi_m : \mathbb{C}^{m+1} \to V, \qquad (c_j) \mapsto \sum_{j=0}^{m} c_j \phi_j.$$

Their actual Mellin identities are M Theta phi_j(s)=2xi(s)s^j. They remain in the original two-moment source and are linearly independent. The source section R_0=s_Z has the exact rank-one defect DR_0-R_0A=Theta phi_* ell_Z, with the unchanged arithmetic remainder functional ell_Z.

Define maps, including the target enlargement,

$$I_m e_j = e_j, \qquad S_m e_j = e_{j+1}, \qquad I_m, S_m : \mathbb{C}^{m+1} \to \mathbb{C}^{m+2}, \qquad D\Phi_m = \Phi_{m+1} S_m.$$

For B in Mat_((m+1) x d)(C), d=dim E_Z, put

$$R_B = R_0 + \Theta\Phi_m B, \qquad K_B = e_0\ell_Z + S_m B - I_m BA.$$

Then

$$DR_B - R_B A = \Theta\Phi_{m+1}K_B, \qquad qR_B = \sigma_Z, \qquad J_Z R_B = 1.$$

Take the actual column map and its Gram matrix

$$\mathcal{T}_m : E_Z \oplus \mathbb{C}^{m+2} \to L^2(\mathbb{R}_+, dx), \qquad (u,c) \mapsto R_0 u + \Theta\Phi_{m+1}c, \qquad M_m = \mathcal{T}_m^*\mathcal{T}_m.$$

M_m is positive definite. Applying J_Z to a vanishing combination gives u=0; Mellin then gives c=0. Its entries are the convergent integrals of the displayed arithmetic functions, with the original measure.

Define

$$X_B = \begin{pmatrix} I_d \\ I_m B \end{pmatrix}, \qquad Y_B = \begin{pmatrix} 0 \\ K_B \end{pmatrix}.$$

The complete control calculation is

$$\boxed{G_B = X_B^* M_m X_B, \qquad W_B = -(X_B^* M_m Y_B + Y_B^* M_m X_B) = A^* G_B + G_B A - G_B.}$$

This restricts representative choice to an actual finite family of theta boundaries. It is not an arbitrary positive matrix ansatz, and it is not asserted to exhaust all continuous source corrections. All packet coordinates, multiplicities, local units and finite source-degree changes are retained.

D.1.7 7. Error transport, duality, and the remaining estimate

With actual enclosures ||M-Mhat||<=eta_M, ||X-Xhat||<=eta_X, ||Y-Yhat||<=eta_Y, put x=||Xhat||, y=||Yhat||, m0=||Mhat||. Product estimates give

$$\eta_G = \eta_M(x+\eta_X)^2 + m_0(2x\eta_X + \eta_X^2),$$

$$\eta_W = 2[\eta_M(x+\eta_X)(y+\eta_Y) + m_0(x\eta_Y + y\eta_X + \eta_X\eta_Y)].$$

These include coefficient errors, not just quadrature errors. Exact certificates for Ghat-eta_G I>0 and

$$\epsilon\widehat{G} \pm \widehat{W} - (\epsilon\eta_G + \eta_W)I \succeq 0$$

imply the actual two-sided control. This is an error-transfer theorem; arithmetic enclosures have not been computed here.

The unchanged residue matrix S satisfies S*=-S and A*S+SA=S. Its exact dual transport is

$$G_B^{\mathrm{D}} = S^* G_B^{-1} S, \qquad W_B^{\mathrm{D}} = -S^* G_B^{-1} W_B G_B^{-1} S.$$

Thus an upper certificate gives a lower certificate on the same arithmetic packet. The Weil form stays S J_g, with metric comparison G_B^-1 S J_g; no positivity is assigned by choosing G_B.

For tensor degree n, retain A_n=sum_j 1 tensor A tensor 1 and G_(B,n)=G_B^(tensor n). The product control is sum_j G_B tensor W_B tensor G_B. Changing a homotopy by gamma alpha q changes the tensor homotopy by its Koszul-signed insertions; its differential is zero, hence it does not change that control form.

The existing orbit construction on this actual tensor image gives

$$G_{B,n,T} = \int_0^T e^{-nt} e^{tA_n^*} G_{B,n} e^{tA_n}\, dt,$$

$$A_n^* G_{B,n,T} + G_{B,n,T} A_n - n G_{B,n,T} = e^{-nT} e^{TA_n^*} G_{B,n} e^{TA_n} - G_{B,n}.$$

Every occurrence of G_(B,n) now has the explicitly constrained source in section 6.

A nonzero boundary need not have nonzero control. The exact calibration D_amb=diag(1/2+i,1/2+2i), Theta(v)=(v,0), A=1/2+2i, R(u)=(u,u) has boundary Theta(-i), Gram 2, and control zero. It is an algebraic calibration, not a zeta packet. The real-linear map B->-(R*B+B*R) explains which nonzero boundaries can have zero control.

Conversely every eigenvector Av=rho v obeys

$$\frac{v^* W_B v}{v^* G_B v} = 2\Re\rho - 1.$$

This ensures that changing representatives has not changed a spectral coordinate. The analytic target is now to bound the actual matrices from section 6, or their tensor-orbit endpoints, with all packet and tensor dependence retained. No uniform purity estimate, RH, GRH, or global spectral-kernel vanishing is proved here.

D.1.8 8. Validation and attribution

The attachment supplies the homotopy-classification theorem and the uncompiled draft. This note verifies its proof and adds the continuous/twisted interface, supported reconstruction, coherent-family condition, and the connection to the actual control matrices. The original owner-directed SplitZero construction remains the source.

The independent checker has sixteen exact unittest methods, including finite parameter families. They passed normally and under python -O with identical successful JSON records. Intentional negative controls failed in both modes. The previous control checker also passed its sixteen methods. These finite checks cover algebraic models and polynomial-times-exponential integral calibrations, not evaluated zeta packets or analytic/Lean certification.

General references: The Stacks Project, Tags 0A8H and 0115 for the Hom-complex differential and homotopy, and 010I and 06XP for extension classes. The formulas above are proved on the specified programme objects; those references are not imported as a number-field weight theorem. Only authored notes, test code and validation records are published; no private transcript, source-literature corpus, or font files are included.

E PR 13: Orthogonal boundary control

Source attribution. Web contribution, pull request #13, Zeta research repository.

Exact revision. e7ba4c29d4503bf20d06cb70c93c3347f199f1b4

Source file. `workbenches/tau-orthogonal-boundary-control/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9d364d9cc2f6840af8eb39555e4b89622acd7714fbcab7ca135e8ec0ce25ab0b

E.1 Orthogonal theta boundaries and rank-two arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #11, d37eade4f9e6a82c1a88 aeff710a3b851a34eb49. This is an additive research contribution with written proofs for review, not a new Lean certificate for the analytic results. The conversation delivery contains a longer numbered LaTeX/HTML version, the full reading record, and exploratory arithmetic moments.

E.1.1 1. Reviewed input and exact types

The supplied other-session theorem is correct: for injective Theta:V->B, an R-linear automorphism F, Q=B/Theta V, and kappa:B->V with kappa Theta=0, all homotopies satisfying BOTH dH=Theta kappa and Hd=0, for d(v,w)=Theta(v-Fw), are

```
H_alpha(b)=(kappa b+alpha qb, F^-1 alpha qb),   alpha:Q->R V.
```

Writing H=(u,v) gives u-Fv=kappa. Since d(x,0)=Theta x, Hd=0 gives H Theta=0; hence Fv descends uniquely to alpha. Substitution proves the converse. The difference is gamma(alpha-beta)q with gamma(v)=(v,F^-1v). The solution set is affine relative to H_0, not generally a vector space with the zero homotopy as origin.

The attachment's uncompiled-draft status has now been superseded by the other session's successful PR #12 at 5d2772ea0a16d177ac3e01ff70394b90ba95a232. The actual SplitZeroTauHomo topy.lean, completed job 103535438942, and its full log were read. Both cochain equations and the inverse equivalence passed the specified kernel checks. Retraction and quotient-topology interfaces also passed with their explicit hypotheses. This is the other session's formalization; neither the actual analytic theta construction nor the new Gram/control results below are certified by it. No local Lean/Lake execution is claimed. See COORDINATION_UPDATE.md for exact scope. PR #11 already supplies the continuous specialization, the Fourier twist over C[t], support-changing lifts, coherent dilations and representative-change formulas. Those are retained inputs, not new results here.

Keep the original tau base and the square G(Z)->G(C) over Z->C, with p:G(Z)->Z still infinite, e=ofR(0), and tau the external scalar zero. No Boolean localization replaces the base quotient. All quotient relations below use the original internal coequalizer and preserve the support label.

Retain V, B and the actual operators of PRs #8-#11:

```
V={even Schwartz phi on R: phi(0)=0 and integral(phi)=0};
B={smooth F on R_+: sup x^b |D^k F(x)|<infinity for all integer b and k>=0};
D=-x d/dx; Theta phi(x)=2 sum_(n>=1) phi(nx);
M F(s)=integral_0^infinity F(x)x^s dx/x;
phi_*=(4 pi^2 x^4-6 pi x^2)exp(-pi x^2);
phi_j=D^j phi_*; f_j=Theta phi_j=D^j f_0;
M f_j(s)=s^j g(s), g=2 xi.
```

The Hilbert observation is H=L2(R_+,dx), conjugate-linear in the first argument. Polynomial packet coordinates are retained, not changed to orthonormal coordinates.

For an actual finite packet with all multiplicities, let E=C[t]/h_Z, A=M_t. The previous analytic constructions supply

```
R_0:E->B, J_Z:B->E, qR_0=sigma_Z, J_ZR_0=1, J_ZTheta=0,
DR_0-R_0 A=f_0 ell_Z.
```

The exact ell_Z comes from epsilon_Z=j_h(h/(2 xi)); at a simple zero its value on 1 is 1/(2 xi'(rho)). These analytic constructions remain the cited research-proof dependencies, not theorems certified by the finite tests here.

E.1.2 2. An actual minimum over the admitted theta relations

For m>=0 define

```
W_m=span(phi_0,...,phi_m);
Phi_m:C^(m+1)->V, c |-> sum c_j phi_j;
F_m=Theta Phi_m:C^(m+1)->H;
H_m=F_m* F_m; C_m=F_m* R_0; G_0=R_0*R_0.
```

The f_j are independent: Mellin sends a relation to g(s)P(s)=0. Thus H_m is positive definite. Here C_m is a cross-Gram matrix E->C^(m+1), not the complex of that name in the endpoint workbench.

Set

```
B_m=-H_m^-1 C_m;
Pi_m=F_m H_m^-1 F_m*;
R_m=R_0+Theta Phi_m B_m=(1-Pi_m)R_0.
```

The correction is a finite original theta boundary, so R_m lands in B, qR_m=sigma_Z, and J_ZR_m=1 at every m. Direct multiplication proves

```
F_m*R_m=0;
G_m=R_m*R_m=G_0-C_m*H_m^-1 C_m >0.
```

Positive definiteness follows from J_ZR_m=1. For any B:E->C^(m+1),

```
(R_0+F_m B)*(R_0+F_m B)
  =G_m+(B-B_m)*H_m(B-B_m).
```

This is the minimum of the Hilbert Gram within the stated source family, not yet a minimum of its relative weight error. Also (1-Pi_m)(R_0+F_m B')=R_m: changing the initial representative by a relation already in W_m leaves the result unchanged.

E.1.3 3. The complete control form has rank at most two

Let I_m,S_m:C^(m+1)->C^(m+2) be I_m e_j=e_j and S_m e_j=e_(j+1). Define

```
K_m=e_0 ell_Z+S_m B_m-I_m B_m A.
```

Then

```
DR_m-R_m A=F_(m+1)K_m=Theta(Phi_(m+1)K_m).
```

No boundary component is discarded in this identity. Put b_m=e_m*B_m:E->C, and z_m=R_m*f_(m+1):C->E. Integration by parts on the stated source functions gives

```
<DF,H>+<F,DH>=<F,H>,
```

with vanishing boundary term [x conjugate(F)H]_0^infinity. Hence

```
W_m=A*G_m+G_m A-G_m
   =-(R_m*F_(m+1)K_m+K_m*F_(m+1)*R_m)
   =-z_m b_m-b_m*z_m*.
```

The last equality uses F_m*R_m=0 and the exact last row of K_m, which is b_m. Thus rank(W_m)<=2 for every packet dimension and every finite m. The pairings with the first m+1 theta directions vanish; those supported vectors themselves are retained.

Define the actual scalar quantities

```
u_m=z_m*G_m^-1 z_m >=0;
v_m=b_m G_m^-1 b_m* >=0;
c_m=b_m G_m^-1 z_m.
```

The nonzero spectrum of G_m^-1 W_m is the nonzero spectrum of

```
-[[c_m,v_m],[u_m,conjugate(c_m)]].
```

This follows by factoring through the two-column map [z_m,b_m*] and using the equality of nonzero spectra of the two compositions. Its trace is -2 Re(c_m), determinant |c_m|^2-u_m v_m. Therefore the least symmetric control allowance is

```
epsilon_m=|Re(c_m)|+sqrt(u_m v_m-(Im(c_m))^2),
-epsilon_m G_m <= W_m <= epsilon_m G_m.
```

Cauchy-Schwarz makes the radicand nonnegative. The formula also covers rank one. A reflection-stable full packet has

```
trace(G_m^-1 W_m)=2 Re(trace A)-dim E=0,
```

so Re(c_m)=0. This is an exact trace identity, not a choice of coordinates. For every eigenvector Av=rho v,

```
(v*W_m v)/(v*G_m v)=2 Re(rho)-1.
```

Thus the relative control still detects precisely the original spectral weight.

E.1.4 4. The next relation is a rank-one update with its own retained coordinate

Set

```
a_(m+1)=(1-Pi_m)f_(m+1)=Theta psi_(m+1),
psi_(m+1)=phi_(m+1)-Phi_m H_m^-1 F_m*f_(m+1),
eta_(m+1)=||a_(m+1)||^2 >0.
```

Then R_m*a_(m+1)=z_m, and orthogonal decomposition gives

```
Pi_(m+1)=Pi_m+a_(m+1) eta_(m+1)^-1 a_(m+1)*;
R_(m+1)=R_m-a_(m+1) eta_(m+1)^-1 z_m*;
G_(m+1)=G_m-eta_(m+1)^-1 z_m z_m*.
```

The control update is obtained by applying the same real-linear operator X |-> A*X+XA-X to the last identity.

The original quotient transport is

```
Q_Wm=B/Theta W_m -> Q_W(m+1)=B/Theta W_(m+1),
kernel=Theta W_(m+1)/Theta W_m ~= W_(m+1)/W_m.
```

That kernel is one-dimensional. The square E->Q_Wm by R_m and E->Q_W(m+1) by R_(m+1) commutes. The new relation becomes the supported zero only through the actual next-level quotient; it is not another scalar adjunction at the bottom.

There is a form on the entire quotient:

```
Hcal_m([F],[G])=<(1-Pi_m)F,(1-Pi_m)G>.
```

It is positive definite since the kernel of 1-Pi_m in B is exactly Theta W_m. Let lambda_(m+1)[F]=<a_(m+1),F>. This is well defined on Q_Wm and

```
Hcal_m(x,y)=Hcal_(m+1)(qx,qy)
            +eta_(m+1)^-1 conjugate(lambda(x))lambda(y).
```

The map (q,lambda):Q_Wm -> Q_W(m+1) plus C is a linear isomorphism. Its inverse is

```
([F],c) |-> [F-a eta^-1<a,F>+a eta^-1 c]_Wm.
```

Changing F by a relation at level m+1 changes this expression only by a relation at level m. This supplies the exact metric comparison between levels instead of treating the quotient transports as isometries.

Every displayed linear map has its original fixed-support lift. On the full source chain, reconstructed addition uses the join max(m,n) and the above transports. The scalar e sends (m,[F]) to (m,[0]); tau sends it to external absence. The two-leg primitives are (k_m u,0) and (0,-Fourier^-1 k_m u), where k_m=Phi_(m+1)K_m. Synchronization by the original E_sync=(1,e) supplies their common joint support. Their difference remains the H0 cycle (k_m u,Fourier^-1 k_m u). The other session's Hom(Q,H0) freedom leaves their differential and hence W_m unchanged.

E.1.5 5. Actual arithmetic moments and their truncation allowance

The matrices H_m come from the original arithmetic functions. Since JF(x)=x^-1 F(1/x) is unitary on L2(dx), Jf_0=f_0 and JD=(1-D)J,

```
H_ij=integral_1^infinity [conjugate(f_i)f_j
             +conjugate(f_i^ref)f_j^ref] dx,
f_j^ref=(1-D)^j f_0.
```

This retains both multiplicative ends and permits absolutely convergent Gaussian sums on x>=1. Write

```
phi_j(x)=P_j(x)exp(-pi x^2),
P_0=4 pi^2 x^4-6 pi x^2,
P_(j+1)=2 pi x^2 P_j-x P_j',
P_j=sum_r c_(j,r)x^(2r),
c_(j+1,r)=2 pi c_(j,r-1)-2r c_(j,r).
```

For f_j^N=2 sum_(n=1)^N P_j(nx)exp(-pi n^2x^2), put

```
C_j=sum_r |c_(j,r)|sqrt(Gamma(2r+1/2)/(2 pi^(2r+1/2))).
```

Then

```
||f_j-f_j^[N]||_L2([1,infinity))
  <= 2 C_j (N+1)^(-1/2) exp(-pi(N+1)^2/2)
    /[1-exp(-pi(2N+3)/2)].
```

Proof: on x>=1, 2 pi n^2x^2 >= pi n^2+pi n^2x^2. Integrating each monomial's squared bound gives n^-1/2 exp(-pi n^2/2) times the displayed gamma constant. Minkowski and a geometric bound on consecutive n terms give the formula. The reflected derivative retains the absolute binomial sum of these allowances.

The finite integral entries use exactly

```
integral_1^infinity x^(2k)exp(-c x^2) dx
  =Gamma(k+1/2,c)/(2 c^(k+1/2)), c=pi(n^2+n'^2).
```

For truncation tail norms delta_i,delta_j, each inner-product error is bounded by delta_i||f_j^[N]||+||f_i^[N]||delta_j+delta_i delta_j, plus its reflected contribution. Numerical evaluation errors for the finite expression still require their own enclosure. The optional conversation computation is explicitly exploratory, not interval-certified.

The exact consistency check is H_(i+1,j)+H_(i,j+1)=H_ij. Cross-Gram columns F_m*R_0 remain their actual source integrals and are not replaced by arbitrary matrices.

E.1.6 6. The same source proves a strong topology comparison

For this actual f_0,

```
closure span{D^j f_0:j>=0} in L2(R_+,dx) = H.
```

Proof. The explicit unitary Mellin-line map sends F to M F(1/2+it) in L2(R,dt/(2 pi)). It sends D to 1/2+it and f_j to (1/2+it)^j g(1/2+it). The log-space function h(z)=exp(z/2) f_0(exp z) is holomorphic in |Im z|<pi/4 by its Gaussian series. For |Im z|<=a<pi/4 it decays superexponentially as Re z tends to positive infinity. Poisson gives h(z)=h(-z), so the same holds at the other end. Fourier contour displacement therefore gives |g(1/2+it)|<=C_a exp(-a|t|).

The measure dmu=|g(1/2+it)|^2dt/(2 pi) has a positive exponential moment. If an L2(mu) function is orthogonal to all polynomials, multiplying mu by its conjugate gives a finite complex measure with a smaller positive exponential moment. Its Fourier transform is analytic in a strip and all derivatives at zero vanish. Analyticity and uniqueness of the Fourier transform imply that measure is zero. Thus polynomials are dense in L2(mu). Multiplication by g is unitary from this weighted space onto L2(dt/(2 pi)), because the nonzero analytic function g on the line is nonzero almost everywhere. This proves the assertion without RH.

As a result, for a fixed finite packet,

```
R_m -> 0 in Hom(E,H), G_m -> 0 and W_m -> 0,
while qR_m=sigma_Z and J_ZR_m=1 for every m.
```

This specifies why unscaled smallness of W_m cannot be used alone as weight control: its denominator G_m is also shrinking. The exact relative formula in section 3 must be retained.

The corrective map is the Hilbert/jet graph:

```
closure{(F,J_ZF):F in B} = H plus E;
closure{(Theta phi,0):phi in V} = H plus 0.
```

Indeed finite theta polynomials give every (h,0) in the closure, and the sequence (R_m u,u) gives (0,u). For a specified positive metric on E, completion in ||F||_2^2+||J_ZF||_E^2 is this graph closure, and the quotient by the boundary closure is E by [(h,u)]|->u. The added metric is retained supplied data, not an implied positivity theorem for the Weil pairing.

Its split observation is

```
((R_m u)^bullet,u^bullet) -> (e_H,u^bullet)
```

in the product of the stated support-fibre topologies. The arithmetic component survives above a supported Hilbert zero. External absence is (tau,tau).

E.1.7 7. Duality, trace and scope

The original residue map has matrix S with S*=-S and A*S+SA=S. Its pullback of the dual metric is

```
G_m^D=S*G_m^-1 S,
W_m^D=-S*G_m^-1 W_m G_m^-1 S.
```

Thus the upper-to-lower comparison is retained on the same arithmetic packet. The Weil pairing stays S J_g, with explicit metric comparison G_m^-1 S J_g. Its positivity is not imposed by choosing a positive Gram matrix.

All representative changes above are original theta boundaries. Their extension class j_h(h/(2 xi)) is unchanged; the induced finite cochain projectors differ by the homotopy (bJ_Z,0). The trace difference is zero because J_ZTheta=0. Degree-one and tensor Koszul signs remain those of the source complex. On a raw tensor metric, the inserted W_m terms give the copied allowance n epsilon_m; tensoring alone does not improve it.

The new results are the actual orthogonal source choice, rank-two boundary formula, exact relative certificate, rank-one source-kernel recurrence, arithmetic moment tail bound, and the full Hilbert/jet topology comparison. No uniform arithmetic purity estimate, global spectral-kernel vanishing, RH or GRH is asserted.

The thirteen finite exact tests check the encoded algebra and declared calibration models, not infinite analytic arguments or actual zeta zeros. The complete cochain-homotopy theorem remains the other session's result, now supported by the independently inspected successful PR #12 run. Its certificate does not extend to the new analytic control results. Existing Lean modules and earlier validation receipts are untouched.

References: live PR #11 at the pinned head; successful PR #12 and source/job references in COORDINATION_UPDATE.md; the supplied Tau_Deligne_Control and Tau_Global_Comparison source notes; the supplied other-session homotopy proof; the previously retained Weil II 3.3.4-3.3.6 and 6.2 duality excerpts. Standard orthogonal-polynomial context is NIST DLMF 18.2; the density argument needed here is proved above for the actual arithmetic measure.

F Derived Split-Zero mathematics

Source attribution. Web contribution retained on the pre-integration main revision.

Exact revision. 14c69d604044b848d64318751a7bee19edfd9aba

Source file. `formal/splitzero/DERIVED_MATHEMATICS.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 6a290e193161eb09f49b6faba578193fdaf28b5ae3a97340d1c41a40aa99e695

F.1 SplitZero: reconstruction, internal homology, and balanced finite jets

Owner-directed formalization continuation, 12 September 2026.

F.1.1 Scope and provenance

This contribution extends formal/splitzero on the owner's Zeta workbench. Its base is the synchronization PR #5, commit 236f0a8370477991e2d0cf334ad1ebfe4f3f8b17, itself based on the structural PR #4. The original scalar core and all five preceding extension sources remain unchanged. There is no Sneed-side submission or integration in this continuation.

The mathematical source is the owner's SplitZero programme, including the supplied continuation's internal homology and balanced-Mellin sections. The implementation below distinguishes source-derived structure, ordinary reusable algebra, and the new explicit comparison example. It makes no global priority claim and does not treat the number of Lean declarations as a number of mathematical discoveries. No private conversation or source-literature corpus is republished.

The exact build certificate for a revision is the completed SplitZero\derived\constructions workflow on that commit, including direct source checks and its generated AuditDerived.lean. The PR records the accepted execution revision and run. A failed earlier development run is not a certificate for later source.

F.1.2 1. Reverse reconstruction, with actual scalar action

Let R be a commutative ring, L a join-semilattice with bottom, and let D consist of R-modules V_i with linear transition maps

```
rho_ij : V_i -> V_j  (i <= j),
rho_ii = id,  rho_jk rho_ij = rho_ik.
```

No finiteness, field, injectivity, or nontriviality assumption is imposed. Construct

```
Total(D) = disjoint union of V_i over i in L.
```

The global zero is (bottom,0), and

```
(i,x) + (j,y) = (i join j, rho_i,i-join-j(x) + rho_j,i-join-j(y)).
tau . (i,x) = (bottom,0).
ofR(r) . (i,x) = (i,r x).
```

LinearDiagram.totalAddCommMonoid and LinearDiagram.totalModule construct the actual instances and prove every law. For associativity, transport both sides to i join j join k. Linearity and transition composition reduce both to the same three-term sum. The scalar laws follow in the supported case from the R-module laws and in the absent cases from the chosen global zero. The supported-zero scalar sends (i,x) to (i,0).

The bottom fibre is allowed to be nonzero. Specializing it to the zero module recovers the source's carrier consisting of one absent point and the non-bottom fibres. This is an explicit generalization, not an assumption that the bottom fibre in every later dual diagram vanishes. Opposite-index duality itself is not constructed here.

Fixed-index natural families of linear maps produce actual G(R)-linear maps between reconstructed totals. Identity and composition are proved. More substantially, Recovery.recoverEquiv reconstructs the intrinsic support fibres of **any** existing G(R)-semimodule M and gives a full G(R)-linear equivalence back to M:

```
(1,x) |-> x,        m |-> (e m,m).
```

Its addition proof uses the original absorption identity m + e m = m. It is not merely a bijection of carriers. A bundled equivalence of the two full categories, including arbitrary changes of support index and both natural inverse comparisons, is a separate library interface not claimed by this implementation.

F.1.3 2. The internal quotient is characterized by a universal property

Let B_i be transition-stable submodules of V_i. The quotient diagram has fibres V_i/B_i and the genuinely induced quotient transition maps. The projection

```
q : Total(D) -> Total(D/B),  q(i,x) = (i,[x])
```

is a surjective G(R)-linear map. The code proves the exact equality criterion

```
q(i,x)=q(i,y)  iff  x-y is in B_i.
```

Different labels cannot become equal under q. In particular a relation maps to its own fibre zero.

The main point is stronger than that observation. Let N be **any** G(R)-semimodule, not assumed additively cancellative, and let f:Total(D)->N be split-linear. If

```
f(i,b)=f(i,0)  for every b in B_i,
```

then there exists a unique split-linear g:Total(D/B)->N with g q=f. The proof does not subtract elements in N. If x-y is in B_i, use the identity y+(x-y)=x inside V_i and then apply additivity of f. This proves constancy on q-fibres. Surjectivity constructs the descended function; linearity and uniqueness follow by lifting representatives.

Relations.coequalizer_universal states this against the actual inclusion of the relation diagram and its supported-zero companion. Thus the universal property is proved against arbitrary semimodule targets, not only against a selected class of fixed-support maps. The implementation gives the complete unbundled universal property; it does not merely define the desired quotient and label it a coequalizer.

F.1.4 3. Actual homology and the exact transported-class kernel

At a degree i of a cochain complex, use the window

```
C^(i-1) --dPrev--> C^i --dNext--> C^(i+1),  dNext dPrev=0.
```

Window.boundaryToCycles constructs the incoming map into ker(dNext). Boundaries are its actual linear range, and H is the module quotient of cycles by that range. Window.ofCochain imports these data at every integer degree of a Mathlib CochainComplex; ChainMap.ofCochain imports an actual cochain map without replacing its differentials.

For a chain map f:C->D, the code constructs

```
f_Z : Z(C) -> Z(D),
H(f) : Z(C)/B(C) -> Z(D)/B(D).
```

It proves identity, composition, and

```
H(f)([z])=0  iff  there exists y with dPrev_D(y)=f_i(z).
```

Define the module of killed representatives and its original relation submodule:

```
Krep = {z in Z(C) : f_Z(z) is in B(D)},
Bold = B(C), viewed as a submodule of Krep.
```

There is a constructed linear equivalence

```
Krep/Bold  ~=  ker H(f).
```

Forward map: the class of z goes to the kernel element [z]. It is well-defined because Bold consists exactly of original source boundaries. It is injective because equality of source classes is exactly a difference in Bold. It is surjective because every kernel class has a cycle representative whose image is a boundary. This is ChainMap.homologyKernelEquiv, obtained from a general quotient-kernel equivalence proved in the same file.

The calculation retains actions as well. Given chain endomorphisms a of C and b of D satisfying f a=b f, homology_intertwines proves H(f) H(a)=H(b) H(f). kernelAction constructs the restriction of H(a) to ker H(f). Thus passing to the comparison kernel does not discard a compatible operator or require asserting that the kernel is zero.

F.1.5 4. Internal homology on the complete support diagram

For coherent support-indexed chain windows, the code constructs the diagrams of preceding terms, middle terms, following terms, cycles, boundaries, and homology. Let R_L denote the reverse reconstruction.

The differential identity is typed as

```
R_L(dNext) R_L(dPrev) = z_L : R_L(C^(i-1)) -> R_L(C^(i+1)),
z_L(l,x)=(l,0).
```

The source and target degrees are explicit. A theorem shows that z_L differs from the constant global-zero map whenever L has a non-bottom element. Cycles satisfy the actual equalizer equation d(x)=e.d(x).

The internal homology projection

```
q : R_L(Z(C)) -> R_L(H(C))
```

is the coequalizer of the **original incoming differential into cycles** and its supported-zero companion. homology_coequalizes and homology_coequalizer establish both parts of that universal property against every split-linear target. This is not an assumed homology compatibility of reconstruction.

For every transition i<=j,

```
rho^H_ij([z])=0  iff  rho_ij(z) is a boundary at j.
```

If z is not a boundary at i, its retained class (i,[z]) is not (i,0), regardless of whether that transition kills it. The quotient-kernel equivalence of section 3 computes all such classes as a module, not just a yes/no predicate.

F.1.6 5. A concrete comparison that detects more than the support labels

Fix any nontrivial commutative ring k. In the displayed degree let

```
C: k --0--> k --0--> k,
D: k --a |-> (a,0)--> k^2 --0--> k.
```

Use zero maps in the outside degrees. The two middle chain maps are

```
f(a)=(a,0),           g(a)=(0,a).
```

The code proves that the source class [1] is nonzero. Its f-image is the **nonzero vector** (1,0), but that vector is a boundary in D, so H(f)([1])=0. In fact H(f)=0 on every source class. The g-image (0,1) cannot be a boundary, because every boundary has second coordinate zero. Hence H(g)([1]) is nonzero and H(f) is not H(g).

The ordinary calculation gives H(C)=k and H(D)=k (the latter through second-coordinate projection). These identifications explain the example; the Lean acceptance tests directly prove the nonzero source class, the nonzero chain image, the zero induced map, the surviving class, and the distinction of induced maps rather than relying on a dimension computation.

Consequently the source and target complexes, individual fibre homology, and terminal homology can be identical while the transition kernel differs. This example is parametric over all nontrivial commutative rings, not a finite numerical experiment. Together with the general assembly theorem it supplies an explicit test of the information retained by support-indexed homology. It is an algebraic test case, not a claimed model of the global theta comparison.

F.1.7 6. Balanced finite jets, including the original split quotient square

Let k and B be commutative rings and fix a k[t]-algebra structure on B. Let u:k[t]->B denote that map and

```
I_(rho,m)=((t-rho)^m),    m>=0.
```

There is an explicit B-algebra equivalence

```
B tensor_(k[t]) (k[t]/I_(rho,m)) ~= B/((u(t)-u(rho))^m).
```

It maps b tensor [p] to [u(p)b], with inverse [b] to b tensor 1. No flatness assumption is used for this quotient base-change theorem. This is a specialization of the existing Mathlib theorem Algebr a.TensorProduct.quotIdealMapEquivTensorQuot, not a claim to have independently developed general tensor-product quotient theory.

splitBaseChange lifts this equivalence to the original G-construction on both sides and proves the commutative square with reflect. It preserves tau and e separately and is a genuine semiring equivalence, not just an amplitude comparison. Thus every element of the finite-jet quotient has an inverse image before forgetting split support.

For any k[t]-module V and v in V, the exact balancing calculation is

```
(1-u(t)) tensor (t.v) - u(t) tensor (v-t.v) = 0.
```

Expanding gives 1 tensor (t.v)-u(t) tensor v, which vanishes by the tensor balancing relation. This is the algebraic equation of the source's Mellin repair. The choice u(t)=s and t.v=Dv is an application once those analytic module structures and intertwining maps have been constructed. It does not assert that the entire global balanced kernel vanishes, or that analytic scalar extension is flat without proof.

F.1.8 7. Where Deligne enters the intended RH route

None of sections 1-6 imports a Deligne purity theorem. They provide the algebraic maps with which a proposed realization can now be specified and its kernel retained.

For this programme, the subsequent dependency diagram should distinguish the following mathematical obligations:

1. Construct the actual operator-balanced theta complex at the relevant support labels, and its action-compatible analytic comparison beta to the spectral sheaf O/(2 xi). Identify its kernel or cone using the same maps; a finite-jet isomorphism is not a global injectivity argument.
2. Construct the geometric compact-support and ordinary complexes, their actual comparison c, their correspondence action, and a compatible realization identifying the required spectral sheaf with the image of H^1(c). Any additional kernel/cone contributions to the trace must be kept or proved to vanish.
3. Establish two-sided weight control on that **same image**. In Deligne's geometric setting the compact-support upper estimate and the duality-derived ordinary-cohomology lower estimate meet on the image. The dualizing operation, degree and Tate twists, tensor-power maps, and hypotheses of the estimates are part of this step, not consequences stipulated in the definition of support.
4. Transport the spectral normalization without changing the arithmetic function. For a dilation parameter a>1, an eigenvalue a^rho has modulus a^(Re rho). A proved weight-one modulus a^(1/2), for the representation identified with every relevant spectral zero, then forces Re rho=1/2. This last calculation does not supply the missing weight estimate; it states the precise final implication of this route.

These are dependencies of the specified route, not a claim that every possible proof of RH must follow it. The newly implemented kernel action is useful in steps 1-2 precisely because it keeps the part killed by a comparison available to the operator and trace calculation.

The supplied Connes-Consani article has an actual quotient-sheaf realization with a specified closure and scaling action. Its Theorem 1.2 realizes **critical zeros**. That theorem alone cannot be substituted for an identification representing all nontrivial zeros in the desired purity argument. This is a concrete source/target distinction for a future comparison map, not a dismissal of the construction.

F.1.9 8. Reading record and references

The six supplied S20 chunks were checked against their individual declared hashes and concatenated in order. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256 e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The current top-level readers/english_standalone.html and readers/source_language.html were used for the selected readings, not the nested zero-accepted salvage corpus. The inspected windows include the tensor-square improvement in 3.2.13, the upper-bound statement in 3.3.1, and the duality/image argument in 3.3.4-3.3.6. This is not an independent audit of every page or of the entire historical proof. The Connes-Consani introduction and Theorem 1.2 were read in the supplied volume, printed pages 83-85 (physical PDF pages 94-96). No copyrighted source corpus is included in this contribution.

Deligne, P. (1980). La conjecture de Weil. II. *Publications Mathematiques de l'IHES, 52*, 137-252. https://doi.org/10.1007/BF02684780

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and zeta-cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83-102). American Mathematical Society.

Mathlib contributors. (2026). *Mathlib4*, revision fabf563a7c95a166b8d7b6efca11c8b4dc9d911f. In particular, `LinearAlgebra/Quotient/{Defs,Basic}.lean`, RingTheory/TensorProduct/Quotient.lean, Algebra/Homology/HomologicalComplex.lean, and Algebra/Category/ModuleCat/Basic.lean.

F.1.10 9. Reproduction

From formal/splitzero on this branch, with elan and Python installed:

```
python3 prepare.py
python3 check_derived.py --prepare
python3 -m unittest -v test_derived_checker.py
lake update
# Verify the resolved Mathlib revision before fetching its cache.
test "$(git -C .lake/packages/mathlib rev-parse HEAD)" = fabf563a7c95a166b8d7b6efca11c8b4dc9d911f
lake exe cache get
lake -KmaxJobs=1 build
for name in SplitZeroReconstruction SplitZeroRecovery SplitZeroHomology SplitZeroInternalQuotient
↪  SplitZeroComplex SplitZeroHomologyExample SplitZeroFiniteJets; do
  lake env lean --trust=0 -DwarningAsError=true "$name.lean"
done
lake env lean --trust=0 -DwarningAsError=true AuditDerived.lean > AuditDerived.log
python3 check_derived.py AuditDerived.log
```

The manifest enumerates the selected declarations, including the principal structures, universal properties, equivalences, and example proofs. Missing, duplicate, unexpected, or nonstandard axiom reports fail. Source checks reject unfinished proofs and native-reduction escapes. The Python tests validate this checking machinery; the mathematical certificate is the successful Lean execution. Build-job totals include dependencies and are not theorem counts.